Multiplexed CRISPRi Perturb-seq in Yeast

Methods, Theoretical Limits, and Genome-Scale Cost

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Abstract

Pooled genetic screens read out by single-cell RNA sequencing map perturbations to their transcriptional consequences at dense, transcriptome-wide resolution. This work establishes what such a screen costs in *S. cerevisiae*, what sets the ceiling on how many genes can be perturbed in one cell, and which parameter is binding. It then asks how the design scales along its two real axes, perturbations per cell and environments per experiment, and what changes when the host is *P. kudriavzevii* rather than *S. cerevisiae*. It serves two purposes: selecting a method, and explaining the chemistry and the statistics well enough that the relevant passages can be reused in a methods write-up. Numbers are regenerated from committed scripts rather than transcribed, and every claim not backed by a hash-pinned artifact in the library mirror carries a visible flag, listed in the key below.

How to read this. Almost every subsection ends in a boxed statement of what it established. Reading only those boxes, in order, gives the whole argument and every number a decision depends on; the surrounding text is the evidence for them. The exceptions are the ranked list in section 6.1, the plan in section 6.3 and the ordering in section 7.4, which are already summaries and should be read whole.

Working notes: `notes/experiments.024-perturb-seq-costing.*.md`

Generating scripts: `experiments/024-perturb-seq-costing/`

References: Zotero `torchcell/microbe-perturb-seq` (group ∪ personal)

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Provenance: ▲ not in the library mirror; verify at the linked source ● from a review's summary table, not the primary paper

Contents

1 Introduction ✗

Metabolic engineering builds strains by stacking perturbations, but the design decisions are made from data on single perturbations measured one at a time. Pooled CRISPR interference screens read out by single-cell RNA sequencing (**Perturb-seq**) close that gap: a library of guide RNAs (**gRNA**s) is transformed into a population, every cell is sequenced, and each cell's gRNA identity is recovered from its own transcriptome, so one experiment yields a matrix from perturbation to full transcriptional response rather than a list of fitness hits [1]. The genome-wide MAGIC library this design would screen returns exactly such a list, scored by growth [2]; the same library read out by Perturb-seq returns the matrix instead. That matrix is the highest-density source of genotype-to-expression training data available.

Two properties of the readout are worth separating, because running them together is what makes the advantage sound larger than it is. The *transcriptional* readout is what scores perturbations a growth screen cannot see: a gene can be fitness-neutral and still move hundreds of transcripts. *Single-cell* resolution is what makes a pooled library readable at all, because it assigns each transcriptome to the gRNA that produced it. The effect estimate is not per cell: cells sharing a gRNA are summed into one pseudobulk profile, so precision is set by how many cells share a gRNA rather than by how deeply any one of them was read (section 4.4).

Three developments have made the yeast version tractable. A split-pool barcoding chemistry removes the instrument from the cell-isolation step [3]; pooled CRISPR activation and interference have been joined to single-cell transcriptomics in bacteria, the first single-cell CRISPR screen in that domain [4]; and a genome-scale yeast deletion collection has been rebuilt so its genotype tag is transcribed and therefore visible to the assay [5]. What follows establishes what such a screen costs in *S. cerevisiae*, how many genes can be perturbed simultaneously before the measurement stops being interpretable, and which parameter is actually binding.

Two further questions decide what gets run first, and a cost table answers neither. Section 7 asks how the design grows along the two axes a metabolic engineering campaign cares about, more perturbations per cell and more environments per experiment. They do not cost alike: the cells needed to observe a *named* gene combination grow quadratically in the size of the panel, while environments cost linearly, so an environment is by far the cheaper way to enlarge a dataset. That is an argument for spending on environments rather than a verdict, because a gene combination and a growth condition are different kinds of information and a model that proposes edits needs both.

Section 6.2 asks what changes when the host is *P. kudriavzevii*, and splits the answer in two. The perturbation tool is close to hand: a reference genome and catalytic Cas9 are published for this organism, and a working CRISPRi system, along with a strain assembly from JGI, is held in house and unpublished▲ [external: in-house, unpublished], which makes it a validation task rather than a research project. Delivery and scale are not close to hand. Transformation efficiency is three to five orders of magnitude below *S. cerevisiae*, and section 4.8 shows library scale is set by transformation, so a pooled library-scale screen in this host is out of reach until that moves.

The first experiment therefore carries no perturbations at all. Part of that is forced by the paragraph above, and the more useful part is chosen: nobody on this project has run the technology before, so an unperturbed run buys footing. It answers the three things that are genuinely uncertain about moving host, wall digestion, UMIs recovered per cell and ribosomal fraction, while depending on no genetics at all (section 6.2). If one unperturbed run looks wasteful for the cells it consumes, the way to spend it better is across several environments, not by adding guides to it.

2 Measurement Principles and Terminology ✗

This section fixes the vocabulary. Everything after it assumes these terms.

2.1 Molecular Identifiers ✗

Table 1 is the reference: every term the document uses in a technical sense, with a link to the section that treats it and, where an entry asserts a number, the source of that number. It is worth a first pass rather than only being consulted, because the field's naming is genuinely treacherous and three conventions are adopted here to keep it from spreading.

- **One thing is called an index**: the Illumina index read added at the final PCR. Anything written into a cell is a *barcode*, of which there are four kinds. The field's method name "combinatorial indexing" is the one place the word means barcode, and it is flagged where it appears.
- **Screen budgets are quoted in cells per *target gene***, never cells per guide. At six guides per gene the two differ by a factor of six, and reading one as the other is the most common way a budget here gets misread.
- q **is the per-guide detection probability** and nothing else. Where a source uses q for something else, its symbol is displaced and the original recorded at the point of definition.

The prose below then develops the four identifiers properly.

Table 1: Controlled vocabulary, alphabetical. The last column links to the section that treats each term in depth, so an entry is a pointer rather than a replacement for the argument. An entry that asserts a *number* ends in the citation that number comes from; entries with no citation are definitional. Generated from `experiments/024-perturb-seq-costing/scripts/glossary.py`.

Term	Definition	See
Additivity assumption	That two perturbations in one cell move a gene by the sum of what each moves it alone, in log space. Every pooled decoding needs it, and it is a model rather than a fact. Yao et al. do not claim interactions are absent but that they cancel over random combinations; measured in yeast they do not cancel, they buffer, with a double producing 0.62 of the sum of its singles. What that costs an analysis is a systematic direction rather than noise: main effects pooled over multiplexed cells are biased toward zero as k rises, so a $1/k$ saving in cells is optimistic by whatever the buffering is at that k, which nobody has measured past $k = 2$.	section 4.7
Barcode collision	Two distinct cells receiving the same full barcode and being merged into one apparent cell. Rate is $1 - ((B-1)/B)^{C-1}$ for C cells in a barcode space of size B, so it is controlled by adding rounds or sublibraries, not by sequencing harder.	section 5.4
Biological floor	The cells a perturbation needs once sequencing has stopped being the limitation. It is not independent of sequencing at any real depth; it is what the requirement converges to as depth grows, the φ_j term of eq. (5) with the shot-noise term driven to zero. Cell-cycle phase and metabolic state have to be averaged over, and reading one cell more deeply measures that one cell better rather than averaging anything. The published field standard is 100 cells. [4]	section 4.4
Biological overdispersion (φ_j)	Variance in a gene's expression between genetically identical cells over and above what counting alone would produce. The *over* is the comparison against shot noise, which fixes the variance at μ; φ_j measures the excess. From cell-cycle phase, metabolic state and transcriptional bursting, so reading each cell more deeply does not reduce it and only more cells do. Equal to the negative-binomial dispersion, so $\sqrt{\varphi_j}$ is the biological coefficient of variation of the RNA-seq literature.	section 4.1
Blocking strand	Added after each ligation round to outcompete the linker and retire that well's barcode, so it cannot participate in a later round. Without it, leftover oligos scramble the barcode sequence.	section 3.1
Cell barcode (CB)	A sequence shared by every cDNA molecule from one cell and ideally unique to it; what makes the data single-cell rather than bulk. Delivered pre-synthesized on a bead in droplet methods, built up across rounds of well-specific ligation in split-pool.	section 2.1
Cells per target gene	Cells carrying a guide against that gene, pooled over all of its guides, and the unit every budget here is quoted in. *Not* cells per guide and *not* cells per plasmid: at six guides per gene the two differ by a factor of six.	section 4.4
Channel	One microfluidic lane of a droplet chip, and the unit droplet reagent is sold in. A Chromium chip runs several channels side by side; each is loaded and priced independently and recovers up to ∼20,000 cells, so more cells means proportionally more channels and proportionally more money. That is what "cost scales with cells" means concretely. A split-pool plate has no equivalent – it is priced per plate however many cells pass through it, which is a cost advantage and not a throughput one: what a plate can carry is capped by cells per RT well and by barcode collisions, both of which bind well before price does (section 5.4).	section 3.2
Combinatorial fluidic indexing (scifi)	Preindexing on a plate followed by droplet capture, so a cell's identity is the pair (round1 well, round2 droplet). Neither round identifies a cell on its own. It composes the two isolation principles rather than choosing between them, and it is what removes the Poisson loading limit from droplet throughput.	section 3.3
Combinatorial split-pool barcoding	No isolation at all. Fixed, permeabilized cells act as their own reaction vessels; the population is split across a plate, barcoded in situ, pooled, and re-split. After R rounds of 96 wells there are 96^R possible paths. Cost scales per well, not per cell, so a plate costs the same for ten thousand cells or a million.	section 2.2
Composite sample	One measurement of a linear combination of perturbation effects rather than of a single perturbation: a cell carrying m guides, or a droplet holding m singly-perturbed cells. Conventional screening is the case $m = 1$. The unit compressed sensing counts, and the reason its sample requirement can be smaller than the number of perturbations.	section 4.7
Depth sufficiency (d^*)	The per-cell depth $d^* = 1/(\varphi_j p_j)$ at which shot noise and biological overdispersion contribute equally, where p_j is the target gene's transcriptome share and φ_j its overdispersion. Below it a screen is measurement-limited and deeper reads help; above it the requirement is within a factor of two of a floor no sequencing can cross.	section 4.9

Table 1, continued

Term	Definition	See
Doublet	Two cells reported as one, and each isolation principle fails into it differently. In plain droplet capture it is two cells in one partition, governed by Poisson loading (10x quote ∼8% at 20,000 cells recovered). In split-pool it is a barcode collision, two cells taking the same path through the plates. Under combinatorial fluidic indexing it is neither of those: several cells per droplet is the operating point, so a doublet is the narrower event of two cells sharing *both* their round1 well and their round2 droplet, which is why overloading stops being ruinous.	section 2.2
Droplet isolation	One cell co-encapsulated with one barcoded bead in an emulsion droplet; the droplet wall is what keeps a cell's molecules together. Reagent cost scales linearly with cells, because you pay per channel.	section 2.2
Droplet overloading	Deliberately loading a droplet generator far above the doublet-avoiding concentration. Ruinous without preindexing, since every multiply-occupied droplet becomes an unresolvable doublet; the operating point with it, since those droplets resolve into several usable cells. Measured at 95.5% droplet occupancy and 9.6 nuclei per droplet at 100-fold overloading. [6]	section 3.3
Effect matrix	Perturbations by genes, each entry the log fold change that perturbation causes in that gene. It is the object a screen estimates, the $\mathbf{\Theta}$ of eq. (1), and an expression compendium of single deletions already is one – which is what makes its rank and sparsity measurable before any screen is run.	section 4.7
FR-Perturb	The decoder for a compressed screen: factorizes the observed expression matrix by sparse PCA, then recovers per-perturbation effects by LASSO against the pooling design. The recovered effects are estimates, so power depends on the design matrix as much as on cell count.	section 3.4
Guide barcode (GBC)	An expressed, polyadenylated sequence that stands in for the gRNA itself. It rides on the same plasmid as the guide, transcribed from a separate Pol II cassette, and the design that removes a whole class of bookkeeping error is to make the barcode *be* the guide's own 20 nt spacer placed in that cassette's 3′UTR. It is needed because a Pol III guide cassette carries no poly(A) tail and is therefore invisible to 3′ capture.	section 3.6
Guide-detection probability (q)	The chance that one gRNA present in a cell is actually seen in that cell's reads. A k-plex cell is usable only if all k are seen, so the usable fraction is q^k and q is the one parameter that enters the multiplex ceiling exponentially rather than linearly. The published range is very wide, 21–25% in bacteria against >71% in the genome-scale yeast screen, and which end applies here is the single most valuable thing the pilot can measure. [5]	section 3.6
Guide-pooling vs cell-pooling	Two ways to compress a screen. Guide-pooling puts several guides in one cell; cell-pooling combines separately perturbed cells before sequencing. Both rely on the perturbations composing additively, and both are decoded rather than read directly.	section 3.4
Guide RNA (gRNA)	The RNA that targets dCas9 to one gene, and the perturbation itself in a CRISPRi screen. Written *gRNA* throughout; *sgRNA* appears only inside quotations and vendor documentation, where it means the same molecule. In yeast it is transcribed from a Pol III promoter, so it is neither capped nor polyadenylated, which is the whole of why perturbation identity is hard to read out (section 3.6).	section 3.6
Linker strand	A splint oligo pre-annealed to a barcode, positioning it against the 5′ end of the growing strand so T4 DNA ligase can seal the nick. Ligation is templated by the linker, which is why a stray barcode from an earlier well would corrupt the address.	section 3.1
Main effect	The average effect of perturbing one gene, pooled over every cell carrying a guide against it regardless of what else that cell carries. The pooling is what makes it cheap and is also what makes it blind: averaging over every genetic background a cell might have is exactly the operation that integrates epistasis away, so a main-effect estimate cannot report an interaction and a screen powered only for main effects is not powered for one. Named interactions need cells carrying the specific combination, which is eq. (3).	section 4.6
Perturb-seq	A pooled genetic screen read out by single-cell RNA-seq, so each cell yields both a perturbation identity and a whole-transcriptome phenotype. The transcriptome, not a growth rate, is the phenotype.	section 2
PhiX spike-in	A fraction of the loaded library that is control phage DNA. At ≥10% it is a split-pool protocol requirement rather than hygiene, and the reason is the read structure rather than the organism: read 2 carries fixed linker sequence at defined cycles whatever cell the library came from, so base diversity collapses at those cycles and patterned flow cells mis-register clusters without a diverse library mixed in. It applies to a yeast split-pool run exactly as it does to a bacterial one. Budget it as a tax on usable reads. [7]	section 5.3
Plex (k)	gRNAs delivered per cell, so k genes are perturbed in that cell at once. *Under the additive model* raising k divides the cell requirement for main effects by k, because each cell then reports on k genes instead of one; the division is a property of that model, not of the biology, and it is what section 4.7 tests. Raising k does not make genome-wide pairwise interactions reachable either way.	section 4.6

Table 1, continued

Term	Definition	See
Preindexing	One round of split-pool barcoding done before the cell reaches any container: a well-specific barcoded primer reverse transcribes in situ, so the cell arrives carrying a label of its own. Several preindexed cells can then share a droplet and still be told apart (section 3.3).	section 3.3
Protocol run	One complete pass through the barcoding workflow – fix, permeabilize, three rounds, split into sublibraries. Consumes one withdrawal from each barcode plate. Capacity is a range, not a number, because it is set by how much of the round-1 plate is loaded: Brettner et al.'s published run filled 48 of 96 wells at ~5,000 cells each and so carried ~240,000 cells, while a full plate at the same density carries ~480,000. A unit of *bench work*, not of sequencing: the sublibraries it yields are sequenced and priced separately. [3]	section 5.1
Pseudobulk	Summing UMIs across all cells sharing a perturbation, to estimate that perturbation's mean expression. Under multiplexing the pool is every cell carrying a guide against that gene *whatever else it carries*, not only cells with an identical guide set, which is what makes the estimand the main effect. It is what the whole design is powered on: single cells are too shallow individually, so the quantity that matters is cells × depth.	section 4.4
Rank and sparsity (r, ν)	The two properties that decide whether an effect matrix can be recovered from fewer samples than it has columns: r how many components carry its variance, ν how many genes one perturbation moves. Measured in yeast as $r \approx 177$ at 90% of variance and $\nu \approx 269$ of 6,169 genes. Yao et al. write ν as q; it is renamed here because q is the per-guide detection probability throughout this document and a document carrying two q's has to warn about it on every use.	section 4.7
Read pair	One paired-end fragment, and the unit every sequencing price in this document is quoted in. Both method families need paired-end runs, one read carrying the cDNA and the other the barcodes, but they put the barcode on opposite reads.	section 5.3
rRNA depletion	Removal of ribosomal cDNA from the amplified library, after barcoding and before fragmentation and indexing. It does not improve capture; it stops ~94% of purchased reads being spent on rRNA, and is the single highest-leverage change available to a split-pool screen.	section 3.5
Sample index	The Illumina index read added at final PCR. Its ordinary job is to let several libraries share one flow cell: each is tagged, sequenced together, and separated computationally afterwards, which is what makes a lane divisible between experiments. See *sublibrary index* for the extra job it does in split-pool.	section 5.3
Semipermeable capsule (SPC)	A hydrogel shell, pore cutoff around 30 nm, that confines cells and their nucleic acids while letting enzymes diffuse in. A fourth isolation principle alongside droplet, plate and split-pool: like a droplet it is a physical container, but unlike a droplet it survives buffer exchange, so a capsule can be washed, sorted and carried through multi-step chemistry. Small molecules leak, which is what rules it out for a metabolomic readout.	section 6.4
Shot noise	The irreducible randomness of *counting*: a run reads a random sample of the library, so a gene whose expected count is μ UMIs comes back scattered around μ with variance also μ, hence relative error $1/\sqrt{\mu}$. Note that μ counts molecules of one gene pooled over every cell sharing a perturbation, not molecules in one cell. The only noise buying more reads removes. Also called counting or sampling noise.	section 4.1
Source plate / working plate	A source plate is the pre-diluted barcode-oligo stock as shipped; working plates are aliquots cycled in daily use. The distinction is economic: source plates are retired by freeze–thaw damage rather than by depletion, so they behave as a capital asset.	section 5.1
Spheroplast	A yeast cell whose β-glucan wall has been digested away, here with zymolyase under osmotic support. Required because no commercial chemistry can get reagents through an intact fungal wall; the trade is that a wall-less cell bursts in a normal buffer.	section 2.3
Splint	The same molecule as the linker strand, named for what it does: bridge two DNA ends so ligase can seal the nick between them. Ligation, not polymerization, which is the whole difference from a template-switching oligo, where a polymerase copies the oligo rather than a ligase joining to it. The two are not rival chemistries and not a split-pool against droplet split: split-pool ligates its barcodes on and template-switches its second handle on, in the same protocol (fig. 3). Sources call the jig a linker strand or a bridge oligonucleotide; splint is the general name.	section 3.1
Sublibrary	An aliquot of barcoded cells (5,000–20,000) split off after the last in-cell round, lysed and prepared as one Illumina library. It is not a droplet channel and has no capacity limit of its own: a run is split into sublibraries because doing so buys barcode space (see *sublibrary index*) and because it is the granularity at which a prep can fail cheaply. The count is bounded below by the barcode-collision target and above by PCR complexity, and several sublibraries are routinely pooled onto one flow cell, where their sample indices separate them again.	section 5.1

Table 1, continued

Term	Definition	See
Sublibrary index	A sample index used as a barcode round. Split-pool divides cells into sublibraries *after* the last in-cell barcoding round, so two cells that took identical paths through the plates end up with identical BC1–BC2–BC3 and are separated only by which sublibrary they landed in. The index therefore multiplies the barcode space by the sublibrary count rather than merely naming a sample, and unique dual indices become mandatory: an index hop moves a cell into another sublibrary's namespace and manufactures a barcode collision rather than mere sample bleed.	section 5.4
Tagmentation	Fragmentation and adapter addition in one step, performed by the Tn5 transposase. Tn5 is loaded with adapter duplexes, cuts double-stranded DNA and ligates an adapter to each cut end in the same reaction, so it replaces a separate shear-then-ligate workflow. Figure 2 and fig. 5 draw it as an enzyme body rather than a sequence block, because it is a protein acting on the molecule, not a piece of the molecule.	section 2.1
Template switch	The reaction that adds a common handle to the far end of the cDNA, giving the molecule PCR primer sites at both ends; everything after it is conventional Illumina library prep. **A 3′ library does this too** – it is how the second handle gets on, and a molecule handled at one end only cannot be amplified. What 3′ and 5′ name is which oligo carries the cell barcode: in 3′ the barcode is on the oligo-dT and the TSO is free and carries only the handle; in 5′ they swap, which is what puts the barcode beside the protospacer and makes a Pol III guide readable (section 3.6).	section 3.1
Template-switching oligo (TSO)	The oligo that adds the second PCR handle, in 3′ and 5′ chemistries alike, and *not* the same mechanism as a linker. It happens *within the single reverse-transcription reaction*, with no melting, re-annealing or second primer extension, which is what distinguishes it from overlap extension: the enzyme reaches the 5′ end of the mRNA and adds two or three non-templated C residues; the TSO, whose 3′ end is rGrGrG, base-pairs with those Cs while the enzyme is still engaged; and the enzyme carries on, now copying the TSO instead of the mRNA. The handle is therefore written by the polymerase starting from the RNA's own 5′ end, which is why a TSO-delivered barcode marks the 5′ end of the transcript while an oligo-dT-delivered one marks the 3′ end.	section 2.1
Transcriptome share (p_j)	The fraction of a cell's mRNA belonging to gene j, so a cell read to d mRNA UMIs yields $d\,p_j$ UMIs of that gene. It is what converts sequencing depth into counts of the gene actually being measured, and it is per-gene and condition-dependent rather than one number. Currently assembled from two sources that do not belong together, which is worth a five-fold spread in cells per perturbation and is the largest unforced uncertainty in the budget.	section 4.2
UMI collision	Two distinct molecules of the same gene in the same cell drawing the same UMI, and being counted once. The comparison that decides whether this matters is *per gene per cell*, not against the whole transcriptome, because collapsing keys on CB, gene and UMI together – two molecules of different genes may share a UMI freely. So a 10 nt UMI's $4^{10} \approx 10^6$ states are set against at most a few thousand molecules of any one gene, and the collision rate is negligible. Not to be confused with barcode collision, which is a cell-level error.	section 2.1
Uninformative sequencing cycle	A sequencing cycle spent reading constant sequence – a ligation overhang, a primer binding site, a Tn5 mosaic end – rather than barcode or transcript. Paid for at the same rate as an informative one. Two thirds of SPLiT-seq's composite-barcode cycles are of this kind, and the reason is the read layout rather than any inefficiency in the run: the linkers sit between the barcodes on the molecule and the sequencer reads straight through in order, so 52 of read 2's 76 barcode cycles are linker. Worked out in section 5.3. [6]	section 5.3
Unique molecular identifier (UMI)	A short random sequence attached to each cDNA molecule *before* amplification, so every PCR copy inherits it and no two original molecules are likely to share one. That is how a copy is told from an original: reads sharing CB, gene and UMI are copies of a single mRNA and collapse to one count, while reads sharing CB and gene but differing in UMI are separate molecules. 10 nt in the split-pool protocols here, 10–12 nt in droplet. [7]	section 2.1
Usable vs sequenced cell	A usable cell clears three filters – enough UMIs to be a cell rather than ambient RNA, a barcode that resolves to exactly one cell, and an identified perturbation – while a sequenced cell is one whose reads were paid for. The two differ by ∼4× in split-pool, which is why a quoted cost per cell understates the real one.	section 5.2

A single-cell RNA-seq read is only useful if you can answer three questions about it: which cell did it come from, was it a distinct original molecule or a PCR copy, and which gene is it. Different chemistries answer these differently, but all of them attach the same four classes of tag.

Cell barcode (CB). A sequence shared by every cDNA molecule from one cell and, ideally, unique to that cell. It is what makes the data *single-cell* rather than bulk. In droplet methods the CB is delivered pre-synthesized on a bead; in split-pool methods it is *built up* across several rounds of well-specific ligation, so the barcode records the trajectory a cell took through a series of plates (Sec. 3.1).

Unique molecular identifier (UMI). A short random sequence – 10–12 nt in the methods here – added to each cDNA molecule *before* any amplification. Two reads sharing a CB, a gene, and a UMI are copies of one original mRNA; two reads sharing CB and gene but differing in UMI are two molecules. Collapsing on the UMI converts read counts, which are distorted by PCR efficiency *and by where fragmentation happened to cut* (section 2.4), into molecule counts, which are the actual measurement. That identity is exact for a pure oligo-dT assay and only approximate for one primed with random hexamers, for the reason section 2.4 gives. The count of distinct UMIs recovered from a cell is the single most important quality number in this document: it is the denominator of every statistical argument in Sec. 4. **UMI collision** – two distinct molecules of the same gene drawing the same UMI – is negligible at yeast expression levels, since a 10 nt UMI has $4^{10} \approx 10^6$ states against at most a few thousand molecules of any one gene.

Sample index. An Illumina index read added during the final PCR. Its ordinary job is divisibility. A flow cell delivers far more reads than any one library needs, so several libraries are tagged, pooled, sequenced together and separated computationally afterwards by their tags; without that, buying a lane would mean buying it for a single sample.

In split-pool the same read does a second job, and it is the one that is easy to miss. Cells are divided into sublibraries *after* the last in-cell barcoding round, so two cells that happened to take identical paths through the three plates carry identical BC1–BC2–BC3 and are indistinguishable by their in-cell barcode alone. If they landed in different sublibraries they are still resolved, because the sample index separates them. The index is therefore a fourth barcode dimension and multiplies the addressable space by the sublibrary count, which is why splitting a run into sublibraries is a barcode decision and not only a handling one (Sec. 5.4). It is also why unique dual indices are mandatory here: an index hop does not merely leak a read between samples, it moves a cell into another sublibrary's namespace and fabricates a barcode collision.

Perturbation identifier. Which perturbation the cell was carrying. This is the hard part in yeast and is treated separately in Sec. 3.6; the short version is that the gRNA itself is usually invisible to the assay, so a proxy has to be engineered.

> **Four tags, and only one of them is hard here.** A read is useful only if you can say which cell it came from, whether it is an original molecule or a PCR copy, and which gene it is. Cell barcode, UMI and sample index are solved by every chemistry on offer. The perturbation identifier is not: a yeast gRNA is invisible to a poly(A) assay, so a proxy has to be engineered or a different capture chemistry used (section 3.6).

2.2 Cell Isolation Strategies ✗

Physical isolation (droplet). A microfluidic device co-encapsulates one cell with one barcoded bead in an emulsion droplet, which isolates every molecule that cell releases. How many cells a droplet receives is not something the machine sets; only the average is, and the count around it is Poisson (section 4.1). A low average occupancy is therefore used to keep **doublets** – two cells in one droplet, reported as one – acceptably rare; 10x quote about 8% at 20,000 cells recovered. Holding occupancy low is what makes the droplet the sole source of cell identity, and it is also what wastes the channel: most droplets are fully reagented and never receive a cell.

Cost scales with cells, and the unit is worth being concrete about because it drives every droplet figure in section 5. Reagent is sold per **channel**: a Chromium chip carries several channels, each loaded and priced separately, and each recovers up to ∼20,000 cells. Two 20,000-cell recoveries are therefore two channels and twice the reagent cost – there is no per-cell dial, only a count of channels. At the UIUC rate a genome-scale screen at 250 cells per target gene needs 137 channels at $2,143 each (section 5.5), which is why library preparation, not sequencing, dominates the droplet column of the budget.

Combinatorial split-pool barcoding. No isolation at all. Cells are fixed and permeabilized so they become their own reaction vessels: their membranes retain the cDNA while enzymes diffuse in. The population is distributed across a 96-well plate, a well-specific barcode is attached in situ, the cells are pooled, redistributed, and barcoded again. After R rounds a cell carries an ordered barcode $\mathrm{BC}_1 \| \dots \| \mathrm{BC}_R$, and with 96 wells per round there are 96^R possible paths. Cells sharing round 1 are overwhelmingly likely to differ by round 3.

The cost consequence is the important one: *you pay per well, not per cell.* A round of barcoding consumes one 96-well plate of oligo and one tube of enzyme whether ten thousand cells or a million pass through it, so the reagent line is flat in cell number and the only per-cell costs left are sequencing and the per-sublibrary prep. Brettner et al.'s barcoding reagent works out to $0.005 per cell and Gaisser et al.'s per-run barcode cost is flat by construction (section 5.1); the entire economic case for split-pool is this one line, and section 5.2 tests whether it survives contact with rRNA waste and discarded cells.

That is a cost property and not a throughput one, and the two are easy to run together. Flat pricing does not mean a plate carries unlimited cells: what caps a run is how many cells a reverse-transcription well tolerates, ∼5,000 in the only yeast measurement (section 3.1), and how many distinct barcodes the rounds supply before collisions bind (section 5.4). What flat pricing does mean is that raising either cap is free at the till, which is why cells per RT well is ranked as the cheapest experiment on the list (section 6.1) and why a droplet run has no equivalent lever.

The two composed (preindexing). The two paragraphs above are usually read as alternatives, and they are not. Barcode a cell in situ once, on a plate, and then load it into a droplet: identity is now the *pair* (well, droplet), so several cells can share a droplet and still be resolved. The Poisson argument that forces low occupancy is an argument about the droplet being the only label, and it stops applying as soon as the cell carries one of its own. Datlinger et al. measured the consequence, loading 100-fold over the manufacturer's recommendation with a stable emulsion and 95.5% of droplets occupied [6]. The cost consequence follows directly from the pricing unit named above: reagent is sold per channel, so putting more cells through one channel divides the dominant term of the droplet budget without touching sequencing (section 3.3, section 5.5).

The landscape between the two poles. Those are the extremes; fig. 1 places the six approaches actually used in yeast between them and pairs each with the library chemistry it can support. The organizing fact is that the isolation choice and the barcode choice are not independent. A method that pools cells before sequencing has to write identity into the molecule, and *which* oligo carries the barcode fixes *where on the transcript* it ends up – an oligo-dT primer puts it at the 3′ end, a **template-switching oligo** at the 5′ end, Tn5 wherever it cuts. Plate methods keep one cell per well until the very end and need no cell barcode at all: the Illumina sample index identifies the well and the well identifies the cell. That index is a sample tag, not a **unique molecular identifier** – it names which library a read came from, whereas a UMI distinguishes two molecules *within* one cell, and a plate method still needs its own UMI to do that.

> **The whole cost comparison reduces to one line: split-pool pays per well, droplet pays per cell.** A barcoding round consumes one plate of oligo whether ten thousand cells or a million pass through it, so its reagent term is flat in cell number. Droplet reagent is sold per channel at ∼20,000 cells, so a genome-scale screen needs 137 of them. Preindexing is what lets the two compose, and it attacks exactly the term that makes droplet expensive.

2.3 Microbial Constraints on Single-Cell Chemistry ✗

Every commercial single-cell chemistry was developed on mammalian cells, and three properties of a yeast cell break its assumptions.

- **A cell wall.** Reagents cannot diffuse in and the cell cannot lyse on the mammalian schedule. Both method families solve this with zymolyase, a β-glucanase. Digestion is not optional – without it no reagent reaches the transcriptome – but *when* it happens decides whether the cell survives to be isolated, because a wall-less spheroplast is fragile and lyses in a normal buffer. That timing is the whole of Jariani et al.'s contribution (Sec. 3.2).
- **Roughly a fifth the mRNA.** A yeast cell holds on the order of 10^4 mRNA molecules against 10^5 for a mammalian cell (Sec. 4.2). Less input means fewer UMIs recovered, which is why per-cell yeast data always looks poor next to a mammalian benchmark. It is also why cells are pooled rather than read individually: at a few hundred UMIs a single cell cannot estimate any gene's expression, but the cells carrying one perturbation are biological replicates of the same genotype, so summing their counts recovers that genotype's mean profile. The single-cell step is doing *assignment* – which cell carries which perturbation – and the pooling is doing the measurement (Sec. 2.5).
- **An overwhelming rRNA background.** About 85% of a yeast cell's RNA is ribosomal. This is a property of the RNA pool, not an unavoidable property of the library: ribosomal RNA is not polyadenylated, so an oligo-dT primer does not prime on it and no separate depletion step is needed. A strictly poly(A)-selective chemistry – 10x's bead oligo, for instance – therefore never sees most of that 85%. The problem arises only when a protocol deliberately adds a non-selective primer. Brettner et al. prime with oligo-dT *and* random hexamers, and hexamers anneal to anything including rRNA, which is why their recovered transcripts are 93.7% rRNA while Jariani et al.'s oligo-dT-only droplet run is not (Sec. 3.5). The hexamer is there to reach transcripts without a poly(A) tail, which is what carries the method to bacteria; in yeast it is largely a tax, and as Sec. 5 shows that single fraction dominates the sequencing bill.

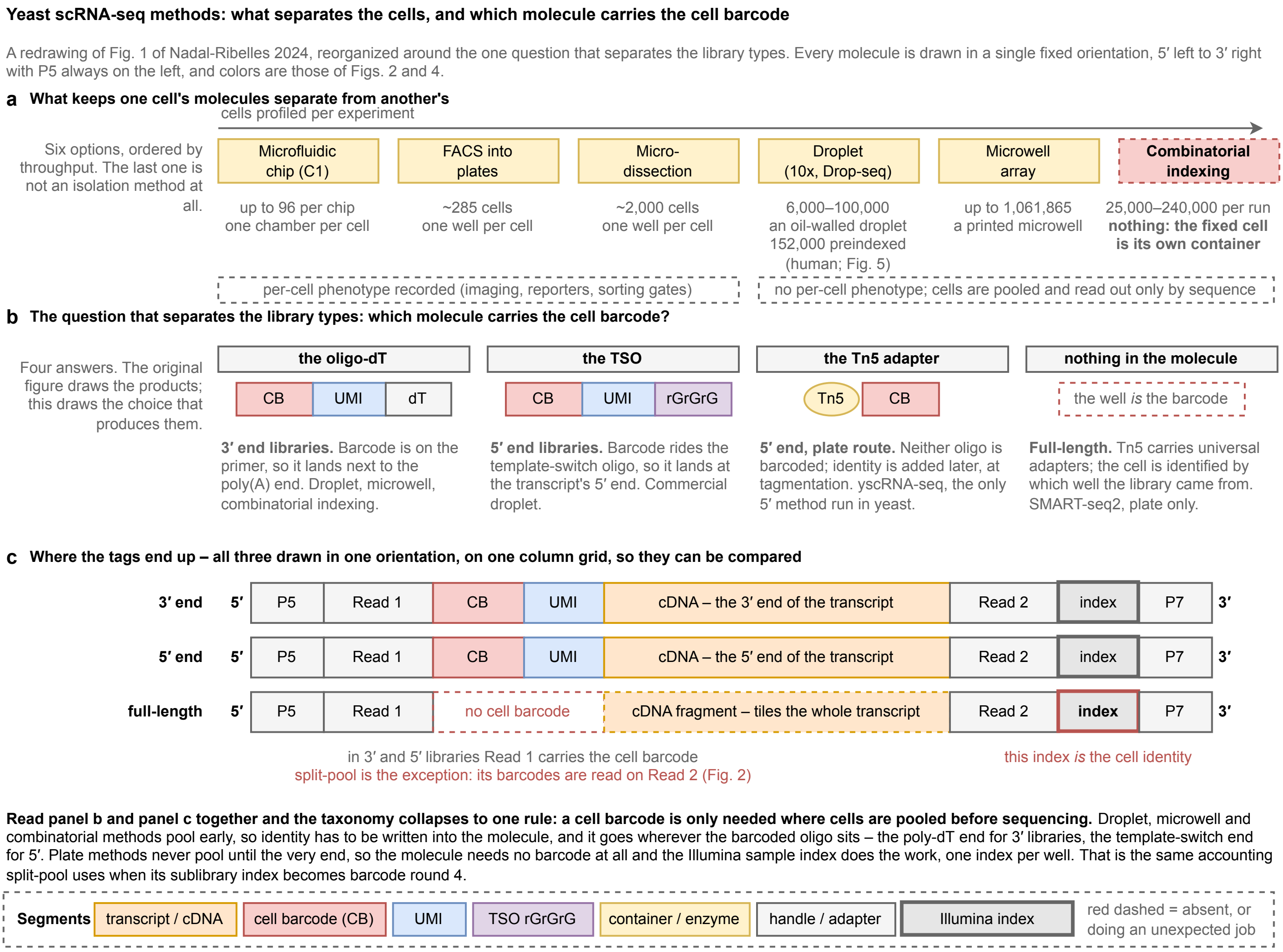

Figure 1: A cell barcode is only needed where cells are pooled before sequencing, and most of the method landscape follows from that one fact. The yeast scRNA-seq landscape, reorganized from the review of [8]. **(a)** What keeps one cell's molecules separate, ordered by throughput and each labeled with the container that does the work. Combinatorial indexing is set apart because it is not an isolation method at all: the fixed cell is its own vessel. The second throughput figure under the droplet column is what preindexing does to that column (section 3.3); it is placed there rather than given a seventh box because preindexing modifies droplet capture rather than replacing the container, and it is marked *human* in the label because it is the one number in a figure of yeast methods that did not come from a microbe. The two dashed boxes beneath the row split the six by whether a per-cell phenotype is recorded, which is what decides whether a method can be paired with a reporter or a sorting gate. **(b)** The question that actually separates the library types, which is which molecule carries the cell barcode. There are four answers: the oligo-dT gives 3′ libraries, the template-switch oligo gives 5′, a barcode loaded into Tn5 gives the plate route used by the only 5′ method yet run in yeast, and in full-length libraries nothing in the molecule carries it. **(c)** Where the tags end up, with all three library types drawn on one column grid in one orientation so they can be compared: 3′ and 5′ differ only in which end of the transcript sits beside the barcode, while full-length carries no barcode anywhere and is resolved instead by the sample index, one per well – the same accounting that makes split-pool's sublibrary index barcode round 4 (Sec. 5.4). Redrawn and reorganized from the text and Table 1 of that review rather than reproduced from its figure, and with the orientation fixed 5′ to 3′ throughout, where the original mirrors its 3′ panel. Every number drawn by hand in this figure is sourced in table 19.

Three properties of a yeast cell break every commercial single-cell chemistry, and only the third is a real cost. The wall means reagents cannot get in and the cell will not lyse on schedule, which is a timing problem both families solve with zymolyase. A fifth the mRNA of a mammalian cell means per-cell yields look poor and cells must be pooled to estimate anything. And ∼85% of the RNA is ribosomal, which is the one that shows up in the sequencing bill rather than in the protocol.

2.4 Fragmentation, and Why Single-Cell Counts Are Not Length-Normalized ✗

Every protocol here ends by cutting long cDNA into short pieces. It is easy to read past as plumbing, and it is not: it is the step that decides whether a count means "molecules" or "molecules times length", and it is what makes the late rRNA depletion of section 3.5 mechanically possible. Figure 6 places it in the whole workflow and draws the bulk against single-cell contrast developed below.

What it is for. After reverse transcription and barcoding, a cDNA copy runs from a few hundred bases to several kilobases. An Illumina sequencer reads roughly 150 bases in from each end and stops, and the bridge amplification that turns one molecule into a readable cluster loses efficiency as molecules lengthen, so long fragments are under-represented or lost. Everything is therefore cut to a size the instrument likes: in the microSPLiT protocol [7], "a tight smear around 450–500 bp."

Two chemistries do it. **Enzymatic fragmentation** uses a fragmentation buffer and a WGS fragmentation mix at 32 °C for 10 min, then 65 °C for 30 min; the first temperature cuts, the second repairs the ragged ends and adds the single A overhang that lets adapters ligate cleanly. **Tagmentation** instead uses a Tn5 transposase preloaded with adapter duplexes, which cuts and inserts the adapter in one reaction – M3-seq fragments "with Tn5 transposase loaded with Nextera read 2 primers." Fewer steps and less input, at the cost of more sequence bias in where the cuts land. Either way a double-sided SPRI size selection follows, discarding both the too-long and the too-short; the too-short are mostly adapter dimers, which is why the protocol tells you to check for "no band at 180–200 bp."

The difference from bulk RNA-seq, which is the whole point. In bulk RNA-seq every fragment of a molecule is sequenceable. Cutting a 3 kb transcript uniformly at random yields about three times as many usable fragments as a 1 kb transcript at equal abundance, so reads scale as length times abundance. That is why bulk expression units divide the length back out: FPKM and TPM exist to undo a bias that uniform random cutting *creates*.

A 3′-counting single-cell assay is immune, and not because the cutting is uniform. It is immune because *only one fragment per molecule can ever be sequenced*. That is worth spelling out step by step, because "the others are lost" is the part that sounds arbitrary and is not.

1. Before fragmentation the cDNA carries the cell barcode and a PCR handle at *one* end only, the end built during barcoding. The far end carries nothing at all.
2. Fragmentation cuts it. A long molecule might give six pieces; exactly one of them contains that barcoded end.
3. Adapter ligation then puts a universal adapter on every freshly cut end, so now every piece has an adapter. This is the step that makes the loss look puzzling: at this moment all six pieces are adapter-flanked.
4. The final PCR is what discriminates. It primes on the *barcode-side handle* at one end and the ligated adapter at the other, so a fragment amplifies only if it has *both*. The five pieces that lost the barcoded end carry ligated adapter on both ends and no barcode handle, so that PCR never amplifies them and they never reach the flow cell.

So the surviving fragment is not the one that happened to be cut well. It is the one that still carries the cell's identity, and it is unique by construction because the barcode was only ever attached once. Cut a long cDNA into six pieces and five lose that end; cut a short one into two and one loses it. Exactly one usable fragment survives per amplified copy, whatever the length, and UMI collapse then reduces all copies of one original molecule to a single count. Length drops out by construction.

Formally, if cuts fall as a Poisson process at rate λ, the distance from the barcode to the first cut is Exponential(λ), so the chance that the surviving fragment lands inside the size-selection window $[a, b]$ is $e^{-\lambda a} - e^{-\lambda b}$: a function of the window, not of the transcript's total length. Every transcript longer than the window has the same capture probability.

Single-cell count matrices are never length-normalized, and this is why. It is a structural consequence of one-usable-fragment-per-molecule plus UMI collapse, not a convention or an approximation that happened to work. Applying TPM or FPKM to a 10x or SPLiT-seq count matrix is an error, not a stylistic choice.

Where a cut does hurt, and where it does not. A cut landing inside the barcode block produces *loss, not corruption*. The piece carrying the cDNA has lost its outer handle and cannot amplify; the piece carrying the handle has no insert and is removed by the size selection as a dimer. No molecule is ever reassigned to a different but valid cell, which is the difference between an assay that under-counts and one that lies. A cut leaving too little transcript is caught twice, physically by the size selection and computationally by the aligner's minimum matched length. And because amplification precedes fragmentation, each original molecule exists as many copies cut at different positions, all sharing one UMI, so a badly cut copy costs nothing.

What genuinely survives is *mapping ambiguity*: short reads against repeated sequence map to several places equally well, which is why Gaisser et al. note that "a longer cDNA read (read 1) will improve the mapping quality given the large number of repeats in the bacterial genome." In these organisms that ambiguity is dominated by rRNA, whose operons are multi-copy – a fact with consequences for how rRNA fractions are reported, developed in section 3.5.

The length bias that does not cancel, and it is upstream. Brettner et al. and microSPLiT both prime with random hexamers as well as oligo-dT, and Kuchina et al. measured that hexamers dominate: "most of the rRNA and mRNA molecules were captured by random hexamer primers, whereas tRNA molecules were predominantly retrieved by poly-T primers." Every hexamer priming event starts its own cDNA and collects its own UMI, and the number of priming sites on a molecule scales with its length. So one long mRNA can yield several distinct UMIs. That is genuine length bias, it arises *before* fragmentation, and UMI collapse cannot remove it – those are not PCR copies of one molecule, they are independent priming events with genuinely different UMIs.

Two consequences, pulling opposite ways. It is harmless for perturb-seq, because the estimand of section 2.5 is differential – the same gene, perturbed against control – so a per-gene length factor divides out. It is not harmless for absolute within-cell composition, and it couples to the one free rRNA lever: shifting the hexamer:dT ratio to suppress rRNA (section 3.5) simultaneously shifts the length bias of the mRNA counts. **Hypothesis (untested):** that shift is small relative to shot noise at the depths in section 4.2, but it has not been measured here and the titration should report it.

2.5 The Count Matrix and Estimand

What comes out is a count matrix $\mathbf{Y} \in \mathbb{Z}_{\geq 0}^{N \times G}$ over N cells and G genes, plus an assignment $\mathbf{Z} \in \{0,1\}^{N \times L}$ of cells to the L *library elements* – the L distinct perturbations the library contains, one per gRNA in a CRISPRi screen or one per barcoded deletion strain in a knockout collection. $Z_{i\ell} = 1$ means cell i was found to carry element ℓ, and recovering $\mathbf{Z}$ at all is the problem Sec. 3.6 is about; a cell whose element cannot be read is a row of zeros and is discarded. Because per-cell depth is low, the analysis does not treat a single cell as an observation of a perturbation's effect. Instead cells sharing a guide are pooled into a **pseudobulk** profile, and the estimand is

$$\theta_{\ell j} \;=\; \log_2 \frac{\mu_{\ell j}}{\mu_{\mathrm{ctrl},\,j}}, \qquad \boldsymbol{\Theta} \in \mathbb{R}^{L \times G}, \tag{1}$$

the log fold change of gene j under element ℓ against non-targeting controls. Two things follow, and they shape every later section. Single-cell resolution is used for *assignment and quality control*, not for the effect estimate – so the relevant precision question is how many cells share a guide, not how deep any one cell was read. And $\boldsymbol{\Theta}$ is dense: a genome-scale screen in one condition produces of order 6000×6000 coefficients, a directed map from every perturbation to every transcriptional consequence. That object, not a hit list, is what makes the screen worth running for model training.

> **The output is a dense map, not a hit list, and that is the reason to run it.** A genome-scale screen in one condition produces of order $6{,}000 \times 6{,}000$ coefficients: a directed map from every perturbation to every transcriptional consequence. Single-cell resolution is used for assignment and quality control rather than for the effect estimate, so the precision question is how many cells share a guide, not how deeply any one was read.

3 Method Descriptions

Two method families are candidates. Split-pool is instrument-free, its reagent cost does not scale with cells, and it extends to bacteria. Droplet is the established option, is available as a service at UIUC, and recovers roughly five times more mRNA per cell.

That depth gap sounds decisive and is not, because what sets precision is total UMIs per perturbation – cells $\times$ depth – not depth alone (section 4.3). Five-fold less depth is exactly offset by five-fold more cells, and split-pool buys cells far more cheaply than droplet buys depth. Two things break the symmetry. Below roughly 100 cells per perturbation the biological floor binds and no amount of depth substitutes (section 4.4), so the trade is only available above that floor. And depth is not free of cells either: at split-pool's 410 UMIs per cell the measurement is firmly shot-noise limited, so rRNA depletion – which raises usable depth without touching cell count – is worth more than either (section 3.5). The comparison that settles it is cost per *usable* cell, section 5.2, not depth.

A third option removes the choice. Preindexing writes a split-pool barcode into the cell on one plate and then loads it into a droplet, so the two families compose and the droplet's cost structure changes rather than its chemistry (section 3.3). It is treated here as a candidate on the same footing as the other two in every respect except evidence: it has never been run in a microbe, so every figure that carries it marks it as a projection.

3.1 Yeast-Optimized SPLiT-seq (Brettner et al., 2024) ✗

Brettner et al. [3] adapted Split-Pool Ligation-based Transcriptome sequencing to yeast, profiling 43,388 cells of *S. cerevisiae* (haploid and diploid) and *C. albicans* in total. That total spans several experiments and is not the number to design against; the per-run figures are. One barcoding run loaded 48 wells at ∼5,000 cells each, so ∼240,000 cells entered the chemistry; those were split into ten sublibraries, and the two that were sequenced returned ∼5,500 and ∼10,000 cells passing filter. Read that way the comparison with droplet is direct: one split-pool run barcodes an order of magnitude more cells than one Chromium channel's ∼20,000, and the sublibrary – not the run – is the unit that gets sequenced and priced. It is the reference protocol for the split-pool arm, so it is worth walking element by element: each step exists to solve a specific failure, and the reasons are what transfer to a new organism.

1. Fix before you digest. Cells are fixed in 4% formaldehyde for ∼18 h at 4 °C, then quenched into 100 mM Tris with RNase inhibitor. Formaldehyde crosslinks protein and nucleic acid, which does two jobs at once: it freezes the transcriptome, and it makes the cell rigid enough to survive as a reaction vessel through the later handling. The ordering is not arbitrary. Cell-wall digestion itself triggers the cell-wall-integrity and HOG stress pathways and induces stress-responsive genes within minutes [8], so a protocol that digested first would measure artifacts of its own sample preparation. Fixing first is what isolates the perturbation's signal from the handling.

2. Open the wall, then the membrane, separately. Zymolyase at 0.005 U/µL in 1 M sorbitol with 0.1 M Na_2EDTA, 37 °C for 15 min, digests the β-glucan layer; the sorbitol is osmotic support, since a wall-less spheroplast bursts in a normal buffer. Then 0.4% Triton X-100 on ice for 3 min permeabilizes the membrane. The two steps are tuned against opposite failures: too little and reagents cannot get in, too much and the cell stops retaining its own cDNA and the whole single-cell premise collapses. Brettner et al. arrived at these values by screening 16 fixation/permeabilization combinations in parallel. That is itself an argument for the method: because a well's barcode is just another round-1 barcode, sixteen conditions are sixteen wells of one run, not sixteen runs. A droplet method has no equivalent – cell and reagent meet inside a droplet, so each condition needs its own channel, and the same screen is sixteen channels of reagent and sixteen library preps.

3. Round 1 – barcoded reverse transcription. Cells are counted, diluted, and distributed into 48 wells of a 96-well plate at ∼5,000 cells per well. Neither number is a tuned optimum. The 5,000 falls out of the recipe: a 25 µL reaction at the stated 200,000 cells/mL. The 48 is half the plate, and the paper gives no reason – but it has a consequence worth carrying, because round 1 then contributes 48 addresses rather than 96 and the realized barcode space for that run is $48 \times 96 \times 96 = 442{,}368$, half the 96^3 the protocol can deliver. Collision rates in section 5.4 are computed at the full 96^3; a run that loads half a plate doubles them.

Each well contains a well-specific barcoded primer pair: a barcoded 15-dT primer and a barcoded random hexamer, both at 3 µM, with Maxima H Minus reverse transcriptase and 7.5% PEG-6000. Both primers carry the same well barcode, BC_1 – neither is a **unique molecular identifier**. The random hexamer is six random bases used to *prime* anywhere on a transcript, not to label a molecule; the UMI arrives three rounds later, on the round-3 oligo, which is the only point at which molecules within a cell become distinguishable.

PEG-6000 is a molecular crowder. It occupies volume that enzyme and template cannot, so their effective local concentration rises and the bimolecular steps – reverse transcription here, ligation later – run faster than the absolute concentrations would suggest. The trade is viscosity, and mixing does get worse; the protocol inherits the 7.5% figure from mammalian SPLiT-seq and does not justify it, so it is a candidate for titration rather than a settled value.

The dual priming is a deliberate trade, and it is a *lever* because it is a ratio rather than a fixed requirement. Oligo-dT captures polyadenylated mRNA specifically; random hexamers prime on anything, including rRNA, and buy better 5′ coverage in exchange. Brettner et al. measure the consequence – recovered transcripts are 93.7% rRNA and only 5.75% mRNA – and say plainly what to do about it: "Users who prefer to study mRNA should lower the percentage of rRNA recovered by decreasing the ratio of random hexamer". The hexamer is not optional in general, since it is what carries the chemistry to transcripts with no poly(A) tail and therefore to bacteria; but for a 3′-counting screen in yeast the 5′ coverage it buys is unused, which makes this ratio the first thing to retune (Sec. 6).

4. Rounds 2 and 3 – split-pool ligation. Cells are pooled, filtered through a 15 µm strainer, and redistributed into a fresh 96-well plate. The filtration removes clumps, not contaminants: two cells stuck together take one path through the plates, receive one barcode and are read as a single cell, which is the split-pool equivalent of a droplet doublet. Straining is the only control on it, since there is no partition to over-fill. Each well holds a barcode oligo pre-annealed to a **linker strand**, a splint that positions the barcode against the 5′ end of the existing cDNA so T4 DNA ligase can seal the nick. After 30 min at 37 °C a **blocking strand** is added, which outcompetes the linker and retires that well's barcode so it cannot participate in any later round – without it, leftover oligos would scramble the barcode sequence. Pool, split, repeat for round 3, then terminate with EDTA. Three rounds of 96 give $96^3 = 884{,}736$ distinct paths available to the protocol; the round-3 oligo also carries the 10 nt UMI and a 5′ biotin.

5. Sublibraries, lysis, and a selection step. Cells are pooled and split into sublibraries of 5,000–20,000, which can be frozen.

What a sublibrary is for. Two independent reasons, and the first is the one that is easy to miss. *It is a fourth barcode.* Three rounds of 96 give 884,736 addresses, and a run puts ∼240,000 cells through them – enough that by the birthday argument of section 5.4 a substantial fraction collide. Splitting into ten sublibraries does not change any cell's BC1–BC2–BC3 path, but it does mean two cells sharing a path are still told apart if they were sequenced in different sublibraries, because the Illumina index distinguishes them. Ten sublibraries therefore multiply the effective address space by ten, and table 15 prices that trade directly against adding a fourth ligation round. *And it is the unit of sequencing.* A sublibrary is what gets lysed, prepped, indexed and loaded, so it is the granularity at which cells are converted into reads and into money. Brettner et al. sequenced two of their ten and froze the rest – which is how you match cells sequenced to the budget and the flow cell, and how a failed prep costs one tenth of a run instead of all of it.

The two reasons pull in opposite directions, which is why the size is a range rather than a number: smaller sublibraries mean fewer collisions but more library preps at \$55 each (section 5.4). That \$55 is Brettner et al.'s own rolled-up figure; the two itemized per-sublibrary lines of table 10 sum to \$53, and the \$2 gap is the "added leeway for the unpriced consumables" their sentence says it includes.

Lysis is SDS plus proteinase K at 55 °C for 2 h. The biotin now earns its place: streptavidin beads capture only molecules that received the round-3 oligo, so anything that fell out of the barcoding scheme is washed away rather than sequenced.

How the second handle is added. At this point the molecule has a PCR handle at one end only – the one delivered on the round-3 oligo. The other end is the far end of the cDNA and carries nothing, so the molecule cannot yet be amplified. A **template-switching oligo** supplies the missing handle: when reverse transcriptase runs off the end of its template it adds two or three non-templated C residues, the TSO ends in rGrGrG and base-pairs with them, and the enzyme then copies the TSO. The handle is therefore *written by the polymerase*, not ligated on – which is what distinguishes it from every earlier step in this protocol, where a linker held two ends together for a ligase. Figure 3 draws the two mechanisms against each other, since a protocol that uses both is where the distinction is easiest to lose. The step is also not run for the handle alone: it is a second reverse-transcriptase pass on bead-captured cDNA, needed because formaldehyde crosslinking hobbles the in-cell reaction, and the TSO rides along on it [7]. With both handles present the library is amplified, fragmented, adapter-ligated and index-PCRed; the i7 index is added at that last step and is a separate sequence from either PCR handle, read on its own index read. That final index is barcode round 4 in all but name (Sec. 5.4).

> **What it delivers.** 100–600 genes and 120–700 mRNA UMIs per cell; 25–50% of input cells recovered; barcode collisions well under 5%. Reagent cost is about \$0.005 per cell, but 94% of reads are rRNA and ∼75% of sequenced cells are discarded – so the cost that matters is 20× higher (Sec. 5).

Reading order follows assembly order. Each round ligates its barcode onto the free 5′ end of the growing strand, so later barcodes end up further from the cDNA. Read 2 therefore meets them newest-first: UMI, BC3, BC2, BC1, then cDNA. The reverse order is the intuitive guess and it is impossible – BC1 arrives on the round-1 RT primer, so it cannot sit further out than a barcode added afterwards. Gaisser et al.'s STARsolo offsets [7] pin the layout to the base (table 14).

3.2 Droplet Capture: 10x Chromium with In-Droplet Spheroplasting ✗

Jariani et al. [10] made the 10x Chromium platform work for yeast with a one-line change, and the reasoning behind it is more interesting than the change itself.

The obstacle is that the standard 10x workflow lyses cells inside the droplet on a thermal schedule tuned for a membrane, and a yeast cell wall does not cooperate. The obvious fix – spheroplast the cells first, then load them – is what Jackson and colleagues did, but it means feeding fragile wall-less cells through a microfluidic chip, where premature lysis releases mRNA into the shared buffer and contaminates every droplet downstream. Jariani et al. instead put zymolyase *into the reverse-transcription master mix*, replacing 1 µL of water with 100× enzyme, so cells enter the chip walled and intact

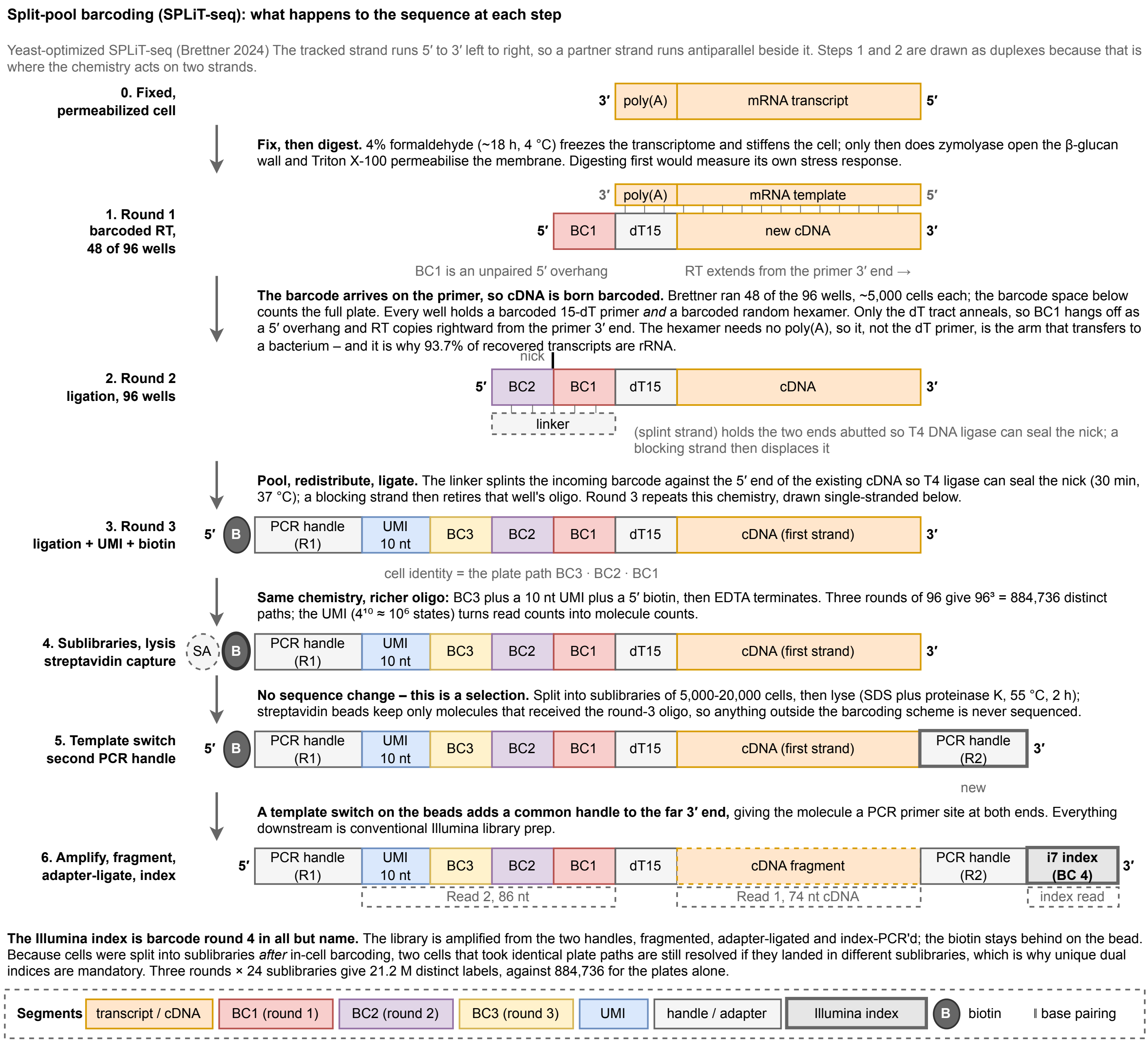

Figure 2: Cell identity in split-pool barcoding is assembled onto the transcript itself, which is what removes the need to ever physically isolate a cell. Seven strand states in the yeast-optimized SPLiT-seq workflow of section 3.1 [3], each arrow annotated with the chemistry that produces the next. Strands are drawn $5'$ left to $3'$ right, so the barcode block grows leftward with each round while the cDNA body stays fixed – the visual point being that a cell's address is written in three installments, and a cell is identified by the *path* it took through three 96-well plates rather than by ever having been alone in a well. That is the whole economic argument for the method: 96^3 distinct paths cost three plates, not 10^6 containers. Two states are drawn double-stranded because that is where the mechanism is easy to get wrong. At round 1 the mRNA template lies antiparallel above the tracked strand, showing BC1 as an unpaired $5'$ overhang, and showing that the random hexamer rather than the poly-dT primer is the arm that carries the method to transcripts without a poly(A) tail – the single change that lets the same chemistry work in bacteria [7, 9]. At round 2 the linker strand base-pairs underneath and a tick marks the nick T4 ligase seals; ligation is templated by that splint, which is why an unretired barcode from a previous well would scramble the address and why a blocking strand follows every round. The $5'$ biotin added in round 3 is not a label but a selection handle: streptavidin capture at the next step discards every molecule that fell out of the scheme, so incompletely barcoded material is removed before it can consume sequencing. Composed from the protocol as described in section 3.1, not traced from any published figure; every parameter shown is sourced there. Drawn at true Nature print size in `notes/assets/drawio/splitseq-barcoding.drawio`. Every number drawn by hand in this figure is sourced in table 21.

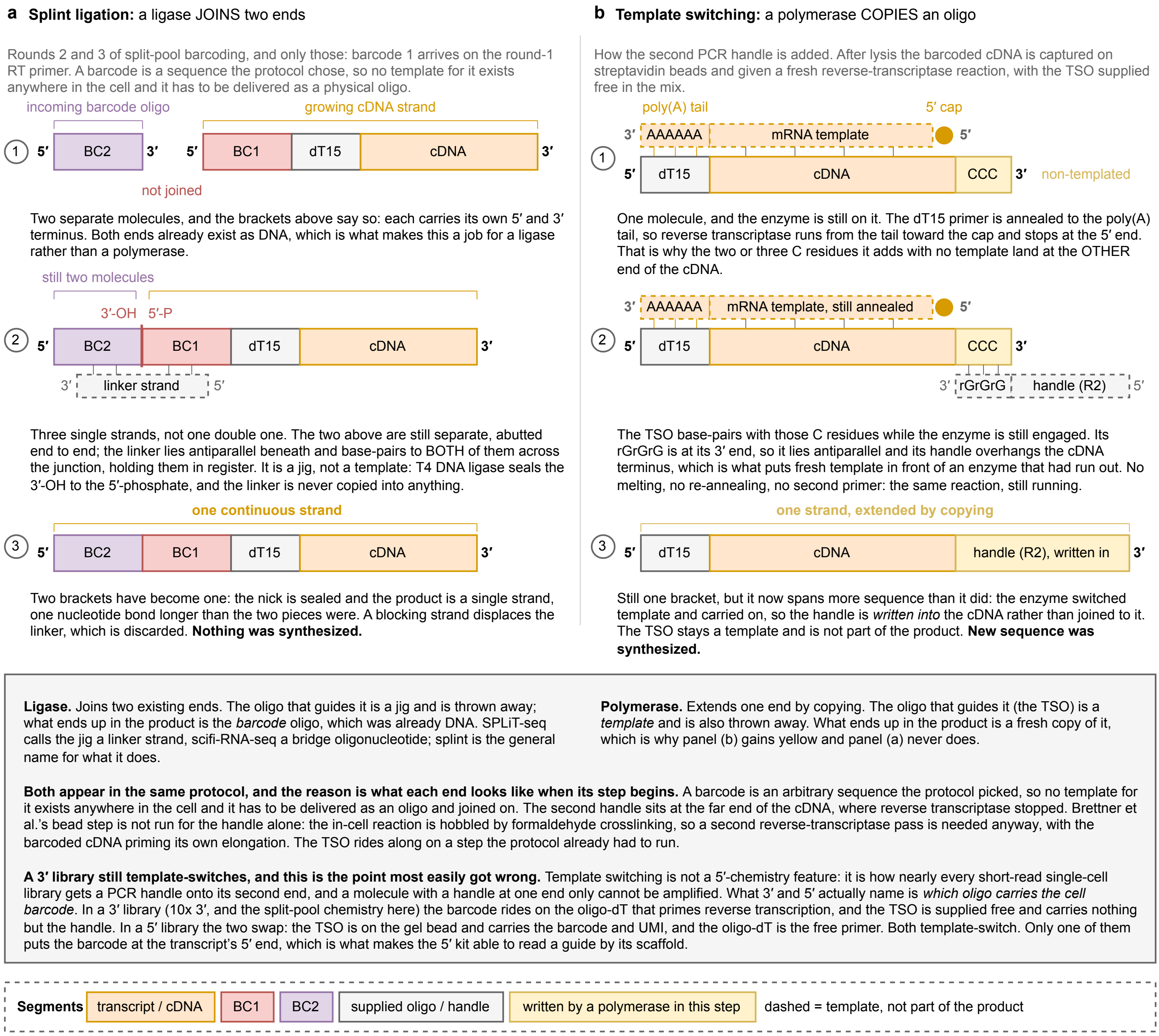

Figure 3: Ligation joins two ends that already exist; template switching copies an oligo into a new one. Both appear in the same protocol, for different ends of the same molecule. The segment colors of fig. 2 carry over, and one is added: **yellow is sequence written by a polymerase during the step drawn**, against gray for an oligo the protocol supplied and orange for nucleic acid that already existed. Panel (a) contains no yellow, and that absence is the point. The key at the foot of the figure gives both systems. Brackets above each strand carry the molecule count, because a row of abutting blocks reads as one molecule however the caption is worded: panel (a) runs two brackets until the ligase acts and one afterwards. **(a)** Rounds 2 and 3 of split-pool barcoding [3]. Three single strands are involved, not one double one. The incoming barcode oligo and the growing cDNA strand are both single-stranded and are abutted end to end; the linker lies antiparallel beneath and base-pairs to *both* of them across the junction, holding them in register so T4 DNA ligase can seal the $3'$-OH to the $5'$-phosphate. The linker is a jig: it positions, it is never copied, and a blocking strand displaces it afterwards. What ends up in the product is the barcode oligo, which was already DNA. Round 1 is not drawn because barcode 1 is not ligated at all, arriving instead on the reverse-transcription primer. **(b)** How the second PCR handle is added [7]. The oligo-dT primer is drawn annealed to the poly(A) tail, which is what fixes the direction of everything else: reverse transcriptase runs from the tail toward the cap, reaches the $5'$ end of the mRNA and adds two or three non-templated C residues at the *far* end of the cDNA. The TSO carries its rGrGrG at its $3'$ end, so it lies antiparallel and its handle overhangs the cDNA terminus, putting fresh template in front of an enzyme that had run out; the enzyme switches template and carries on, so the handle is written into the cDNA rather than joined to it. There is no melting and no re-annealing, which is the detail that separates it from overlap extension: it is one reaction, still running. **The summary panel states the $3'$ against $5'$ point** that section 3.6 turns on. Composed from the protocols as described in section 3.1, not traced from any published figure. Drawn at true Nature print size in `notes/assets/drawio/splint-vs-template-switch.drawio`.

and are digested only after they are already isolated. They verified the timing directly: cell counts hold on ice and through the 6 min of droplet generation, then drop 200-fold within 20 min at 53 °C. Wall digestion, lysis, and reverse transcription all happen in the same droplet, in that order, without the cell ever being vulnerable outside one.

The result was 6,118 cells across five conditions with median transcripts per cell of 528–1,261 – roughly triple Brettner et al.'s per-cell recovery, because oligo-dT-only capture on a bead is far more mRNA-selective than dT-plus-hexamer in situ. Library preparation cost fell to $0.30 per cell against $4.15 for the plate-based methods it replaced. Sequencing saturation was only 70–89%, meaning deeper sequencing would still have recovered more.

The chemistry has moved since. That work used 3′ v2. Version 3 roughly doubles UMI yield on 10x's own matched datasets, and the current generation at the UIUC core is GEM-X v3.1 on a Chromium X, quoted at ∼8% doublets at 20,000 cells recovered with CRISPR guide capture available as a catalog add-on. Any comparison drawn from the 2020 paper therefore understates droplet performance today, and Sec. 5 uses the newer figures.

> **The yeast fix is a scheduling change, not a chemistry one.** Zymolyase goes into the reverse-transcription master mix, so the wall comes off only after the cell is already sealed inside a droplet and can no longer contaminate its neighbors. Cell counts hold on ice and through droplet generation, then drop 200-fold within 20 minutes at 53 °C. That ordering is the whole adaptation, and the core has said they will run it.

3.3 Combinatorial Fluidic Indexing (Datlinger et al., 2021) ✗

scifi-RNA-seq [6] composes the two families above rather than choosing between them, and the composition removes the constraint that sets droplet throughput.

Why Poisson stops binding. A droplet run holds occupancy low so that doublets stay rare, which leaves most droplets empty and most of the reagent in a channel unused: at the maximum recommended load of 15,300 nuclei per channel, 16.4% of droplets held one or more nuclei, an average of 0.2 each. That accounting holds only while a droplet has to establish cell identity by itself. If the cells arriving at the chip already carry distinct barcodes, several can share a droplet and still be resolved. Datlinger et al. loaded 100-fold over the recommendation, 1.53 million nuclei in a single channel, and measured a 95.5% fill rate at an average of 9.6 nuclei per droplet, with the emulsion still monodisperse and the device unclogged.

Two rounds, in two different places. Permeabilized cells or nuclei are preindexed on one 384-well plate, where a well-specific barcoded oligo-dT primer reverse transcribes in situ and writes the round1 barcode, a UMI and a sequencing primer site into the cDNA while it is still inside the cell. The cells are then pooled and overloaded into the Chromium system, where a gel bead delivers the round2 barcode. The join is a ligation onto the 5′-phosphate of the reverse-transcription primer, directed by a bridge oligonucleotide: the splint chemistry of Sec. 3.1, performed inside a droplet rather than in a well. Round1 is shared by every cell from one well and round2 by every cell in one droplet, so only the pair identifies a cell.

What was actually recovered. Occupancy is not yield, and the two are easy to conflate. The yield figure is a separate experiment: 383,000 preindexed nuclei into one channel returned **151,788** single-cell transcriptomes through quality control, a 15-fold increase over a standard Chromium run. Their baseline for that ratio is a ∼10,000-cell run; against the 20,000 cells per channel the UIUC rate card quotes, the same recovery is 7.6-fold, and 7.6 is what every budget in this document uses. That is a recovery, not a capacity claim, which is the distinction that demoted the Gaisser row from fig. 9. Depth did not pay for it: UMI counts and unique-read fractions showed no downward trend in droplets holding up to 15 nuclei, so the reagent in one droplet is not the limiting factor over the range that matters.

What is modeled, and is not a measurement. The barcode-space argument is of a different kind. With the 737,280 round2 barcodes supplied by the Chromium ATAC reagents, a zero-inflated Poisson analysis plus Monte Carlo simulation put 1 million resolvable transcriptomes within reach of 96 round1 wells. Nothing was recovered at that scale. The figure is a modeled bound and is drawn as one in fig. 5, and only the 151,788 is carried into the budget.

Two rounds beat three, on sequencing cost. The comparison against multiround combinatorial indexing is not only about handling. Because the round1 barcode is placed by the reverse-transcription primer and round2 by a single ligation, every cycle spent on the composite cell barcode reads barcode. Multiround protocols interleave constant sequence – ligation overhangs, primer binding sites, Tn5 mosaic ends – between their barcode blocks, so a measured 42% of those cycles are **uninformative** in sci-RNA-seq v1, 13% in sci-RNA-seq v3 and sci-Plex, and **67% in SPLiT-seq**. Those cycles are bought at the same price as informative ones. This is a charge against the chemistry of Sec. 3.1, and section 5.3 works out what it is worth.

What it changes in the budget, and the step nobody has run. Reagent is sold per **channel**, and per-channel reagent is 81% of the droplet column of table 16. Preindexing divides that charge across 7.6 times as many cells at

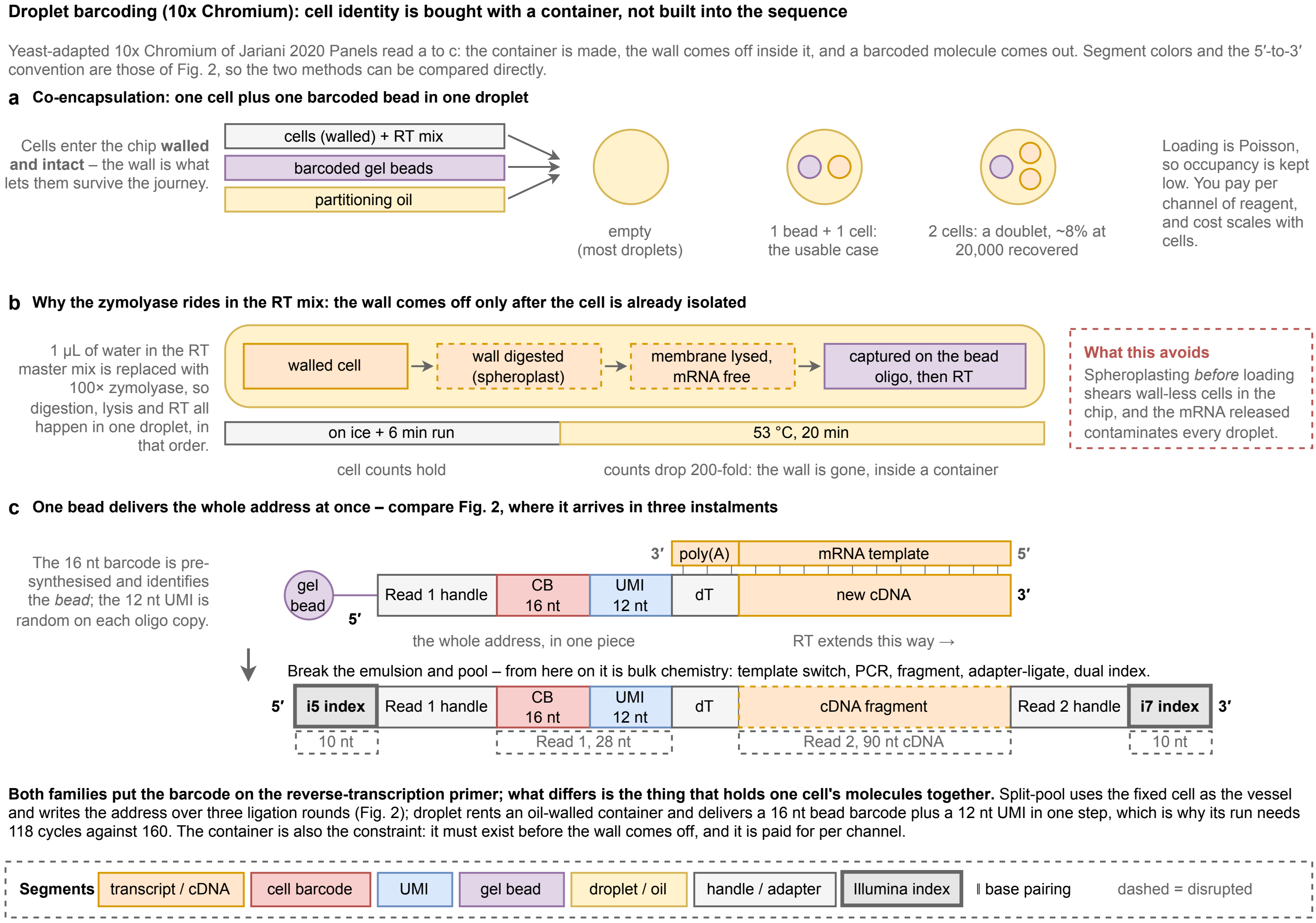

Figure 4: Droplet methods buy cell identity with a physical container, which is what lets the whole barcode arrive at once – and what turns the yeast cell wall into a scheduling problem rather than a chemistry one. The yeast-adapted 10x Chromium workflow of section 3.2 [10]. **(a)** Cells, beads and oil meet at a junction and partition into droplets. Identity here is positional rather than sequential: every molecule tagged inside a droplet carries that droplet's bead barcode because the oil wall keeps it that way. Loading follows Poisson statistics, so occupancy is kept deliberately low – most droplets are empty, and the doublets that survive that choice are two cells reported as one. The container is bought per channel of reagent and its cost scales linearly with cells, which is precisely the economics fig. 2 avoids by never isolating anything. **(b)** Jariani et al.'s change is one line of the protocol and is entirely about *ordering*: 1 µL of water in the reverse-transcription master mix is replaced with 100× zymolyase, so cells enter the chip walled and are digested only once they are already alone. The timing was verified rather than assumed – counts hold on ice and through the 6 min of droplet generation, then fall 200-fold within 20 min at 53 °C, by which point every cell is sealed in its own container. The alternative of spheroplasting before loading feeds wall-less cells through a microfluidic chip, and mRNA leaking from the ones that shear contaminates every droplet downstream, which is a whole-run failure rather than a per-cell one. **(c)** The resulting molecule, drawn in the notation of fig. 2 so the two can be read against each other. The bead oligo *is* the reverse-transcription primer, so as in split-pool the barcode arrives on the primer and only the poly-dT tract anneals – but the entire address, a 16 nt bead barcode plus a 12 nt UMI▲ [external: 10x CG000733, a vendor specification], is delivered in one piece instead of over three ligation rounds. Once the emulsion is broken the chemistry is bulk, and the read layout that results needs 118 cycles against split-pool's 160 (`experiments/024-perturb-seq-costing/scripts/read_structure.py`). That layout is the current GEM-X/v3.1 chemistry priced in section 5; Jariani et al. demonstrated on 3′ v2, whose Read 1 is 26 nt. Composed from the protocol as described in section 3.2, not traced from any published figure. Every number drawn by hand in this figure is sourced in table 23.

the UIUC baseline of 20,000 recovered per channel, which is the only lever available that reduces droplet cost without touching sequencing. Carried through the model, a 250-cells-per-gene screen falls from 137 channels and \$364 k to 18 channels and \$109 k (section 5.5).

Two cautions come before that number is used. Every demonstration is human or mouse – cell lines, primary T cells, isolated nuclei – and no yeast or bacterial run is reported. Preindexing also requires reagents to reach the transcriptome in situ, which in yeast means the fixation and wall digestion of Sec. 3.1 rather than the in-droplet spheroplasting of Sec. 3.2. **Hypothesis (untested):** the two halves compose, since in-cell reverse transcription on fixed yeast and droplet capture of yeast are each established separately. No measurement of scifi-RNA-seq in any microbe exists in the literature held here, which is why the row is marked as a projection everywhere it appears and why section 6.3 adds a checkpoint for it rather than assuming it.

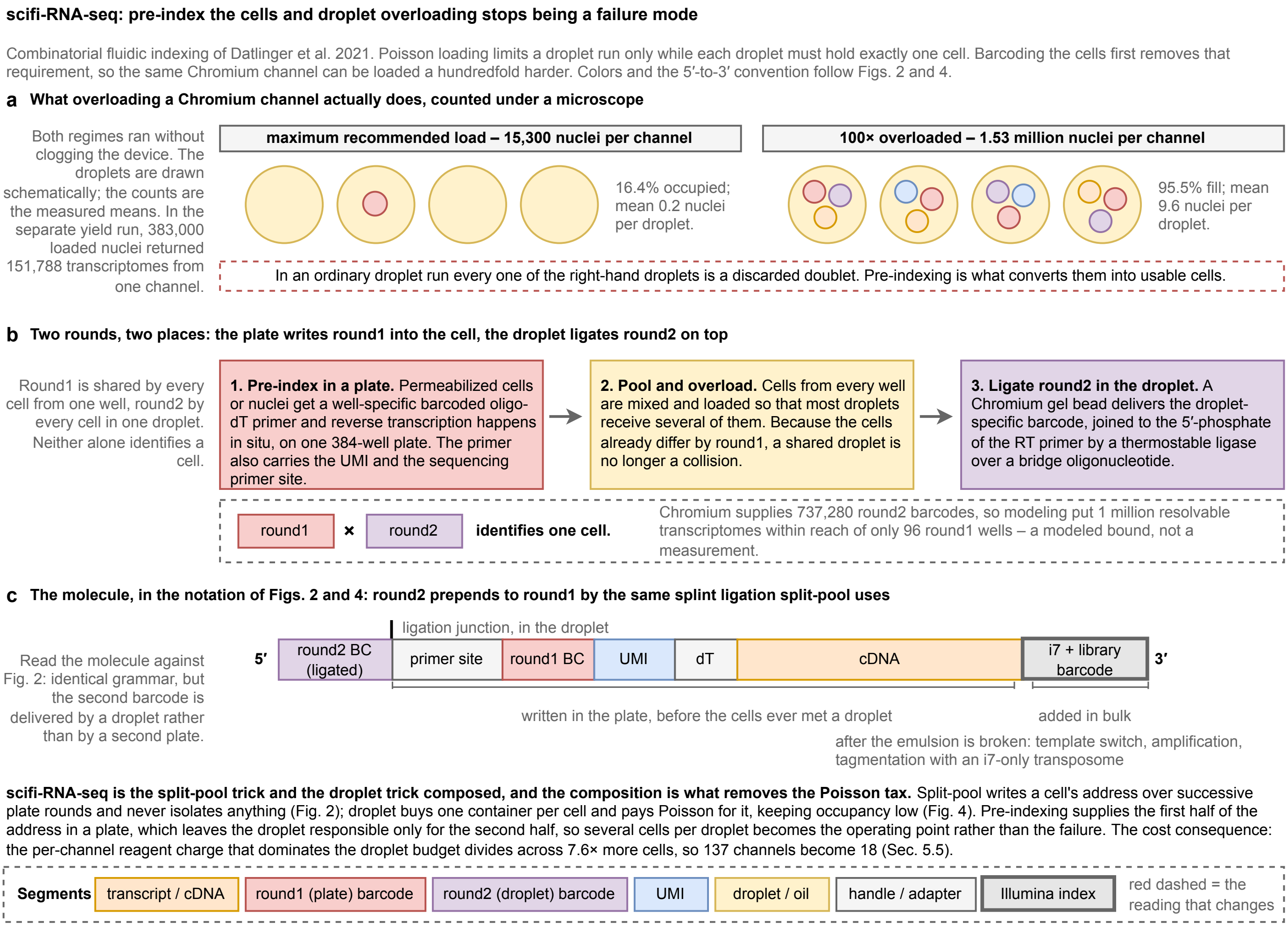

Figure 5: Pre-indexing the cells before they reach the chip converts droplet overloading from the dominant failure mode into the operating point. Combinatorial fluidic indexing of section 3.3 [6]. **(a)** The two loading regimes, counted under a light microscope. At the recommended load most droplets are empty, so most of a channel's reagent is spent on nothing; at 100-fold overloading almost every droplet is occupied and holds several nuclei. In an ordinary droplet run every one of those right-hand droplets is a discarded doublet, and pre-indexing is the single change that converts them into usable cells. The droplet cartoons are schematic; the percentages and means are the measured values. **(b)** Where the two barcode rounds are written. The plate writes round1 into the cell by reverse transcription; the droplet ligates round2 on top, by the same splint chemistry SPLiT-seq uses in fig. 2. Neither round identifies a cell alone, and the barcode-space figure beside them is a modeled bound rather than a measurement. **(c)** The assembled molecule in the notation of fig. 2 and fig. 4, with each span bracketed by where it came from. The molecule is written in three places and reads 5′ to 3′ in that order reversed: the droplet contributes only the round2 barcode at the 5′ end; the plate wrote the whole middle span, from the primer site through the cDNA, before the cells ever met a droplet; and the i7 arrives last, in bulk, after the emulsion is broken. That middle span is what makes the grammar identical to split-pool while the second barcode is supplied by a container. Composed from the protocol as described in section 3.3, not traced from any published figure. Every number drawn by hand in this figure is sourced in table 25. Drawn at true Nature print size in `notes/assets/drawio/scifi-fluidic-indexing.drawio`.

Preindexing is the only published result that moves a method right without moving it down. 383,000 nuclei into one channel returned 151,788 transcriptomes, with no downward trend in UMIs per cell in droplets holding up to 15 nuclei. Against the UIUC baseline of 20,000 cells per channel that is 7.6 times more cells through a priced channel, which is the single largest lever available to the droplet route. It has never been run in a microbe, so every figure carrying it marks it as a projection.

3.4 Compressed Perturb-seq (Yao et al., 2024) ✗

Yao et al. [11] is the direct precedent for deliberately putting *many* perturbations in one cell and recovering individual effects computationally, rather than treating multi-guide cells as a defect to be filtered out. It inverts the assumption underneath Sec. 4.6, so it is worth stating precisely what is compressed and what is assumed.

What is compressed. Not the library – the *samples*. Their framing: "if the perturbation effects are sparse ... then we can measure a small number of random composite samples (comprising 'linear combinations' of individual sample phenotypes) and decompress those measurements to infer the effects of individual perturbations." Two routes generate composites: **cell-pooling**, deliberately overloading droplets with several singly-perturbed cells, and **guide-pooling**, infecting at high MOI so each cell carries a random set of guides. Guide-pooling is the one relevant to us. They ran it at MOI 10 in human THP1 cells over 598 target genes, achieving 2.50 guides per cell, and analyzed the 8,448 cells carrying three or more.

The decoding. FR-Perturb factorizes the expression matrix by sparse PCA, runs LASSO on the resulting matrix of perturbations by factors, then multiplies back out to per-gene effects. The sparsity that makes this work is structural rather than incidental: regulatory effects are modular, so the effect matrix has low rank r and few non-zeros q per module, and the required number of composite samples is $O((q + r) \log n)$ rather than $O(n)$.

The assumption, exactly. Guide-pooling "requires the non-trivial assumption that the effect sizes of guides tend to combine additively in log expression space when multiple guides are present in the same cell." Interactions are not modeled in the first-order fit; they are argued to *cancel*, because each cell gets a random combination, so individual interaction effects average out "so long as there are an equal number of positive and negative interactions" – a condition they justify by citing genome-wide second-order fitness interactions in yeast. That is our own data domain, which makes the assumption more checkable for us than for them.

What it delivers, and what it does not. A stated 10-fold cost reduction for guide-pooling, with performance *increasing* in guides per cell and the best results in cells carrying four or more – they found no ceiling within the range tested and expect further gains from "even higher efficiency with a greater degree of overloading." The honest limits are equally clear. Pairwise interactions were not recovered at all: "none of these effects was significant in either screen due to insufficient power ... even with a lax significance threshold ($q < 0.5$)." And higher-order additivity is essentially untested – "very little is known about the structure of higher-order interaction effects and how they systematically combine within a cell."

Why this matters here. It converts multiplexing from a defect into a design variable, and supplies the constant that Sec. 4.6 needs: ∼400 cells per pairwise interaction, four times the 100-cell first-order floor. But it recovers *main effects* efficiently by assuming interactions cancel – and interactions are what metabolic engineering needs. The two goals pull in opposite directions, which is the tension Sec. 4.6 resolves.

3.5 Ribosomal RNA Depletion ✗

Roughly 85% of a yeast cell's RNA is ribosomal, and Brettner et al. recover 93.7% rRNA against 5.75% mRNA. Ninety-four of every hundred reads therefore buy ribosomal sequence. Since sequencing is 81% of the split-pool budget (Sec. 5.5), this single fraction is the largest cost lever in the document – worth $99,160 at genome scale, the difference between the undepleted and depleted split-pool rows of table 16 – and it is the reason depletion is treated here as a method component rather than an optimization.

Four chemistries, not one. The literature this document draws on uses four distinct operations, and they differ in what they act on, when they act, and what they cost. Treating them as one recipe is the mistake this subsection used to make, and it hid a head-to-head result that argues against the obvious choice. Figure 7 draws all four on one workflow axis.

1. **Cas9 cleavage of rRNA-derived cDNA**, in the finished library. Brandner et al. [4] designed 253 sgRNA spacers – selected from 824 candidates mapping to rRNA – against the 5S, 16S and 23S sequences of *E. coli* and

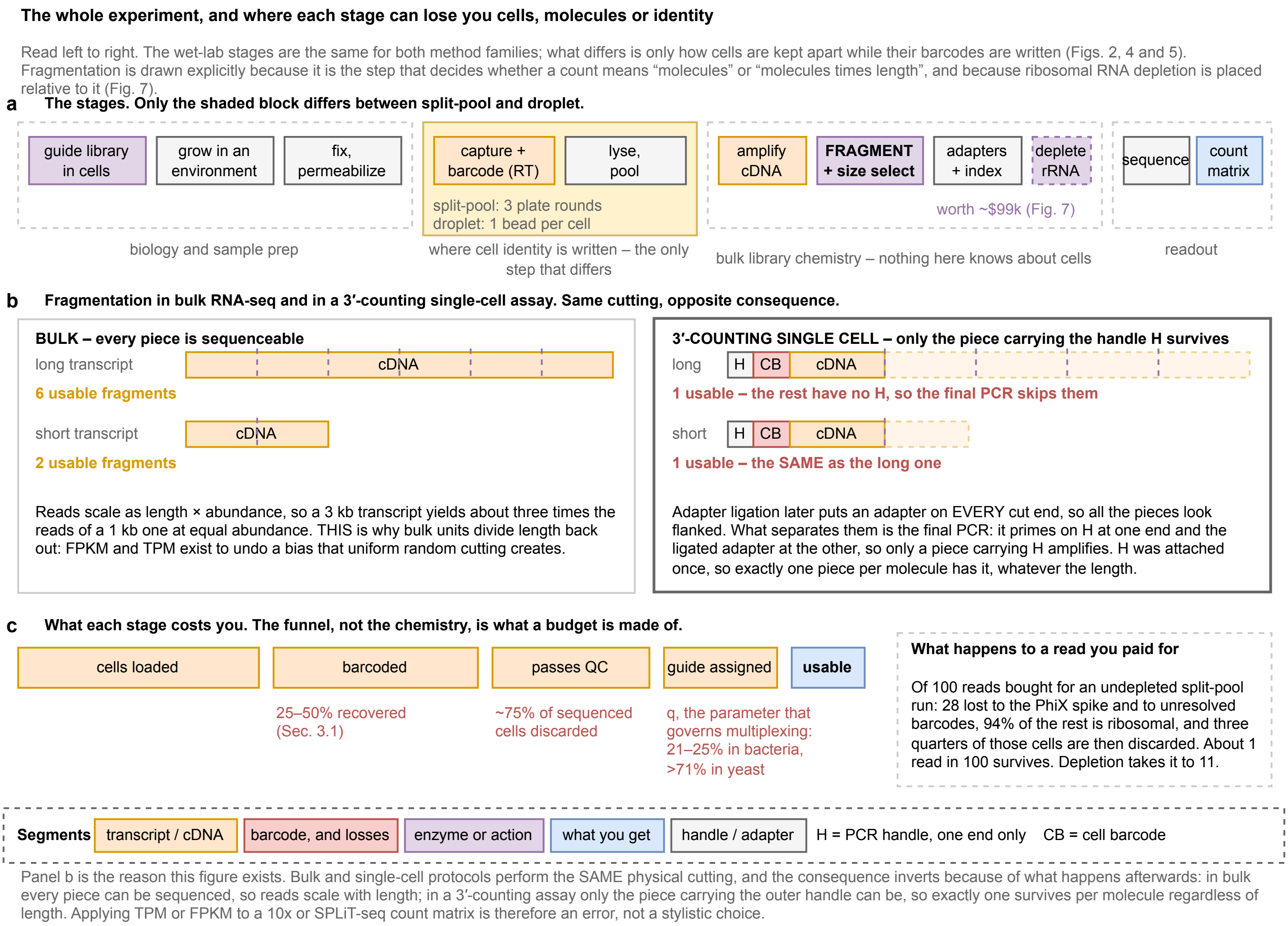

Figure 6: The whole experiment, and where each stage costs you cells, molecules or identity. Two symbols recur in panel (b) and are worth having before reading it: **H** is the PCR handle, attached at one end only during barcoding, and **CB** is the cell barcode. The full color key is repeated at the foot of the figure. **(a)** The stages. Everything outside the shaded block is common to both method families; only the way cell identity gets written differs, which is the argument sections 3.1 to 3.3 make in detail. Fragmentation is drawn as its own stage because its position relative to rRNA depletion is what section 3.5 corrects, and because it determines what a count means. **(b)** The same physical cutting, with opposite consequences. In bulk RNA-seq every piece of a molecule can be sequenced, so a long transcript yields proportionally more reads and the length has to be divided back out, which is what FPKM and TPM are for. In a 3′-counting single-cell assay only the barcode-proximal piece keeps the outer handle, so exactly one fragment survives per amplified copy whatever the length, and UMI collapse reduces the copies to one count. Length drops out by construction; see section 2.4. **(c)** The funnel. A budget is made of these fractions rather than of the chemistry: cells are lost at recovery, at QC and at guide assignment, and reads are lost separately to spike-in, to ribosomal sequence and to cells that are later discarded. Composed from the protocols as described in sections 3.1, 3.2 and 3.5, not traced from any published figure. Drawn at true Nature print size in `notes/assets/drawio/perturb-seq-workflow.drawio`. Every number drawn by hand in this figure is sourced in table 27.

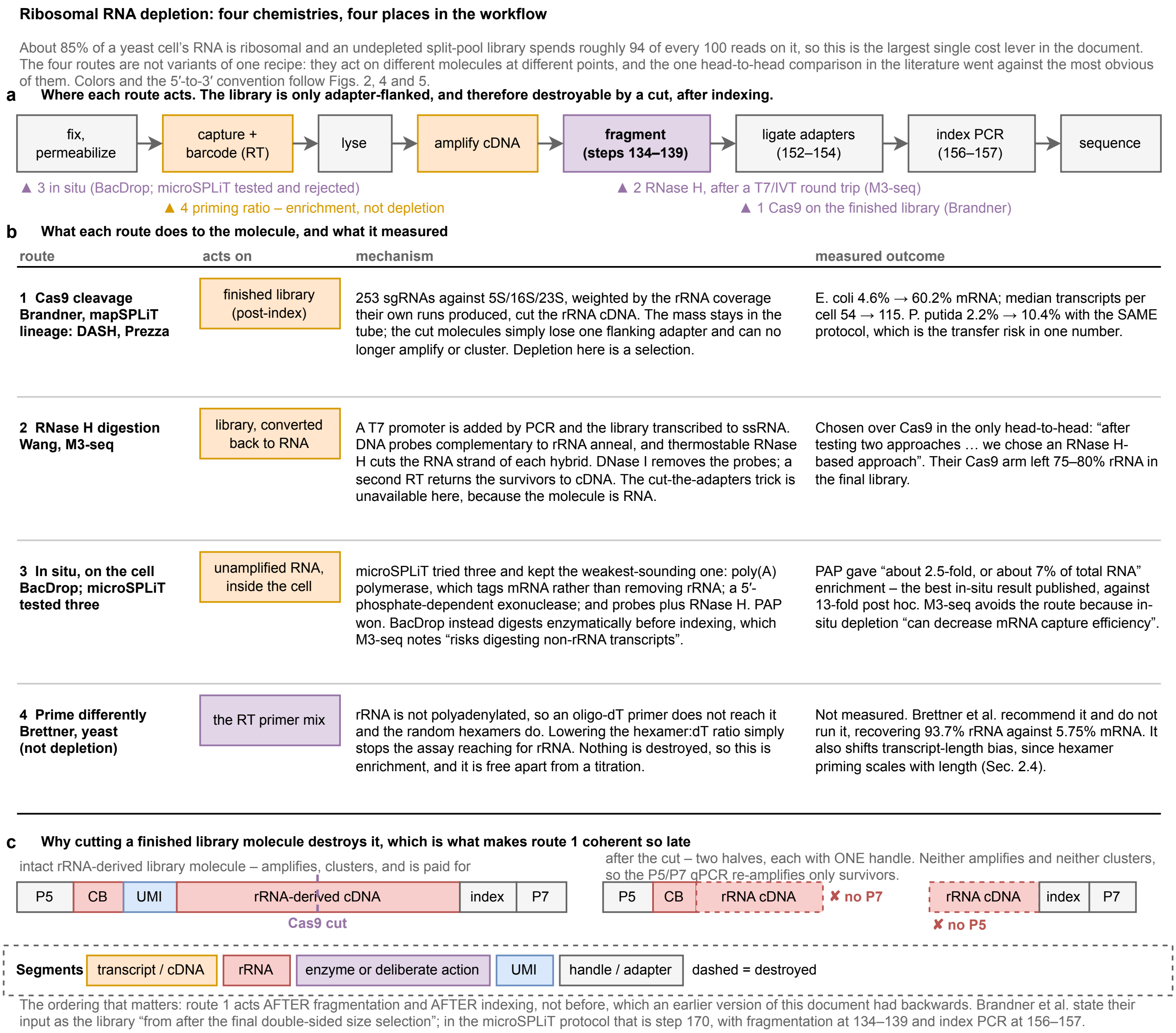

Figure 7: Four rRNA-depletion chemistries, four places in the workflow, and one head-to-head that went against the obvious choice. (a) Where each route acts. The step numbers are the microSPLiT protocol's [7], and the ordering is the correction this subsection makes: Cas9 depletion acts *after* fragmentation and *after* indexing, on a sequencing-ready library, not before either. **(b)** What each route does to the molecule and what it measured. The four differ in what they act on, which is why they cannot be compared as variants of one recipe. **(c)** Why a cut destroys a finished library molecule: both flanking adapters are needed to amplify and to cluster, so each half of a cut molecule retains only one and neither survives the P5/P7 re-amplification. This is what makes depleting so late coherent, and it is the same mechanism section 2.4 uses to explain why a mid-barcode cut causes loss rather than corruption. Composed from the protocols as described in sections 3.1 and 3.5, not traced from any published figure. Drawn at true Nature print size in `notes/assets/drawio/rrna-depletion-options.drawio`. Every number drawn by hand in this figure is sourced in table 29.

P. putida, "weighted proportionally to the regions in the rRNA that appeared more frequently during NGS after microSPLiT." That weighting is the "sequencing-guided" part of their name, and it exists because in-situ reverse transcription gives non-uniform rRNA coverage. 100 pmol of transcribed sgRNA is pre-incubated with 10 pmol spCas9, then 30 ng of library is added and digested at 37 °C for 2 h. The lineage is DASH [12] and its bacterial adaptation [13].

2. **RNase H digestion of rRNA:DNA hybrids**, after a round trip through RNA. Wang et al.'s M3-seq [14] appends a T7 promoter by PCR, transcribes the library to single-stranded RNA, anneals rRNA-complementary DNA probes, and cleaves the RNA strand of each hybrid with thermostable RNase H; DNase I then removes the probes and a second reverse transcription returns the survivors to cDNA.
3. **In-situ enzymatic depletion**, on the cell, before amplification. BacDrop does this. M3-seq describes it as acting "on unamplified transcripts before indexing" and judges that it "risks digesting non-rRNA transcripts."
4. **Not depleting, but priming differently.** Brettner et al. prime with both oligo-dT and random hexamers, and it is the hexamers that pick up rRNA, since rRNA is not polyadenylated. Shifting the hexamer:dT ratio is enrichment rather than depletion – nothing is destroyed, the assay simply stops reaching for it.

Where it happens in the workflow, corrected. An earlier version of this subsection placed depletion before fragmentation and before indexing, and argued that it had to be there. Both halves were wrong, and the correction matters because it changes what a protocol has to look like. Brandner et al. state their input exactly: "the mapSPLiT cDNA library (from after the final double-sided size selection, step 171 in Gaisser et al.)." In the microSPLiT protocol [7] fragmentation is steps 134–139, adapter ligation 152–154, index PCR 156–157, and the final double-sided size selection 158–170, with step 171 the QC that follows it. So the material entering the Cas9 reaction is a fully fragmented, adapter-ligated, indexed, size-selected, sequencing-ready library:

cDNA → fragment → ligate adapters → index → **deplete (Cas9)** → re-amplify P5/P7 → sequence

Why depleting that late works at all. A finished Illumina molecule needs both flanking adapters: P5 at one end, P7 at the other. Cut it anywhere in the middle and each half retains only one, so neither half amplifies exponentially and neither clusters. Cas9 does not remove rRNA from the tube – the mass is still there – it makes those molecules un-amplifiable, and the P5/P7 qPCR that follows selectively re-amplifies the survivors. Depletion here is a *selection*, not a purification, and it is coherent only because the molecules are already adapter-flanked. That is the same mechanism section 2.4 describes for why a mid-barcode cut causes loss rather than corruption.

M3-seq reaches the same end by a different road, and cannot use the cut-the-adapters trick because it converts the library to RNA first. There the rRNA is digested into pieces and the subsequent reverse transcription off a P5 primer simply never regenerates them.

The head-to-head, which argues against the obvious choice. This subsection previously cited M3-seq as corroborating the Cas9 route. It does the opposite. Wang et al. tested both and chose RNase H: "after testing two approaches for depleting ribosomal sequences from bulk libraries (Extended Data Fig. 3a–c), we chose an RNase H-based approach to complete our pipeline." Their Cas9 arm left 75–80% rRNA in the final library. The two published Cas9 results are therefore not consistent with each other – Brandner et al. reach 60.2% mRNA in *E. coli*, Wang et al. reach 20–25% – and the difference is plausibly the guide pool, since Brandner's is coverage-weighted against their own sequencing and Wang's is not described as being. That is a reason to treat guide design as the active ingredient rather than the enzyme.

The in-situ options were tested and lost, on a different axis. Kuchina et al. tried three before settling: polyadenylation with *E. coli* poly(A) polymerase, a 5′-phosphate-dependent exonuclease ("Terminator"), and RT with rRNA-specific probes followed by RNase H. PAP won, "about 2.5-fold, or about 7% of total RNA" enrichment of mRNA reads. Note what that means: the best in-situ result in the microbial literature is a 2.5-fold enrichment, against the 13-fold Brandner et al. get post-hoc. M3-seq's stated reason for avoiding in-situ work is that "depletion of rRNA in situ can decrease mRNA capture efficiency." In-situ depletion is not impossible – BacDrop does it – it is a trade of capture efficiency for read economy, and the published trade is unfavorable.

How much it is worth, and the transfer risk. Brandner et al. took *E. coli* from 4.6% to 60.2% mRNA, a 13-fold improvement, and median mRNA transcripts per cell more than doubled from 54 to 115. But the same protocol in *P. putida* moved only 2.2% to 10.4%. Depletion efficiency clearly does not transfer between organisms for free, since the guide pool is designed against one species' rRNA and against the coverage non-uniformity of one protocol. Yeast should be better than either bacterium – it starts at ∼85% rRNA rather than >95% – but the achieved figure has to be measured, and it is the single most valuable number the pilot can return (Sec. 6).

A reporting convention sits underneath the headline number, and the cost argument survives it. The 93.7% is not purely a biochemical fact. Gaisser et al. warn that comparing mRNA against rRNA fractions across methods "will depend on whether non-uniquely-aligned UMIs were retained during the analysis," because rRNA operons are multi-copy and most aligners discard multi-mappers by default. The swing is enormous: discarding them moves Kuchina et al.'s apparent mRNA fraction from 9.5% to 90.5% in *B. subtilis* and from 3.7% to 28.2% in *E. coli*. Yeast is if anything more exposed, since the rDNA repeat at *RDN1* is a tandem array of order a hundred copies, so nearly every rRNA read is a multi-mapper.

Two things keep the cost argument intact. Brettner et al. retained multi-mappers – their aligner is STARsolo and they state they "used the most basic uniform multi-mapping algorithm" – which is the same convention microSPLiT uses, so the yeast and bacterial figures are on the same side of it and are comparable. More importantly the confound is a *denominator* choice made after sequencing, and the budget question is about *reads purchased*. A read off an rRNA molecule was paid for whether the aligner assigns it uniquely, splits it, or throws it away; discarding multi-mappers removes rRNA from the count matrix, not from the invoice. Kuchina et al. say as much in passing, noting that when they discarded multiply aligned reads "mRNA UMI counts per cell were not strongly affected" even as the ratio moved by an order of magnitude.

The read-level number exists and it corroborates the lever: Gaisser et al. state that the uniquely aligned fraction in a STAR log "typically corresponds to the mRNA proportion in the library" and "usually falls between 3% and 12%," with 11.9% to 12.0% measured on their own *B. subtilis* data under both counting policies. Brettner et al.'s 5.75% sits inside that band, and the invariance across policies is exactly what the argument above predicts.

Which route to try first in yeast. The RNase H route is the better starting point, for a construction reason rather than a performance one. Brandner et al.'s Cas9 pool is species-specific *and* protocol-specific: its 253 spacers are weighted by the rRNA coverage their own runs produced, so porting it to yeast means redesigning the pool and then re-titrating it against yeast coverage that does not exist yet. A probe set against 18S, 25S, 5.8S and 5S is by contrast a design problem with a known answer – yeast rRNA is a small, well annotated set of sequences – and the same probes work regardless of which priming chemistry produced the coverage. The counter-argument is real and worth stating: RNase H costs an IVT and two extra enzymatic steps against Cas9's single 2 h incubation, and Brandner et al.'s route has been run on a split-pool library of exactly the kind proposed here while M3-seq's has not. **Hypothesis (untested):** in yeast the probe route reaches a higher mRNA fraction than a first-pass Cas9 pool, because the yeast rDNA repeat gives fewer distinct target sequences to cover. Nothing measures this yet, and section 6 ranks it accordingly.

> **Treat depletion as mandatory, not optional.** Without it the genome-scale screen costs \$157 k and sequencing dominates; with it, \$58 k and the budget becomes balanced. No other change in this document moves the total by as much.

3.6 Perturbation Identity Capture ✗

Both families above assume the guide can be read out of the cell. In yeast that has been the hardest assumption in the document to satisfy, and this subsection reaches a different conclusion from the one it used to: the obstacle is real for a 3′ assay and may not exist at all for a 5′ one.

gRNAs are transcribed from RNA polymerase III promoters – *SNR52* upstream, *SUP4* terminator – so they are neither capped nor polyadenylated and terminate on a T-tract. A 3′ poly(A) capture assay, which is what both Brettner et al.'s oligo-dT priming and 10x's bead oligo are, sees nothing. Our own MAGIC library uses exactly this architecture and reads out by plasmid amplicon sequencing, not RNA [2]; our reprocessing of it is `experiments/016-lian-magic-reprocess`.

One mammalian workaround has no yeast equivalent at all: CROP-seq [15] relies on the guide cassette being duplicated into a lentiviral 3′ LTR, which is a retroviral trick. The other, 10x's direct capture, turns out to depend on which assay is meant, and that distinction was missed in an earlier version of this subsection.

The 3′ and 5′ assays capture guides by different mechanisms, and only one of them requires rebuilding the library. On the 3′ assay the guide is caught by a capture sequence on the gel-bead oligo, so 10x require that "sgRNAs should be engineered containing either Capture Sequence 1 or Capture Sequence 2"▲ [external: 10x CG000316]. For an existing genome-scale yeast library that means resynthesizing every guide, which is the reading this document has been working from.

The 5′ assay does something else. Capture is by a primer against the sgRNA scaffold, carried in the reverse-transcription mix alongside the poly-dT primer: "For CRISPR screening, a PCR-based approach is used to capture the single-guide RNA (sgRNA), which is reverse transcribed"▲ [external: 10x CG000735]. The stated requirement is architectural rather than a modification: "sgRNAs should be engineered for use with standard Cas9 systems with a protospacer on the 5′ end," and the capture-sequence route is offered as an additional option rather than a prerequisite. A conventional Pol

III sgRNA cassette, including ours, has exactly that architecture. The cell barcode reaches the guide the same way it reaches any transcript on a 5′ assay, by template switching at the 5′ end, which puts the protospacer immediately downstream of the barcode and UMI where read 2 can see it.

This is the most consequential thing in this document that we did not previously know, and the core already runs the kit. If the 5′ scaffold primer reverse transcribes a yeast Pol III guide, then perturbation identity is readable from an *unmodified* library, and section 3.6's construction project collapses into a pilot measurement. Two things have to be true and neither is established. The primer set is proprietary and validated against the standard *SpCas9* scaffold, and 10x say plainly that "alternate sgRNA structures . . . have not been tested"▲ [external: 10x CG000735], so our scaffold has to match theirs. And a yeast Pol III transcript is not polyadenylated, so the poly-dT half of that primer mix contributes nothing and capture rests entirely on the scaffold primer. **Hypothesis (untested):** it works, because the guide is an ordinary RNA with the right 5′ architecture and the assay never needed the poly(A) tail for this molecule. Testing it needs one channel and an existing library, which makes it the cheapest high-value experiment named anywhere in this document.

Two practical numbers follow if it does work. The guide library is sequenced far more shallowly than the transcriptome, at a minimum of 5,000 read pairs per cell against 20,000 for gene expression, and 10x warn against sequencing it on its own▲ [external: 10x CG000735], so it is pooled with the expression library rather than being a separate submission. The reads are cheap; the risk is entirely in whether the chemistry engages.

What 3′ and 5′ actually name, since the whole argument above turns on it. A 3′ and a 5′ library differ in *which oligo carries the cell barcode*, and in nothing else that matters here. Both are built by the same two operations, and fig. 3 draws them against each other because the distinction between them is the one piece of chemistry in this document that is easiest to get backwards.

Template switching is not a 5′ feature. It is how nearly every short-read single-cell library gets a PCR handle onto its second end, and a molecule carrying a handle at one end only cannot be amplified at all, so a 3′ library template-switches exactly as a 5′ one does. What differs is the payload. In a 3′ library the barcode rides on the oligo-dT that primes reverse transcription and the **template-switching oligo** is supplied free, carrying nothing but the handle; in a 5′ library the two swap, and the TSO arrives on the gel bead carrying the barcode and the UMI while the oligo-dT is the free primer [8]. That swap is the entire reason the 5′ kit might read a Pol III guide: it puts the barcode at the transcript's 5′ end, next to the protospacer, where read 2 can reach both.

The split-pool route is a 3′ library and template-switches too, which is worth saying because the two operations sit side by side in it. Barcodes 2 and 3 are *ligated* on with a linker strand; the second handle is *copied* on from a TSO, in a bead-bound reverse-transcriptase step after lysis that the protocol has to run anyway, since formaldehyde crosslinking hobbles the in-cell reaction and the barcoded cDNA primes its own elongation [3, 7].

The gap has closed for deletions, and that changes the risk. The 2024 review states plainly that no genome-scale system in yeast lets a genetic perturbation be traced during scRNA-seq [8]. That was true when written and is no longer, because the same group built one the following year [5]. The construction is worth following closely, because it is the design this section was going to propose.

They rebuilt the yeast knockout collection so its genotype tag is *transcribed*. In the classical YKOC a *TEF1* promoter drives KanMX and two 20 bp barcodes sit outside the transcript, which – in their words – "renders the genotype identity invisible to transcriptomic readouts." They replaced the KanMX marker with *URA3* and shortened the heterologous terminator from 262 nt to 43 nt, which brings the Downtag barcode inside the *URA3* 3′UTR, and added a 5 nt clone barcode after the stop codon. The tag is now a polyadenylated Pol II transcript, so an oligo-dT assay sees it. That is exactly the architecture proposed below, executed at genome scale: 4,162 barcoded genotypes, of which more than 3,500 were recovered across 1,061,865 profiled cells, 710,952 of them passing QC with a genotype assigned.

Two numbers from it matter more than the rest. Genotype identity was assigned for **more than 71% of cells** – against the 21–25% Brandner et al. reach in bacteria – which is the parameter q that governs everything in section 4.6, and it is three-fold better than the figure this document has been budgeting with. And it was not free: the assignment combines the transcriptome with a *separate* one-step PCR amplicon library off the *URA3* 3′UTR. Targeted enrichment is doing the work in both systems, so it should be costed as a second library, not as a primer.

What is still open. Their perturbation is a chromosomal deletion with the tag built into the deletion cassette. Ours is a plasmid-borne CRISPRi guide, so the tag has to ride on the plasmid, and the construct does not exist. The architecture transfers directly, and its bacterial counterpart is also precedented: Brandner et al. [4] built a 316 bp non-coding transcript carrying a 12 bp barcode at its 5′ end, a poly(A) tail and a dedicated enrichment primer site, using an inert gene body as filler, and raised correct assignment from 5.5% to 21–25% by adding the enrichment primer at cDNA amplification. A design choice worth taking: make the barcode *be* the guide's own 20 nt spacer, placed in the 3′UTR of a Pol II reporter, so the guide-to-barcode mapping is known by construction – that removes a linkage sequencing run and a whole class of bookkeeping error.

Still the critical path, but no longer the risky one. Every cost in Sec. 5 assumes guide identity is recoverable, and for a plasmid-borne CRISPRi guide it is still unbuilt. What changed is that the same thing has now been done in yeast at genome scale with a published construct and a 71% assignment rate, so this is a construction and validation task against a working template rather than an open research question. The number to reproduce in the pilot is q, the fraction of cells whose guide identity is correctly assigned, because section 4.6 shows it enters the multiplex ceiling as q^k in the guides per cell k.

4 Theoretical Limits ✗

The question this section answers: given how much mRNA a yeast cell contains and how little of it any method recovers, how many cells must share a perturbation before its mean effect is measurable – and how does that change when a cell carries more than one perturbation?

4.1 Counting Is Poisson, and It Appears Three Times ✗

One distribution has already appeared four times in passing and is about to start carrying weight, so it is worth making precise once rather than in four disguises. A Poisson distribution is what you get whenever many independent things each have a small chance of landing in the same place and you count how many did. Its defining property is that its variance equals its mean, so the only parameter is that mean, conventionally λ: fix the average and the spread is fixed with it. **You cannot set an outcome, only an average, and the outcomes arrange themselves around it.** That is the whole of the idea and it is the part people find counterintuitive.

Three of its appearances set a design parameter, and they pull in two directions.

- **Molecules sampled by a sequencer** (**shot noise**), where the Poisson is over reads and the design goal is to make λ large. This is the one the rest of section 4 runs on.
- **Cells loaded into droplets** (section 2.2), where the Poisson is over cells per partition and the goal is a *small* λ, so that two cells rarely share a droplet. Preindexing (section 3.3) is what makes a large λ safe instead.
- **Plasmids taken up at transformation** (section 7.1), where the Poisson is over uptake events per cell and a large λ is again wanted, because plasmids per cell is what puts several guides in one cell.

The fourth appearance is not a design parameter and is worth naming so it is not mistaken for one: cuts falling along a cDNA molecule during fragmentation are Poisson too (section 2.4), which is what makes capture probability a function of the size-selection window rather than of transcript length.

Figure 8 builds the distribution on the third of the three, because it is the one where the counterintuitive property does real work: selection kills the cells that took up nothing, and deleting the smallest value from a distribution *raises* the mean of what is left. That arithmetic is used in full in section 7.1; everything between here and there uses the first.

Two sources of noise, and only one of them is fixable by sequencing. Every argument below turns on a distinction worth making once, plainly. **Shot noise** – also called counting or sampling noise – is the Poisson above, applied to sequencing. A library holds far more molecules than a run reads, so the run draws a random sample of them, and a gene whose expected count is μ UMIs comes back as a count scattered around μ with variance also μ, hence a relative error $1/\sqrt{\mu}$: 25% at 16 expected UMIs, 5% at 400. Note what μ counts. It is molecules of one gene summed over all the cells sharing a perturbation, not molecules in one cell, so it rises with cells *and* with depth and both routes buy the same precision (section 4.3). Shot noise is a property of the measurement rather than of the cell, and it is the one kind of noise that buying more reads removes.

Biological overdispersion is the rest: two genetically identical cells in the same flask genuinely differ in how much of a gene they are expressing, because of cell-cycle phase, metabolic state and transcriptional bursting. The name records a comparison. Pure counting pins the variance at μ, so a gene whose counts vary across cells by more than that is dispersed *over* what the counting model allows, and φ_j measures the excess. Sequencing each cell to infinite depth measures each cell perfectly and leaves this variation completely untouched; only counting *more cells* averages it down. Keeping the two apart is what section 4.9 is for, since the whole design reduces to which of them is currently binding.

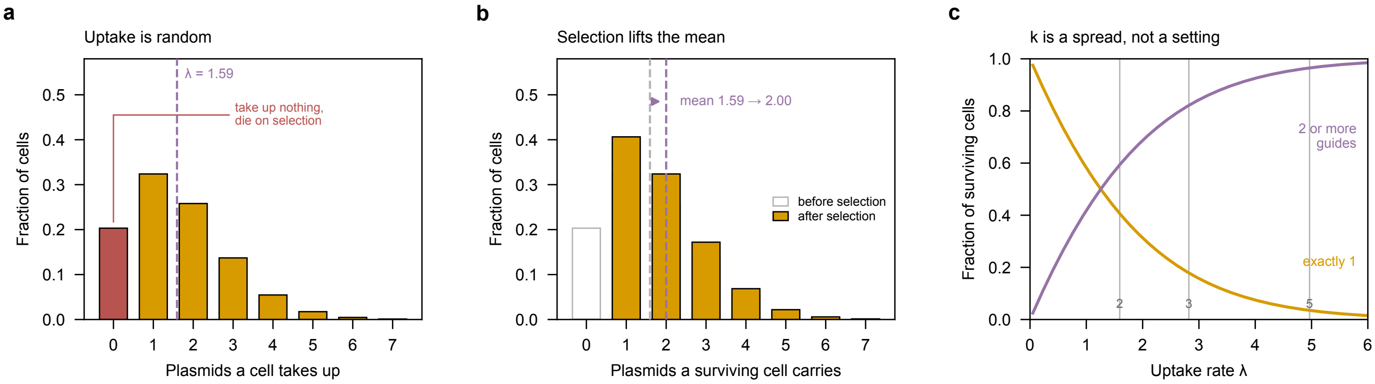

Figure 8: Poisson sampling, built on plasmid uptake at transformation. The setting, since it is developed in full much later: a guide library is transformed into a population, each cell takes up some number of plasmid molecules, and cells that took up none die on the selection marker. That is the mechanism section 7.1 uses to put several guides in one cell, and it is drawn here rather than the sequencing case because one property of it is genuinely counterintuitive. **(a)** A plasmid cannot be handed to a cell. Raising the DNA concentration raises an *average* uptake rate λ, and cells sort themselves around it: at $\lambda = 1.59$ about a fifth of cells take up nothing and die on selection (red), while others take up three or four. Nothing in the protocol picks which. **(b)** Selection deletes that zero class, and deleting the smallest value *raises* the average of what remains, here from $\lambda = 1.59$ to 2.0 plasmids per surviving cell. This is the "zero-truncated" part, and it is why the λ a protocol is set to is always lower than the plasmids-per-cell figure it delivers (table 17 gives the operating points). Gray outlines are the distribution before selection, filled bars after. **(c)** What that means for a design. Vertical rules mark the λ giving 2, 3 and 5 plasmids per surviving cell. Turning λ up to get more multiply-loaded cells necessarily produces triples and quadruples nobody asked for, so k is a distribution to be steered rather than a number to be set, and even at a target of 3 nearly a fifth of cells still carry one guide. The same mathematics governs droplet loading (section 2.2) with the goal reversed: there λ is kept low so that two cells rarely share a droplet, and preindexing (section 3.3) is what makes a high λ safe. Generated by `experiments/024-perturb-seq-costing/scripts/plot_poisson_primer.py`.

One distribution, three design parameters, and two of the three want it pushed the opposite way from the third. Sequencing wants a large mean λ because shot noise falls as $1/\sqrt{\mu}$ in the mean count μ; plasmid uptake wants a large one because that is what puts several guides in a cell; droplet loading wants a small one so that two cells rarely share a partition, and preindexing is the change that lets it want a large one instead. The property none of them escapes is that only the average is set, never the outcome. **And only the first of the three is noise that money removes**: reading a cell more deeply measures that cell better and leaves the differences between cells exactly where they were.

4.2 Transcript Content per Cell ✗

Every capture-efficiency claim needs this denominator, and the literature does not agree on it.

The disagreement is real and is left visible rather than averaged away: the 3,000 genes at 3.5 molecules each implied by the review works out to ∼10,500 mRNA per cell, materially below the 20,000 floor of the range Jariani et al. cites and well below the 40,000–60,000 Jackson et al. give.

This choice is not harmless, and the working value mixes two sources. It is tempting to treat the spread as a detail, and it is not. The design equation reaches this quantity through p_j, the transcriptome share of the target gene, and p_j enters only the shot-noise term – where both candidate platforms sit (section 4.9), so n^* scales with it almost directly. Taking 3.5 molecules per gene against each candidate denominator, at $d = 410$ and the measured effect size, gives 765 cells per perturbation at 10,500 total mRNA, 2,001 at 30,000, and 3,901 at 60,000. That is a five-fold spread, and it is larger than most of the effects this document argues about.

Read the spread, not the level. The level here is not what section 5 budgets against, and the two must not be confused. eq. (5) pairs an effect size with an overdispersion, because φ is pinned by requiring the 100-cell convention to fall out of the equation's floor: at the measured 1.34-fold, $A = 90.9$ and $\varphi = 1.1$; at a nominal two-fold, $A = 16.3$ and $\varphi = 6.1$. Both give a 100-cell floor, and the second pairing gives 219, 441 and 782 for the same three denominators. So the absolute number moves by a factor of three with the effect-size assumption, while the *spread* across denominators stays at five-fold either way. The spread is the finding; it is a property of the unresolved p_j, and it is why measuring p_j in our own pilot data retires more uncertainty than any other cheap measurement in this document.

Worse, the working value is not internally consistent. The document uses $p_j = 3.5/30{,}000$, which pairs the review's per-gene figure with Brettner's total – but the 3.5 was derived *against* the review's own ∼10,500, so used self-consistently

Table 3: mRNA content per cell. This is the denominator for every capture-efficiency claim in the document. [a] BNID 112795, `https://bionumbers.hms.harvard.edu/bionumber.aspx?id=112795` – the one row here that is *not* in the torchcell-library mirror. No mirrored bacterial paper states a total mRNA content, only what its assay detected, which is a different and ∼30× smaller quantity. Every other row is mirror-backed and separately citable. **The yeast rows disagree and are left that way.** Jariani et al.'s 20,000 floor, Jackson et al.'s 40,000–60,000 and the ∼10,500 implied by the review's own 3,000 genes × 3.5 molecules span six-fold. The document works at ∼30,000; because p_j enters eq. (5) only through the shot-noise term, and section 4.9 finds both candidate platforms firmly shot-noise limited, this choice does move the cells-per-perturbation numbers roughly in proportion.

Organism	Quantity	Value	Source
S. cerevisiae	Total mRNA molecules per cell	**20,000–60,000**	Jariani 2020 `(md:31)`
	Working figure (Brettner)	∼30,000	Brettner 2024 `(md:70)`
	Total mRNA molecules per cell	40,000–60,000	Jackson 2020 [16]
	Implied by 3,000 genes × 3.5 molecules	∼10,500	Nadal-Ribelles 2024 `(md:107)`
	Mean UMIs per detected gene, per cell	3.46	Nadal-Ribelles 2019 [17]
	Capture rate, 10x 3′ v2	3–5%	Jackson 2020 [16]
	Total RNA per cell (∼85% rRNA)	0.7–1 pg	Nadal-Ribelles 2024 `(md:49)`
	Protein-coding genes	∼6,000	SGD / R64
E. coli	Total mRNA per cell, **rich (LB)**	**∼7,800**	Bartholomäus 2016[a]
	Total mRNA per cell, minimal medium	∼2,400	Bartholomäus 2016[a]
	Total mRNA per cell, medium growth rate	∼3,000	Taniguchi 2010 [18]
	rRNA share of all transcripts	**>95%**	Brandner 2025 `(md:71)`
	Protein-coding genes	∼4,400	K-12 MG1655
Mammalian	Total mRNA molecules per cell	50,000–300,000	Jariani 2020 `(md:31)`

it would give $p_j = 3.3 \times 10^{-4}$ and cut every cells-per-perturbation figure by a factor of 2.9. The 30,000 pairing is the conservative one, which is why it is kept, but it is a choice and not a measurement, and table 9 accordingly marks p_j derived.

The comparison that does matter is across organisms – a yeast cell holds roughly four times the mRNA of an *E. coli* cell in rich medium over ∼1.4× the gene count, which is why the same split-pool chemistry recovers ∼410 mRNA UMIs per cell in yeast and 235–397 in bacteria.

What would settle it, concretely. The equation does not need total mRNA per cell. It needs p_j, the fraction of a cell's mRNA belonging to the target gene, and the three-way disagreement above exists only because p_j is currently being *assembled* from two sources that do not belong together: a per-gene molecule count from one paper over a per-cell total from another.

A pilot measures p_j directly and skips the assembly. Any run that yields a count matrix gives it: take the UMIs assigned to a gene, divide by the total mRNA UMIs in that cell, and average over cells. That is one line of analysis on data the first experiment produces anyway (section 6.2), it needs no extra sequencing and no extra sample, and it replaces a derived quantity with a measured one for the genes we actually intend to target. Two caveats keep it honest: p_j is per-gene, so it should be reported for a handful of representative targets rather than as one number, and it is condition-dependent, so it belongs to the environment it was measured in.

A denominator nobody has measured moves the whole budget five-fold. How much total mRNA a yeast cell holds is disputed by a factor of six across three sources, and cells per perturbation moves with it. The design equation does not actually need that total: it needs p_j, the target gene's share of the transcriptome, which any count matrix yields directly as gene UMIs over total mRNA UMIs per cell. The first pilot produces one, so this is a line of analysis rather than an experiment, and it replaces the largest derived quantity in the budget with a measured one.

4.3 The Cells-Versus-Depth Frontier ✗

Plotting every published yeast and microbial run on one pair of axes – cells profiled against mRNA UMIs recovered per cell – shows that the field has not found a method that is good at both (fig. 9; per-study values in table 4).[1] Depth does fall as cell count rises, so there is a trade-off, but it is a much weaker one than the picture suggests at a glance. Cells profiled span 3.6 orders of magnitude across these ten studies and depth only 1.6, and moving from the smallest study to the largest multiplies cells by 3,726 while dividing depth by just 9.8. Both endpoints happen to be Nadal-Ribelles et al., six years apart [5, 17].

[1] Plotted studies, in order of cells profiled: Nadal-Ribelles et al. 2019 [17], Urbonaite et al. [19], McNulty et al. [20], Jariani et al. [10], Kuchina et al. [9], Jackson et al. [16], Brettner et al. [3], Brandner et al. [4], Boocock et al. [21] and Nadal-Ribelles et al. 2025 [5]. Tabulated but not plotted, for want of a per-cell molecule count: Gasch et al. [22], Ma et al. [23], Dohn et al., and – below the rule, because it is not a microbial study – Datlinger et al. [6].

The consequence is worth stating plainly, because the figure's diagonals invite the opposite reading. *Total* UMIs is not conserved along this cloud: it spans 3.2 orders of magnitude, which is nearly all of the variation in cell count. Larger studies do not redistribute a fixed amount of signal over more cells; they collect more signal. The published record is therefore not a frontier of equal-value points, and no method here has been shown to be a better bargain than another by sitting further out along a diagonal.

What the diagonals do encode is a statistical fact about *design*, not a summary of what has been published. In the shot-noise limit the precision of a perturbation's estimated mean depends on how many molecules were counted in total, not on how they were distributed over cells, so for a fixed sequencing spend, moving along a diagonal is free and should be decided on other grounds – instrument access, cost structure, whether a perturbation can be read out at all. That is the lever section 5 pulls. It is a claim about the arithmetic in eq. (5), and the studies plotted here neither support nor contradict it, because none of them was designed against a common budget.

Two features carry into the rest of this document, and the second has changed recently. Published runs span 9×10^5 to 1.3×10^9 total UMIs. At the low end that is three orders of magnitude inside what a single modern flow cell delivers (section 5.3), so those studies were bounded by protocol throughput and budget rather than by sequencing capacity – which is why platform choice, not sequencer choice, is what a small screen design turns on. That is no longer true at the top end. Nadal-Ribelles et al.'s 1.06-million-cell atlas needs 1.3×10^9 UMIs, within a factor of three of the 3.2 billion read pairs in one NovaSeq X 25B lane *before* counting duplication or the rRNA fraction, so at genome scale in yeast the sequencer has started to bind. Section 5 prices exactly that regime.

Second, four methods now carry a perturbation readout, and they no longer sit at one end of anything: they run from Jackson et al.'s 38,225 cells to Nadal-Ribelles et al.'s 1.06 million, and from 325 UMIs per cell to 2,809. The largest genetic screen in this figure is a yeast one, and it used neither of the two chemistries this document has been comparing – it used a microwell array (section 3.6). Split-pool versus droplet is therefore not a throughput question at the scales anyone has run; it is a question of depth and of cost per *usable* cell, which is section 5.

The one experiment that moves a point right without moving it down. Every study here is microbial, which is the axis this figure is about, and it is why scifi-RNA-seq is absent from it: Datlinger et al. ran human and mouse material only (section 3.3). The reason to raise it here anyway is that it tests the claim the diagonals invite and the cloud denies. Preindexing took one Chromium channel from ∼15,000 loaded nuclei to 383,000, returning 151,788 transcriptomes, and measured no downward trend in UMIs per cell across droplets holding up to 15 nuclei. Cells rose by an order of magnitude at constant depth, which is a move to the right along no diagonal at all – exactly the "bigger studies collect more signal" behavior the published record already shows, produced deliberately and in one experiment rather than incidentally across a literature. It also localizes what is being traded. What the cloud's weak downward slope reflects is not a law relating cells to depth; it is that cheap-per-cell chemistries have so far also been shallow ones. Preindexing separates those two properties, and section 5.5 prices the separation.

The published cloud is not a frontier, because total UMIs is not conserved along it. Bigger studies collect more signal rather than spreading a fixed amount of it thinner. What the diagonals encode is a fact about design rather than a summary of the literature: at a fixed sequencing spend, trading cells against depth is free, so the choice should be made on instrument access, cost structure, and whether the perturbation can be read out at all.

4.4 Cells per Perturbation ✗

Two requirements act independently, and the larger of the two binds.

Constraint 1 – a biological floor. Averaging out cell-cycle phase and metabolic state takes cells, and no amount of sequencing depth substitutes for them. The field standard is stated explicitly by Brandner et al. [4]: perturbations are retained only with at least 100 cells. Brettner et al.'s own heuristic is 500 cells per condition. The strongest corroboration is a design choice rather than a rule: the genome-scale yeast screen of Nadal-Ribelles et al. [5] landed at an average of 93 and 108 cells per genotype in its two conditions (medians 77 and 87). A million-cell budget spread over 3,500 mutants converges on the same hundred-cell scale that the convention asserts, arrived at independently.

Constraint 2 – pseudobulk depth. Pooling n cells at depth d gives $K \approx ndp_j$ UMIs of gene j. For counts, $\mathrm{Var}(\log K) \approx 1/K$, so resolving a $\log_2$ fold change Δ at power $1-\beta$ and level α requires

$$K \geq \left(\frac{z}{\Delta \ln 2}\right)^2, \qquad z = z_{1-\alpha/2} + z_{1-\beta} = 2.80, \tag{2}$$

which is ∼16 UMIs of that gene for a two-fold change, ∼69 for 1.4-fold, ∼158 for 1.25-fold. This is a floor, not an estimate: it counts only sampling noise and ignores biological variance between cells, which is exactly why the 100-cell floor is a separate and independent requirement.

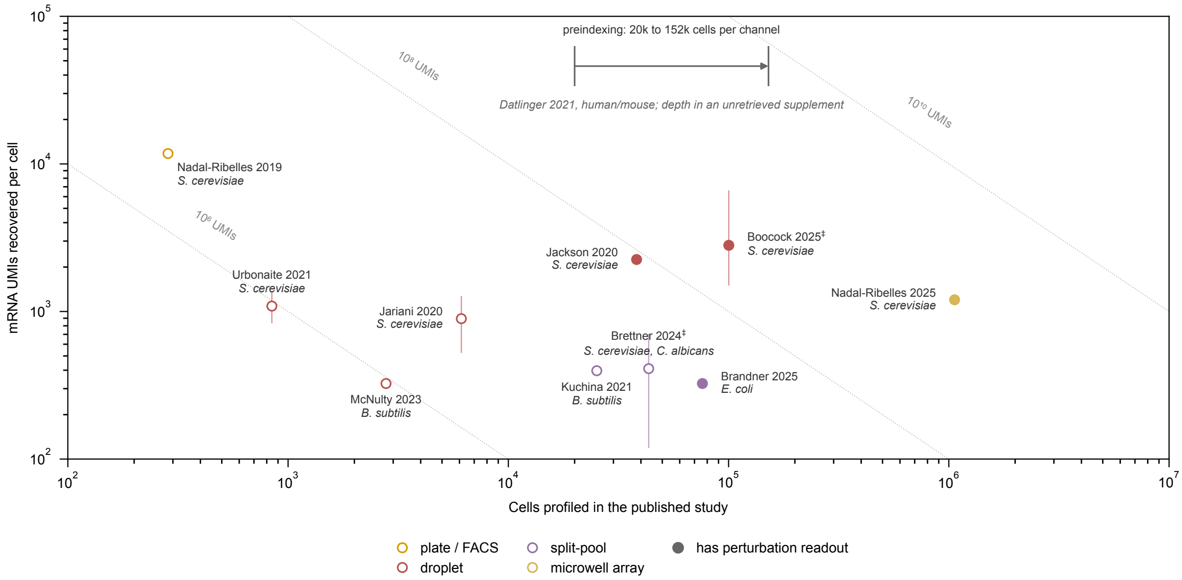

Figure 9: Published microbial single-cell RNA-seq trades cells against depth, but far more weakly than the diagonals suggest. Cells profiled in each published study (x) against mRNA UMIs recovered per cell (y), organism in italic beneath each label; marker color is the isolation principle and filled markers denote a method with a perturbation readout. Across these ten studies cells span 3.6 orders of magnitude and depth only 1.6, and the largest study holds 3,726 times the cells of the smallest at just 9.8 times less depth – so *total* UMIs is not conserved along the cloud, but spans 3.2 orders of magnitude, nearly all of the variation in cell count. Bigger studies collect more signal rather than spreading a fixed amount of it thinner. **The dotted diagonals are therefore a design tool, not a summary of this data.** They are iso-lines of constant total UMIs, and that product is what sets the shot-noise floor on a perturbation's estimated mean, so at a *fixed budget* sliding along one is free and only moving up-and-right buys precision. None of these studies was designed against a common budget, so none of them tests that. **The vertical bars are not error bars.** No plotted study reports a dispersion statistic over cells. Each bar is the min–max range of the *per-condition summary values* a study reports – medians or means across its own treatment conditions – so it describes how far the assay moved between conditions, not how much cells vary within one; ‡ marks a bar that spans separate experiments of deliberately differing depth, which is weaker still. Studies reporting a single value have no bar. **Three methods are absent from the depth axis by design.** Gasch et al. used Fluidigm C1 with SMART-Seq, which carries no UMI; Ma et al. report only genes per cell; and Datlinger et al. measured a per-cell depth but published it out of reach, deferring the performance metrics to a supplementary table that sits behind the journal's paywall and is not in the mirror – every UMI-per-cell reference in the article itself is a plot axis. So for two of the three no molecule count exists, and for the third it exists and is not in hand; in neither case is one invented. Their measured quantities appear in table 4. This matters because the quantity most easily confused with depth here is the gene count, and reading one as the other moves a method bodily along the frontier. **The gray arrow is not a data point.** It is the throughput move preindexing produces, drawn at a height chosen to carry no claim – above every marker, in the empty band below the frame – because scifi-RNA-seq has no coordinate on the depth axis and is not a microbial study. Its two ends are the UIUC baseline of 20,000 cells recovered per Chromium channel and the 151,788 Datlinger et al. recovered from one overloaded channel [6]. It is here because it is the only published experiment that moves a method to the right without moving it down: UMIs per cell showed no downward trend in droplets holding up to 15 nuclei (section 3.3). Published runs span 9×10^5 to 1.3×10^9 total UMIs. The smaller ones sit three orders of magnitude short of the 3.2 billion read pairs a single NovaSeq X 25B lane delivers (fig. 14) and were limited by protocol throughput and budget rather than by the instrument; the largest is within a factor of three of one lane before duplication and rRNA are counted, so at that scale the sequencer does bind. Plate methods buy depth by giving up cell number, bottoming out at a few hundred cells; droplet methods run deeper than split-pool at comparable cell counts, the largest being the ~100,000-cell eQTL run of Boocock et al. [21]; split-pool (Brettner et al., microSPLiT, mapSPLiT) sits at 10^4–10^5 cells and ~300–400 UMIs, on the order of 1% of the 20,000–60,000 mRNA molecules a yeast cell carries; and the microwell array of Nadal-Ribelles et al. [5] reaches 1.06 million cells at 1,200 UMIs. That last point is a genome-scale yeast Perturb-seq – 3,500 barcoded deletion mutants read out in one atlas – and it is the largest genetic screen here (section 3.6). ‡ Boocock et al.'s seven datasets were sequenced to deliberately different depths, so the bar spans the per-dataset medians (1,514–6,559) and the marker is placed at their cell-weighted mean, 2,809 UMIs, from Supplementary file 1 Table S4 – weighting by cells is what keeps the point describing the same 100,220 cells its x-coordinate claims. Nadal-Ribelles et al.'s depth is the one derived value (3,399 genes per cell × 3.46 UMIs per detected gene = 11,760); the source states both factors but never their product. Generated by `experiments/024-perturb-seq-costing/scripts/plot_method_landscape.py`.

Table 4: Published single-cell RNA-seq runs in yeast and other microbes, plus one non-microbial comparator below the rule. †● [via the Nadal-Ribelles review, not primary] values taken from that review's summary table because the primary paper is not in our mirror, so they are second-hand and should be verified before being cited. ‡ the quoted UMI range spans separate *experiments* of deliberately differing depth rather than conditions within one run. **No range in the UMI column is a dispersion statistic over cells**; all are min–max ranges of per-condition or per-experiment summaries. **A dash in the UMI column is deliberate**: Gasch et al. used a protocol that carries no UMI and Ma et al. report only genes, so for those two no molecule count exists. Datlinger et al. is a different case worth distinguishing: a per-cell depth was measured, but the article defers its performance metrics to a supplementary table held behind the journal's paywall and absent from the mirror, so the value exists and is not in hand. No value is invented in either case, and where a measured gene count exists it appears under Genes/cell instead. **The row below the rule is not microbial.** scifi-RNA-seq was run on human and mouse material only; it is tabulated because section 3.3 and the budget both turn on it, and it is separated because a mammalian cell count is not evidence about microbes. Its 151,788 is a recovery through quality control from one Chromium channel, not the 1.53-million-nuclei loading demonstration, which is a feasibility ceiling. It is absent from fig. 9 for want of a depth coordinate; the throughput move it demonstrates is drawn there as an arrow instead. **Basis** states how the single plotted value was obtained: *reported* = stated by the source; *midpoint* = midpoint of a reported range, a declared convention where no per-condition value exists; *median-of* = median of the reported per-condition values; *weighted* = cell-weighted mean of reported per-dataset values; *derived* = computed from two quantities the source states separately, never printing the one needed here. "Pert." = has a perturbation readout.

Study	Platform	Isolation	Cells	mRNA UMIs/cell	Genes/cell	Pert.	Basis
Gasch 2017	Fluidigm C1	plate	163	–	735–5,437	–	rep.
Nadal-Ribelles 2019	FACS + STRT-seq	plate	285	11,760	3,399	–	deriv.
Jackson 2020	10x 3′ v2	droplet	38,225	2,250	695	yes	rep.
Jariani 2020	10x 3′ v2 (in-drop zym.)	droplet	6,118	528–1,261	–	–	med-of
Urbonaite 2021	yeastDrop-Seq	droplet	844	840–1,336	443–619	–	midpt.
Dohn 2021†	mDrop-seq	droplet	74,814	–	193–967	–	rep.
Boocock 2025‡	10x 3′ v2 + v3	droplet	100,220	1,514–6,559	–	yes	wtd.
Nadal-Ribelles 2025	Singleron GEXSCOPE HD (microwell)	microwell	1,061,865	1,200	550	yes	rep.
Brettner 2024‡	SPLiT-seq (yeast)	split-pool	43,388	120–700	100–600	–	midpt.
Kuchina 2021	microSPLiT	split-pool	25,214	397	230	–	rep.
McNulty 2023	ProBac-seq	droplet	2,784	325	241	–	rep.
Ma 2023	BacDrop	droplet	44,140	–	30–127	–	rep.
Brandner 2025	mapSPLiT	split-pool	76,068	325	161	yes	rep.
For comparison, not a microbial study:							
Datlinger 2021	10x ATAC + preindex.	preindexed-droplet	151,788	–	–	yes	rep.

Table 5: Cells per perturbation at the standard pseudobulk tier (2×10^5 mRNA UMIs), and what multiplexing does to the total. The 100-cell biological floor and the depth requirement are independent; the larger binds. **Cells/pert.** does not depend on k: a perturbation needs the cells it needs, whatever else those cells carry. What k changes is the *total*, because a cell carrying k guides informs k different main effects, so the library total divides by k (section 4.6). k **is guides per cell**, each against a different gene – not the six guides per gene the library carries. These totals are main effects only; named gene pairs scale quite differently and are table 18.

				Millions of cells for 6,000 genes, at k guides per cell				
Method	mRNA UMIs/cell	Cells/pert.	Binds on	$k = 1$	$k = 2$	$k = 3$	$k = 5$	$k = 8$
SPLiT-seq as published	410	**488**	depth	2.93	1.46	0.98	0.59	0.37
SPLiT-seq + rRNA depletion	861	**233**	depth	1.40	0.70	0.47	0.28	0.17
10x v2 (Jariani)	894	**224**	depth	1.34	0.67	0.45	0.27	0.17
10x v3 (Jackson)	2,000	**100**	biological floor	0.60	0.30	0.20	0.12	0.07

Two things follow. The 488 cells that the depth constraint demands of Brettner et al.'s chemistry reproduces his own 500-cell heuristic from an independent derivation, which is reassurance that the framework is calibrated. And depth buys you out of cells only until the 100-cell floor binds – past 10x v3 chemistry, extra depth is wasted on this question and should be spent on more perturbations instead.

Two independent constraints, and the larger one binds. A perturbation needs enough cells to average out cell-cycle phase and metabolic state, and enough total molecules to beat counting noise. Depth buys you out of cells only until the 100-cell biological floor binds, so past roughly 10x v3 depth extra reads are wasted on this question and belong on more perturbations instead.

4.5 Effect Size of a Single Knockdown ✗

Equation 2 takes the fold change Δ as given, but Δ is an empirical property of yeast regulatory biology, and it is the parameter the whole design is most sensitive to. It is measured here rather than assumed, from two datasets already in the library mirror: Kemmeren et al.'s single-deletion expression compendium [24] and Sameith et al.'s double-deletion compendium [25], both built locally and analyzed below.

What counts as a responding gene, stated before it is used. A gene is called *responding* to a deletion when $|\log_2 \mathrm{FC}| > \log_2 1.25$ against wild type, that is, when its expression moved by more than 1.25-fold in either direction. No significance test sits behind that, and none is available: the compendia report a log ratio per gene per strain, so the criterion is a magnitude cut, applied identically to all three datasets. The cut is a choice, it is the one this document argues from, and fig. 10c shows what every alternative choice would have given.

Table 6: Measured response of the transcriptome to a gene deletion. **"Responding" means** $|\log_2 \mathbf{FC}| > \log_2 1.25$, that is, a gene whose expression moved by more than 1.25-fold in either direction relative to wild type, on the array platform of the source compendium. There is no significance test behind it and none is available per gene per strain: the criterion is a magnitude cut on the reported log ratio, applied identically to all three datasets. A single deletion moves a few hundred genes by that criterion but only ∼10–15 by 2×, so designing at two-fold targets a few percent of the response. **The 1.34× in the last column is conditioned on that cut and is not independent of it.** The $|\Delta|$ distribution falls off monotonically (fig. 10a), so the median of whatever upper tail is selected sits just above the cut that selected it: the same statistic is 1.16× at a 1.1× cut, 1.36× at 1.25×, 1.73× at 1.5× and 2.57× at 2× (fig. 10c, right axis). What the column establishes is therefore not a natural effect size but a consequence of choosing 1.25×, and the choice is what to argue with. It matters because the power coefficient A scales as $|\Delta|^{-2}$, so at 1.34× every cell requirement is ∼5.6 times the two-fold figure. Doubling the perturbation count roughly doubles the responder count at 1.25× and trebles it at 2×, without shifting the median response – more perturbations make a cell noisier, not its individual effects larger. Generated from `experiments/024-perturb-seq-costing/scripts/effect_size_analysis.py`.

Compendium	Strains	Median responding genes at 1.25×	1.5×	2×	Median $\lvert\Delta\rvert$, responders $\log_2$	fold
Kemmeren 2014, single deletion	1,484	269	55	15	**0.42**	1.34×
Sameith 2015, single deletion	82	220	42	10	**0.42**	1.34×
Sameith 2015, **double** deletion	72	460	112	30	**0.44**	1.36×

Three findings, and the first two change the arithmetic of the whole document.

A deletion moves hundreds of genes, but almost none of them by two-fold. The median single deletion moves 220–269 genes by more than 1.25×, but only 10–15 by more than 2× (table 6). Designing a screen at a two-fold threshold therefore targets a few percent of the transcriptional response and discards the rest. Kemmeren's 1,484 single deletions and Sameith's 82 agree closely, which is reassurance that this is a property of yeast rather than of one platform.

The median responding gene moves 1.34-fold, and that number inherits the 1.25 cut. Among genes clearing the 1.25× criterion, the median $|\Delta|$ is 0.42 in $\log_2$ units, i.e. 1.34-fold, and the three compendia agree to within 0.02 despite differing in size by a factor of eighteen. This is the number Eq. 2 wants.

Where it comes from is worth being exact about, because 1.34 looks like a measured property of a yeast deletion and is not one. The $|\Delta|$ distribution decays monotonically from zero (fig. 10a), so selecting its upper tail at any cut returns a median sitting just above that cut. Recomputed across the ladder, the median response among responders reads 1.16× at a 1.1× cut, 1.36× at 1.25×, 1.73× at 1.5× and 2.57× at 2× (fig. 10c, right axis) – close to a straight line against the cut itself. So 1.34 is the arithmetic consequence of calling 1.25× a response, and the defensible content of this subsection is not the value but the pair: at a 1.25× definition, a deletion moves a few hundred genes and the typical one of them moves 1.34-fold.

That pair is what a design has to be built on, and both halves matter, because the two move in opposite directions. A stricter cut raises $|\Delta|$, which makes each surviving gene cheaper to resolve, and cuts the responder count, which

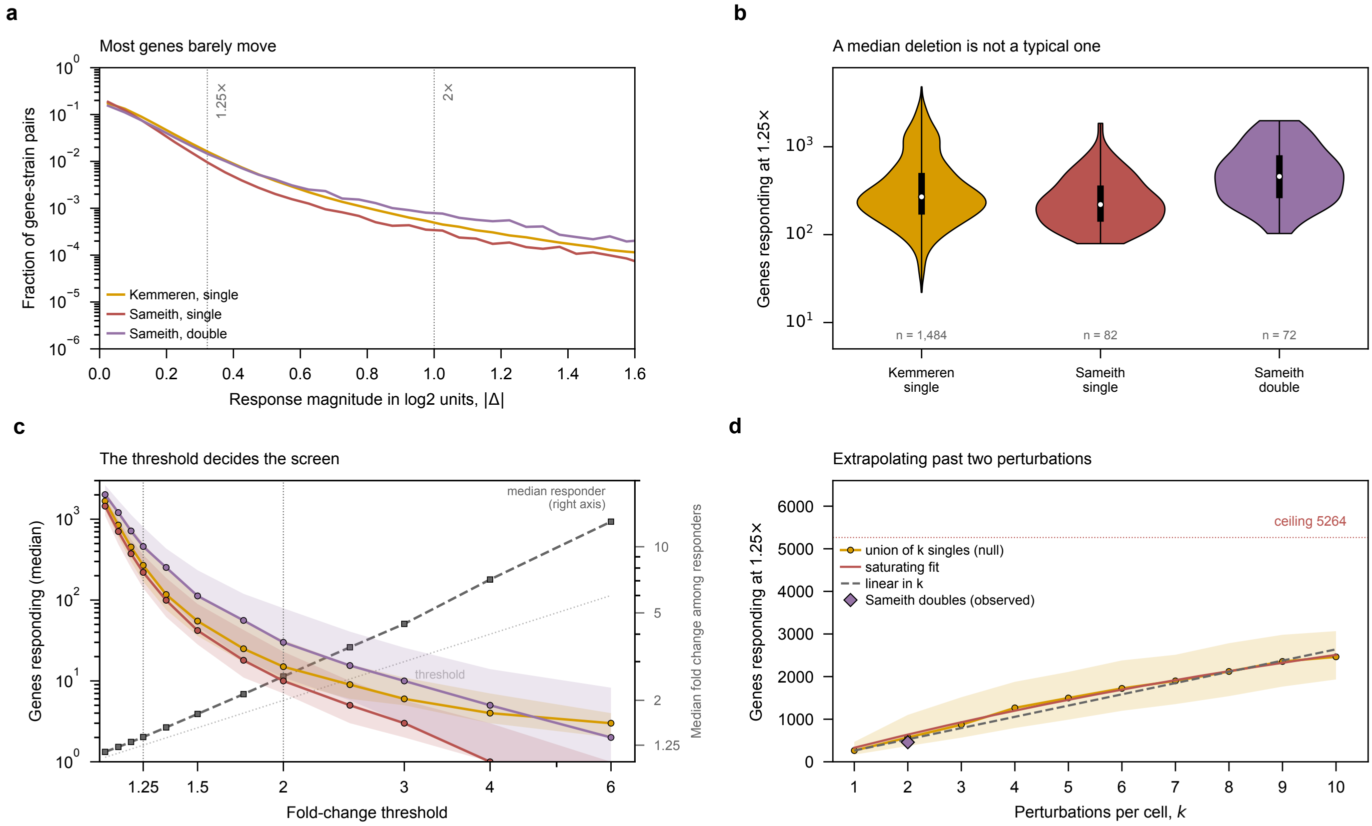

Figure 10: What differential expression after a gene deletion actually looks like, and why a two-fold design target is the wrong one for this biology. All four panels are the Kemmeren and Sameith compendia, singles and doubles shown separately. (a) Distribution of response magnitude over every gene-in-every-strain pair. It is a steep decay: the mass sits below $|\Delta| = 0.5$, and the two thresholds the document argues about fall far out on a thinning tail, so the threshold chosen decides how much of the response is visible at all rather than merely how it is filtered. The double-deletion curve sits above the singles across the whole range, but the shape is the same – more perturbation moves more genes, it does not change the character of the response. (b) Responders per strain at 1.25×, as a full distribution. Violin width is a kernel density over $\log_{10}$ counts – these distributions are strongly right-skewed as counts (skew 1.3–3.3) but nearly symmetric once logged (0.08–0.56), so log space is where a density is a fair summary; the black bar is the interquartile range, the thin line the observed full range, and the white dot the median. The interquartile spread *within* each group is wider than the gap between singles and doubles, and the range runs from 22 genes to 4,810 – a factor of 18 above the median at the top end, with 12% of Kemmeren's strains moving more than a thousand genes. A screen is not designed for the median deletion, because the median deletion is not a typical one, and cells-per-perturbation has to be set for the quiet end. **The two Sameith violins end in a blunt horizontal edge and are not capped**; n is printed under each for that reason. A violin body is clipped at the observed extremes, and with 72–82 strains the kernel density is still appreciable when it reaches the largest one, so it stops flat. Kemmeren has 1,484, so its density has decayed to nothing by its maximum and comes to a point. The flat edge is where the data stops, not where it was cut. (c) The design curve – median responders against the threshold, shaded to the interquartile range. Between 1.25× and 2× the count falls by roughly 15–20-fold in every dataset. That collapse is the cost of a two-fold target stated in the currency that matters: choosing it discards about 95% of the genes that actually responded. **The dashed gray series on the right axis is the other half of the same choice**: the median fold change *among* the genes that clear each threshold. It rises with the threshold, close to a straight line above the dotted identity, which is what makes the headline 1.34× a consequence of the 1.25× cut rather than an independent measurement of yeast biology (section 4.5). Read the two series together: moving the cut right buys larger effects and fewer of them, and the design pays for the first and is paid by the second. (d) Extrapolating past two perturbations. Sameith gives only $k = 1$ and $k = 2$, and any two points are fitted exactly by a line, a power law or a saturating curve, so a fit to them cannot choose a model. Instead each of Kemmeren's 1,484 single-deletion responder *sets* is known, so drawing k at random and taking their union measures directly what k combined perturbations would move if their effects did not interact – with the real overlap structure, which matters because stress and ribosomal genes respond to almost anything. Shading is the interquartile range over 2,000 draws. The saturating form $R(k) = G_{\mathrm{eff}}(1 - (1 - p)^k)$ fits with $G_{\mathrm{eff}} = 5{,}264$ and $p = 0.063$; a purely linear extrapolation is shown for comparison and the two are indistinguishable to $k \approx 8$, diverging only as the ceiling begins to bind. Sameith's observed doubles (diamond) sit 8% above the same null built from Sameith's own singles, which is the size of the epistatic correction at $k = 2$. Generated by `experiments/024-perturb-seq-costing/scripts/plot_effect_size.py`.

makes the screen see less. Because the power coefficient scales as $|\Delta|^{-2}$, adopting the 1.25× pairing multiplies every cells-per-perturbation figure in this document by ∼5.6 against the nominal two-fold. It is the single largest correction in the analysis, it moves in the expensive direction, and a reader who prefers a different definition of responding can read its consequence off the same panel rather than having to redo the work.

More perturbations make a cell noisier, not its effects larger. Sameith's doubles move roughly twice as many genes as its singles at 1.25× (460 against 220) and three times as many at 2× (30 against 10), while the median response among responders is unchanged (0.44 against 0.42). A second deletion recruits additional genes into the response rather than amplifying the existing ones.

One caveat on which way this is likely to be biased. Sameith et al. did not sample gene pairs at random: the double-deletion panel is built from genes with regulatory roles, chosen because they were expected to interact. Its responses should therefore sit *above* what a random pair of genes would produce, and its singles above a random single. Kemmeren's 1,484 deletions are the closer thing to an unbiased draw, and its median response, 0.42, is essentially identical to Sameith's 0.42 – so the bias does not appear to move the median *effect size* much. Where it should be assumed to matter is the responder *counts*, and the doubles in particular: 460 responding genes per double deletion is an upper estimate for a randomly chosen pair, not a typical one. That is the favorable reading for multiplexing: the per-gene effect a screen must resolve does not shrink as k rises, so eq. (6) may treat $|\Delta|$ as fixed in k. The comparison is made *within* Sameith – same lab, platform and noise floor – because comparing Kemmeren's singles against Sameith's doubles would confound perturbation count with study.

> **The design target is 1.34-fold, and it is expensive.** Every cells-per-perturbation number derived under a two-fold assumption understates the requirement by ∼5.6×. What this does *not* do is invalidate the 100-cell convention – the opposite, as section 4.9 shows: at the measured effect size that convention corresponds to an ordinary biological overdispersion rather than an implausible one.

What is still missing. These compendia are bulk microarray, so they measure the *effect size* $|\Delta|$ and say nothing about φ_j, the between-cell variance, which needs single-cell data. Of the three unmeasured inputs to eq. (5), this settles one.

4.6 Multiplexing and Interaction Coverage ✗

The end goal is multiplexed screens, because engineered strains carry stacked perturbations. The arithmetic splits cleanly into a part that helps and a part that does not.

Main effects get k times cheaper, and the saving is a property of the model. Under an additive model with random k-subsets of T genes and N cells total, the cells informing gene g's main effect number Nk/T, independent of how many distinct combinations the library contains. Every cell carrying g contributes, whatever else it carries. So k-plexing divides the cell requirement by k.

Two things about that division are worth stating before it is used, because both are easy to read as free. It holds *because* the other $k-1$ perturbations in each cell are assumed to add a background that averages out over random combinations; if they instead interact, the pooled estimate is not the main effect of g but the main effect of g averaged over whatever backgrounds the library happened to deliver. Section 4.7 measures that in yeast and finds it is not zero: two deletions produce about 60% of the sum of their singles, so the assumption fails in a systematic direction and the $1/k$ saving is optimistic by whatever the buffering is at that k. And the main effect is the wrong estimand for epistasis by construction rather than by accident: pooling over every background is exactly the operation that integrates interactions away, so a design powered for main effects is not under-powered for interactions, it is not measuring them at all. Those need cells carrying the specific pair, which is eq. (3) and is where the rest of this subsection goes.

Named interactions scale far worse. The probability that a specific pair $\{g, h\}$ both appear in a random k-subset is $k(k-1)/\left(T(T-1)\right)$. Requiring c cells on every pair among T targets therefore needs

$$N = c\binom{T}{2}\Big/\binom{k}{2}, \tag{3}$$

pairs needed over pairs delivered per cell. Yao et al. [11] derive the same expression and, importantly, supply the constant: $c = 400$, being "the number of cells needed to learn a single second-order interaction effect with the same power as our conventional screens had to learn first-order effects (∼100 cells/perturbations), assuming that four times as many cells are needed to estimate a second-order interaction effect as a first-order effect." Our implementation of Eq. 3 reproduces their published worked cases to within rounding – 100,000 cells at $k = 3$ powers all pairs among 39.2 targets against their stated ∼39, and 10^6 cells at $k = 8$ gives 374.7 against their 374 – which is the check that the constant and the combinatorics are being applied correctly.

Table 7: What multiplexing buys, and what it does not. **Plex** k = gRNAs delivered per cell, each against a different gene, so k target genes are knocked down in that cell. It is not the six guides per gene the library carries. **Main effects** = the average effect of one gene, pooling every cell that carries a guide against it whatever else it carries. **Pairs** are pairs of *target genes*, not plasmids: a cell informs a pair only if it happens to carry guides against both. Column 2 targets 250-cells-per-gene precision over 6,000 genes; columns 3–4 apply Eq. 3 at Yao's 400 cells per pair; column 5 inverts it – how many target genes can have *all* their pairs powered inside one 300,000-cell run.

Guides per cell (k)	Cells for main effects	Cells for all pairs among 6,000 genes	Cells for all pairs among 200 genes	Genes fully paired in 300k cells
1	1,500,000	–	–	–
2	750,000	7,198,800,000	7,960,000	39
3	500,000	2,399,600,000	2,653,333	68
5	300,000	719,880,000	796,000	123
8	187,500	257,100,000	284,286	205
10	150,000	159,973,333	176,889	260

The design that follows. Genome-wide pairwise epistasis is out of reach at any plex: 2.4×10^9 cells at plex $k = 3$, still 2.6×10^8 at $k = 8$. But a *focused* library inverts the picture, because Eq. 3 is quadratic in the panel size T and quadratic in the guides per cell k. A 200-gene metabolic sub-library at $k = 8$ needs ~284,000 cells for *every* one of its 19,900 pairs – one genome-scale-sized run. That is the single most useful number in this document for the multiplexed screening goal.

So the program splits cleanly: run genome-wide at $k \approx 3$ for main effects, where compressed decoding applies and interactions are assumed to cancel; and run a focused metabolic sub-library at high plex for named epistasis, where they are the signal.

Three costs the cell arithmetic hides. Guide detection compounds: assigning k guides at per-guide detection q succeeds at q^k, and the two published rates bracket a very wide range. At Brandner et al.'s 25% in bacteria, even after enrichment, a 3-plex assigns 1.6% of cells and an 8-plex is arithmetically hopeless; at the >71% Nadal-Ribelles et al. reach in yeast, the same figures are 36% and 6%. This – not sequencing – is what actually caps plex here, and which end of that range applies is the single most valuable thing the pilot can measure (Sec. 3.6). Barcode collisions change character: in a singles screen a collision attenuates a main effect slightly, but in a multiplex screen it *manufactures a gene pair that was never constructed*, and pairs are precisely what is being measured. And Yao et al.'s own screens, despite being designed for this, recovered no significant pairwise interaction even at $q < 0.5$ – a caution that the 400-cell constant is a power calculation, not a demonstration.

4.7 Is the Response Matrix Compressible? ✗

Sec. 3.4 states compressed Perturb-seq's premise – that the number of composite samples needed scales as $(\nu + r) \log n$ rather than n, for a library of n targets, an effect matrix of rank r and ν non-zeros per module – and leaves all three unplannable. Two of them are measurable from compendia already in the mirror, because Kemmeren et al.'s 1,484 single-deletion profiles *are* a perturbation-by-gene effect matrix for yeast: the same object a compressed screen would be recovering, in our organism rather than in THP1 cells.

A note on ν. Yao et al. write this quantity as q. It is renamed here because q is the per-guide detection probability everywhere else in this document (eq. (6), section 3.6, section 6), and carrying two q's would mean warning about the collision at every use. The two are unrelated: an expression compendium measures ν and says nothing whatever about q, since it contains no guides.

The matrix is strongly low rank. Four components carry half the variance of the whole compendium and 177 carry 90%, against 1,484 rows; the participation ratio, which needs no cut-off, is 8.4 (fig. 11a). A deletion screen in yeast is not 6,000 independent experiments, it is a few dozen recurring programs sampled 1,484 times, which is the property compression exists to exploit and it is present.

The sparsity is real per perturbation and absent per component. One deletion moves a median of 269 genes of 6,169 at the 1.25× criterion of section 4.5 – 4% of the transcriptome, genuinely sparse. A singular component is not sparse at all: it takes 2,155 genes to carry 90% of one's loading mass (fig. 11b). That gap is not a defect in the measurement, it is the reason FR-Perturb factorizes by *sparse* PCA rather than by PCA, which section 3.4 reports as a design choice without saying what it buys. Sparsity is in the rows of this matrix, not in its singular basis, and a method has to go looking for it.

Compression pays only above a library size, and the panel is below it. Evaluated at $\nu = 269$ and $r = 177$,

Table 8: Parameters of the compressed-screen bound, measured from the expression compendia. Yao et al. give the sample requirement as $O((\nu + r)\log n)$ without a constant, writing ν as q; it is renamed here because q is the per-guide detection probability throughout (table 9). The fourth row is a *shape* evaluated at unit constant and not a budget; what survives the unknown constant is that the requirement is nearly flat in n while the conventional one is linear. **The last row is the assumption, not a parameter.** Guide-pooling requires effects to combine additively and argues that interactions cancel; in yeast they do not cancel, they buffer, and a double produces about 60% of the sum of its singles. Sameith et al.'s pairs were chosen because they were expected to interact, so 0.62 is a lower bound on additivity rather than an estimate of it (section 4.7). Generated from `experiments/024-perturb-seq-costing/scripts/compression_analysis.py`.

Symbol	Quantity	Value	Status	Basis
n	targets in the library	200–6,000	**a choice**	the panel of section 4.6, or the genome
ν	genes moved per perturbation	269	measured	IQR 169–504 over 1,484 deletions
r	components in the effect matrix	177	measured	4 at 50% of variance, 451 at 95%; no knee
$(\nu{+}r)\log n$	composite samples, unit constant	3,880	derived	at $n = 6{,}000$; $0.65n$, so compression barely pays
slope	observed / additive, double deletions	0.62	measured	median over 72 Sameith doubles

$(\nu+r)\log n$ crosses n near $n \approx 3{,}600$ (fig. 11d). At genome scale it is $0.65n$; for the 200-gene focused panel of section 4.6 it is $11.8n$, so on a panel compression costs an order of magnitude *more* samples than measuring each target directly. The constant is unknown and shifts the crossing, but not the existence of one: the compressed requirement is nearly flat in n and the conventional one is linear, so they must cross, and the only question is where. Plain guide-pooling, which divides samples by m and assumes nothing about sparsity, sits below both across the whole range this document contemplates (fig. 11e).

And the assumption underneath it does not hold in the additive direction. Guide-pooling requires that effects combine additively in log expression space, and Yao et al. do not claim interactions are absent – they claim interactions *cancel*, because each cell draws a random combination and positive and negative average out. That is a claim about a mean, and Sameith et al.'s doubles test it directly. Over the 72 doubles whose two singles were profiled by the same lab on the same platform, the observed response correlates well with the additive prediction (median Pearson $r = 0.87$) but is systematically *smaller*: the median slope of observed on predicted is 0.62 (fig. 11c). Two deletions produce about 60% of the sum of their singles. Interactions in yeast do not cancel here; they buffer.

Compression is a poor fit for the screen this document is designing, and the data says so in three independent ways. The library size that makes it pay ($n \gtrsim 3{,}600$) is above the focused panel where interactions are actually reachable; plain guide-pooling beats it at every n considered; and the additivity its decoding assumes is violated in yeast in a systematic direction, with doubles at 0.62 of additive rather than scattering around 1. What the measurement *does* support is the simpler half of the same idea: the matrix is low rank, so pooling and averaging recovers main effects efficiently, which is what fig. 15f already prices.

Three reasons not to over-read the 0.62. Sameith et al. did not sample gene pairs at random – the panel is built from regulatory genes chosen because they were expected to interact, so its deviation from additivity should sit *above* what a random pair would show, and 0.62 is therefore a lower bound on additivity rather than an estimate of it. Seventy-two pairs is a small sample. And $m = 2$ is the only multiplicity anyone has measured: whether attenuation compounds, saturates or reverses at $m = 10$ is exactly the question section 4.5 could not answer for responder counts either. **Hypothesis (untested):** buffering at $m = 2$ predicts a shortfall at higher m, which would make pooled main effects weaker than the $1/m$ scaling of fig. 15f assumes. The two-guide pilot of section 6.3 is what would settle it, and this is a second reason to run it.

4.8 Library Construction Limits ✗

Multiplexing is limited by cloning and transformation before it is limited by sequencing, and the binding number comes from our own library's paper.

Lian et al. [2] put yeast transformation at 10^6–10^7 independent clones and set their own construction standard at $> 10^6$ clones with *at least 30-fold redundancy*. Those two numbers decide the question. A dual-guide library over 6,000 genes has $\binom{6000}{2} = 1.8 \times 10^7$ pairs, which at 30-fold redundancy needs 5.4×10^8 independent clones – 54 to 540 times beyond the ceiling. Genome-scale pairwise coverage is not merely expensive to construct; it is unreachable by about two orders of magnitude, and no sequencing budget changes that.

The precedent already ran this experiment. sMAGIC is the combinatorial version of our own library, and it hit exactly this wall: "due to the limitations in transformation efficiency ($\sim 10^6$–10^7 for yeast transformation), only ~0.01% to ~0.1% of the total combinations ($\sim 10^5 \times 10^5 = 10^{10}$) were covered." That is the same arithmetic seen from the other side, and it carries a design lesson: they reached those combinations by *co-transforming* single-guide libraries

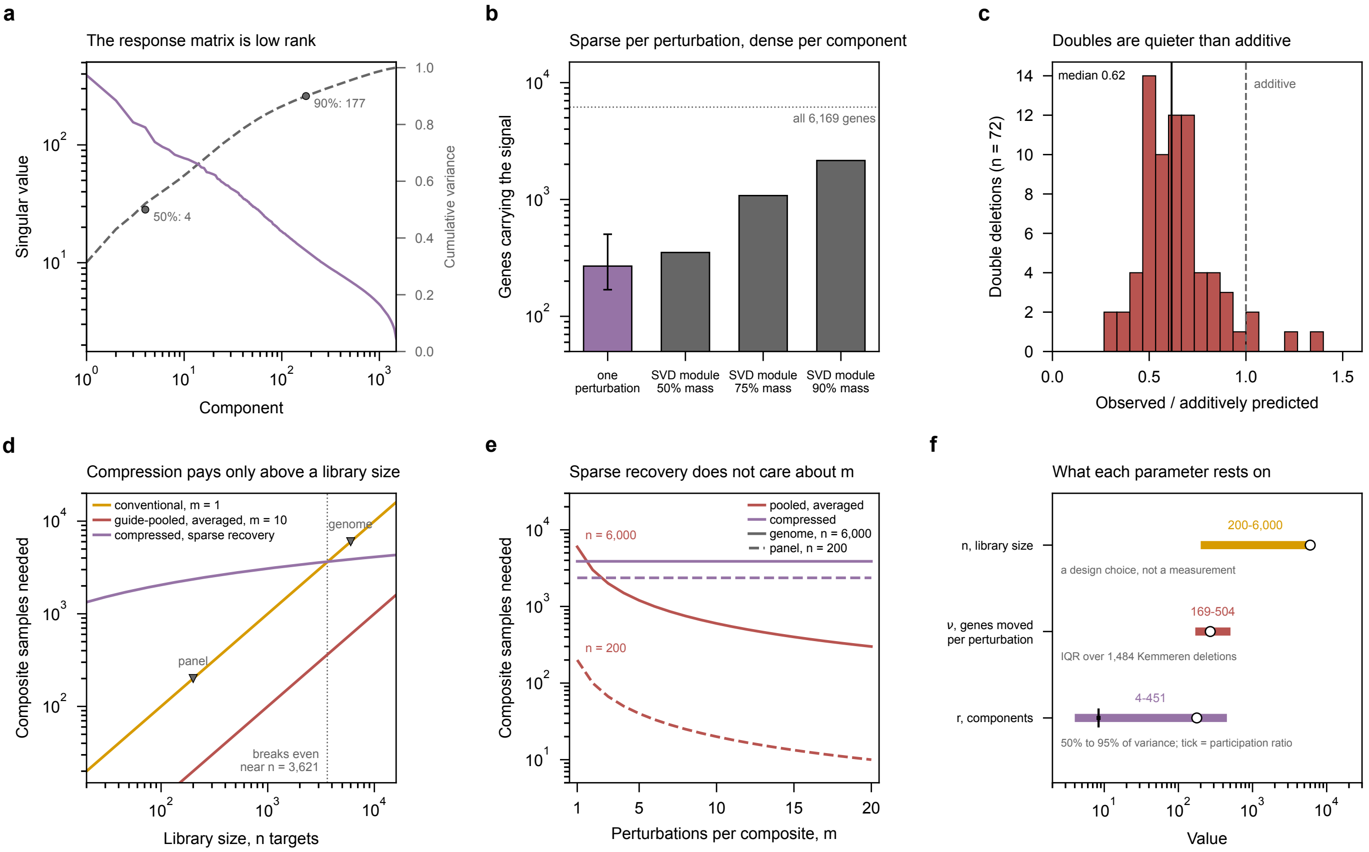

Figure 11: The yeast response matrix is compressible, and compression is still the wrong tool for a focused screen. All panels are Kemmeren et al.'s 1,484 single deletions [24] and Sameith et al.'s doubles [25], the same compendia as fig. 10. Color is the design throughout: amber conventional, red pooled-and-averaged, blue compressed. **(a)** Singular spectrum of the effect matrix (left axis) and cumulative variance (right). Four components carry half the variance and 177 carry 90%; the participation ratio, a cut-off-free reading of the same spectrum, is 8.4. **(b)** Two sparsities, and only the first is the one the bound uses. One perturbation moves 269 genes of 6,169 (bar, with interquartile range); a singular component needs 2,155 to carry 90% of its mass, because an SVD component is dense by construction. The gray bars are an upper bound on what a sparse factorization would need, not a measurement of it. **(c)** The additivity assumption, tested. For each double deletion whose two singles were profiled by the same lab, the slope of observed response on additively predicted response, over genes either vector moves. Additive is 1; the median is 0.62. Restricting to moved genes matters – a correlation over all 6,000 would be dominated by the ~5,700 sitting at zero in both vectors and would report near-perfect additivity for any pair of sparse vectors. **(d)** Composite samples needed against library size, at the measured ν and r. **Every constant here is 1**: Yao et al. state an order, and a compressed-sensing constant depends on the measurement ensemble and the recovery guarantee wanted, so the panel compares shapes – flat in n against linear in n – and the crossing, whose location moves with the constant but whose existence does not. **(e)** The same quantity against perturbations per composite. The compressed curves are horizontal because sparse recovery's requirement does not depend on m at all, only on what is being recovered; that is the whole difference between the two decodings. **(f)** What each parameter rests on. n is a design choice; ν is an interquartile range over 1,484 measured deletions; r has no single value because the spectrum has no knee, so the bar spans the 50% and 95% variance cuts and the tick marks the participation ratio. Generated by `experiments/024-perturb-seq-costing/scripts/plot_compression.py`, from `compression_analysis.py`.

rather than by cloning pairs. Co-transformation converts a construction problem into a Poisson sampling problem – diversity becomes free, but which pairs you get is random, so coverage of a *named* pair is governed by eq. (3) rather than by the library design. Assembly is not the obstacle either way: Golden-Gate efficiency is "nearly 100%".

Restricting the panel is what makes it constructible. The same arithmetic that rules out the genome-scale library admits the focused one. A 200-gene metabolic sub-library has $\binom{200}{2} = 19{,}900$ pairs, needing 597,000 clones at 30-fold redundancy – comfortably inside a single transformation. This is the same conclusion section 4.6 reaches from cell counts, arrived at independently from cloning: panel size, not plex, is the variable that decides whether named epistasis is reachable, and it binds in both the wet lab and the budget.

Figure 12 sweeps both variables rather than evaluating the two points above, and the crossing is sharper than the genome-scale conclusion suggests. Cloning pairs caps the panel at 258 genes at 10^6 clones and 816 at 10^7; cloning triples caps it at 59 and 126. So the reachable panel is not somewhat smaller than the genome, it is two orders of magnitude smaller, and it shrinks by roughly another order for each additional guide cloned onto the plasmid.

Which is the argument for co-transformation, stated as a picture. Panel (b) is the one worth looking at twice. Co-transforming single-guide libraries needs $30T$ clones *whatever* k *is*, because the clone count scales with the number of guides rather than with the number of combinations. That is 180,000 clones to cover all 6,000 genes, inside one transformation, at any plex. The combinations then come from Poisson uptake (section 7.1) instead of from cloning. Nothing about this makes a *named* pair cheaper to observe – eq. (3) still governs that, and section 7.2 shows what it costs – but it removes construction from the list of reasons multiplexing is hard, and leaves only the measurement.

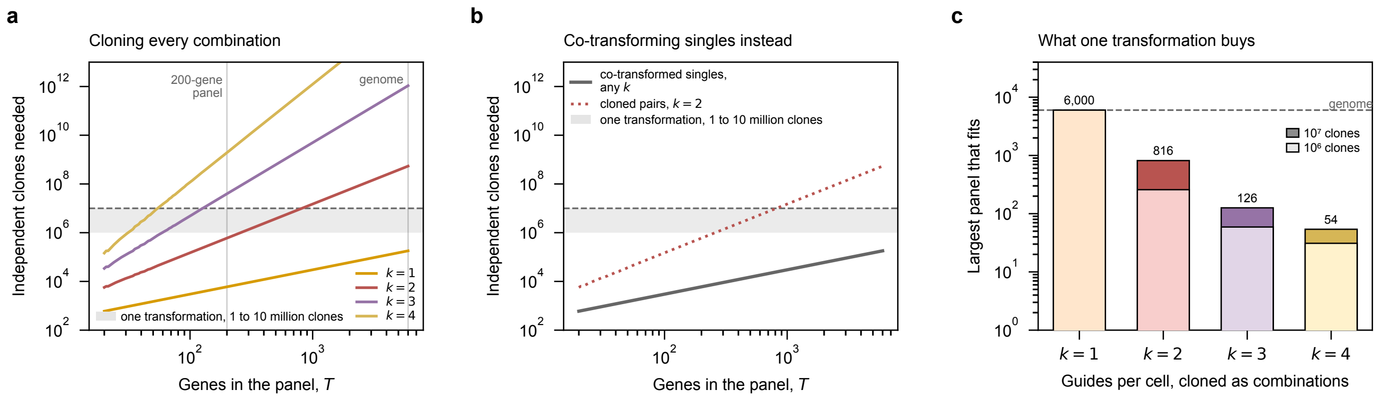

Figure 12: Cloning combinations caps the panel two orders of magnitude below the genome; co-transforming singles does not. **(a)** Independent clones needed to cover every k-combination of a T-gene panel at the 30-fold redundancy our own library's standard demands [2]. The keyed gray band in (a) and (b) is what one yeast transformation delivers, 10^6 to 10^7 clones, and the dashed rule is its optimistic edge. Each additional cloned guide costs roughly another order of magnitude of panel size. **(b)** The same question for co-transformed single-guide libraries, where one clone is needed per *guide* rather than per combination, so the requirement is $30T$ and does not depend on k at all: 180,000 clones cover all 6,000 genes, inside one transformation, at any plex. Combinations then arise from Poisson uptake (section 7.1) rather than from cloning. **(c)** The direct answer: the largest panel whose k-combinations fit in one transformation. Each bar runs to the optimistic 10^7 ceiling and its pale lower part to the conservative 10^6 one. Pairs reach 816 genes at best and triples 126, so the 200-gene panel of section 4.6 is constructible at $k = 2$ and not at $k = 3$ unless the guides are co-transformed. Generated by `experiments/024-perturb-seq-costing/scripts/plot_library_ceiling.py`.

> **Genome-scale pairwise coverage fails at the cloning bench, not in the budget.** A dual-guide library over 6,000 genes needs 5.4×10^8 independent clones at the 30-fold redundancy our own library's standard demands, which is 54 to 540 times beyond what yeast transformation delivers. A 200-gene panel needs 597,000 and sits comfortably inside a single transformation. Panel size, not sequencing money, is what makes named epistasis constructible.

4.9 A Unified Design Equation ✗

The preceding subsections give three rules that look independent: a shot-noise requirement (eq. (2)), a 100-cell biological floor, and a combinatorial penalty for multiplexing. Sections 4.4 and 4.6 combine the first two by taking whichever is larger. They are not independent, and the max is an artefact of having modeled the count as Poisson.

This is not a new result. It is the standard negative-binomial sample-size calculation from the bulk RNA-seq literature, applied to a pseudobulk pool. Equation (5) appears in essentially this form in Hart et al. [26]: a sum of a sampling term and a dispersion term over the squared log fold change. The parameter φ_j is the negative-binomial dispersion of McCarthy et al. [27], whose biological coefficient of variation is $\sqrt{\varphi_j}$ – so published BCV values transfer directly as priors for the one input we cannot measure – and the variance parameterization itself is that of Robinson and Smyth [28]. The decomposition into Poisson sampling on top of biological variation is the single-cell UMI noise model of Grün et al. [29], and using a plain negative binomial rather than a zero-inflated one follows Svensson [30]. Treating the estimand as a pseudobulk profile rather than a per-cell test follows Squair et al. [31]; the cells-versus-depth optimum that fig. 13a traces under a fixed budget is the subject of Zhang et al. [32]; and the largest published Perturb-seq screen, whose empirical cells-per-perturbation choice section 4.4 is measured against, is Replogle et al. [33]. What is specific to this document is the multiplexing term of eq. (6) and the reading of the 100-cell convention below.

Setup. Pool n cells carrying one perturbation, each read to depth d mRNA UMIs, and consider gene j with transcriptome share p_j. The pseudobulk count is $K = ndp_j$. Treating it as negative binomial, with φ_j the gene's biological overdispersion between cells – the variance in excess of sampling –

$$\mathrm{Var}(\log K) \;\approx\; \underbrace{\frac{1}{ndp_j}}_{\text{shot noise}} \;+\; \underbrace{\frac{\varphi_j}{n}}_{\text{biology}}\;. \tag{4}$$

Claim. Requiring a $\log_2$ fold change Δ to be resolvable against controls at level α and power $1-\beta$ gives one requirement,

$$n^* \;=\; A(\Delta)\left[\frac{1}{d\,p_j} + \varphi_j\right], \qquad A(\Delta) = \frac{\kappa\,(z_{1-\alpha/2} + z_{1-\beta})^2}{(\Delta \ln 2)^2}, \tag{5}$$

where $\kappa = 1$ when the non-targeting control pool is large enough that its own variance is negligible – the usual case, since controls are pooled over many guides – and $\kappa = 2$ for a control of the same size as the perturbation. and with k guides per cell over a panel of T targets the total cells for an order-r effect are

$$N^* \;=\; \rho_r\, n^*\, \frac{\binom{T}{r}}{\binom{k}{r}} \cdot \frac{1}{s\,q^r}, \tag{6}$$

with ρ_r the contrast variance inflation, s the QC survival fraction and q the per-guide detection probability. ρ_1=1 by definition. ρ_2=4 is *quoted* from Yao et al. rather than derived: it is the variance of a full 2×2 factorial contrast over four equally sized groups. Under the same large-control assumption used for main effects the exact factor is 3, so Yao's 4 is conservative by about a third; it is kept because the 400-cells-per-pair constant of section 4.6 is built on it and the two must not disagree.

Consequence. Equation (5) contains both earlier rules as limits. Setting $\varphi_j \to 0$ leaves $n^* = A/(d\,p_j)$, that is a required pseudobulk count $K = n^* d\,p_j = A$; at $\kappa = 1$ this is *identically* eq. (2), so section 4.4 is the shot-noise corner of this equation rather than a separate rule. Letting $d\,p_j \to \infty$ gives $n^* \to A\varphi_j$, a floor no amount of sequencing can cross – which *is* the biological floor, now with a number attached. At 80% power $A = 16.3$ at two-fold and $A = 90.9$ at the measured 1.34-fold, so a 100-cell floor is precisely the claim $\varphi_j \approx 6.1$ or $\varphi_j \approx 1.1$ respectively.

That number depends on Δ, and this is where measuring the effect size pays off. At the nominal two-fold the 100-cell floor implies $\varphi_j \approx 6.1$, a between-cell coefficient of variation of 2.5 – bursty even by single-cell standards, and grounds for suspecting the convention was really guarding against something other than sampling variance. At the *measured* 1.34-fold (section 4.5) the same floor implies $\varphi_j \approx 1.1$, a coefficient of variation of about one, which is an entirely ordinary single-cell overdispersion. The tension dissolves: the 100-cell convention is consistent with unremarkable yeast biology once it is evaluated at the effect size screens are actually built to detect. The convention was right; the two-fold assumption was wrong.

Table 9: Inputs to the design equation. Two of the seven are assumed rather than measured, and those two are what separate a structure from a predictor. **Status**: *measured* = from data we hold; *quoted* = stated by a source; *derived* = computed from measured values; **assumed** = a placeholder, and every number downstream of it inherits that status. One qualification on d: the 410–2,000 range is measured per platform, but the preindexed droplet of section 3.3 has no published per-cell depth in any microbe and is held at the 10x value throughout, so for that platform alone d is **assumed**.

Symbol	Quantity	Value	Status	Source
$A(\Delta)$	power coefficient, Eq. (5)	90.9	derived	at the measured Δ; scales as Δ^{-2}
Δ	$\log_2$ fold change to resolve	0.42	measured	median responder at a 1.25× cut, table 6
p_j	transcriptome share of target gene	$1.2{\times}10^{-4}$	derived	3.5 molecules / 30,000 mRNA, table 3
φ_j	biological overdispersion between cells	0.5–10	**assumed**	swept; sets the floor. Not yet measured
d	sequencing depth, mRNA UMIs per cell	410–2,000	measured	per platform, table 4
ρ_2	variance inflation, second order	4	quoted	Yao et al.; the 400-cells-per-pair constant
q	per-guide detection probability	–	**assumed**	needs the construct of section 3.6

Interpretation – what would make this predictive. As a structure eqs. (4) to (6) are complete, and every design question in section 5 is a substitution into them. As a *predictor* they are not usable yet, because two of their inputs

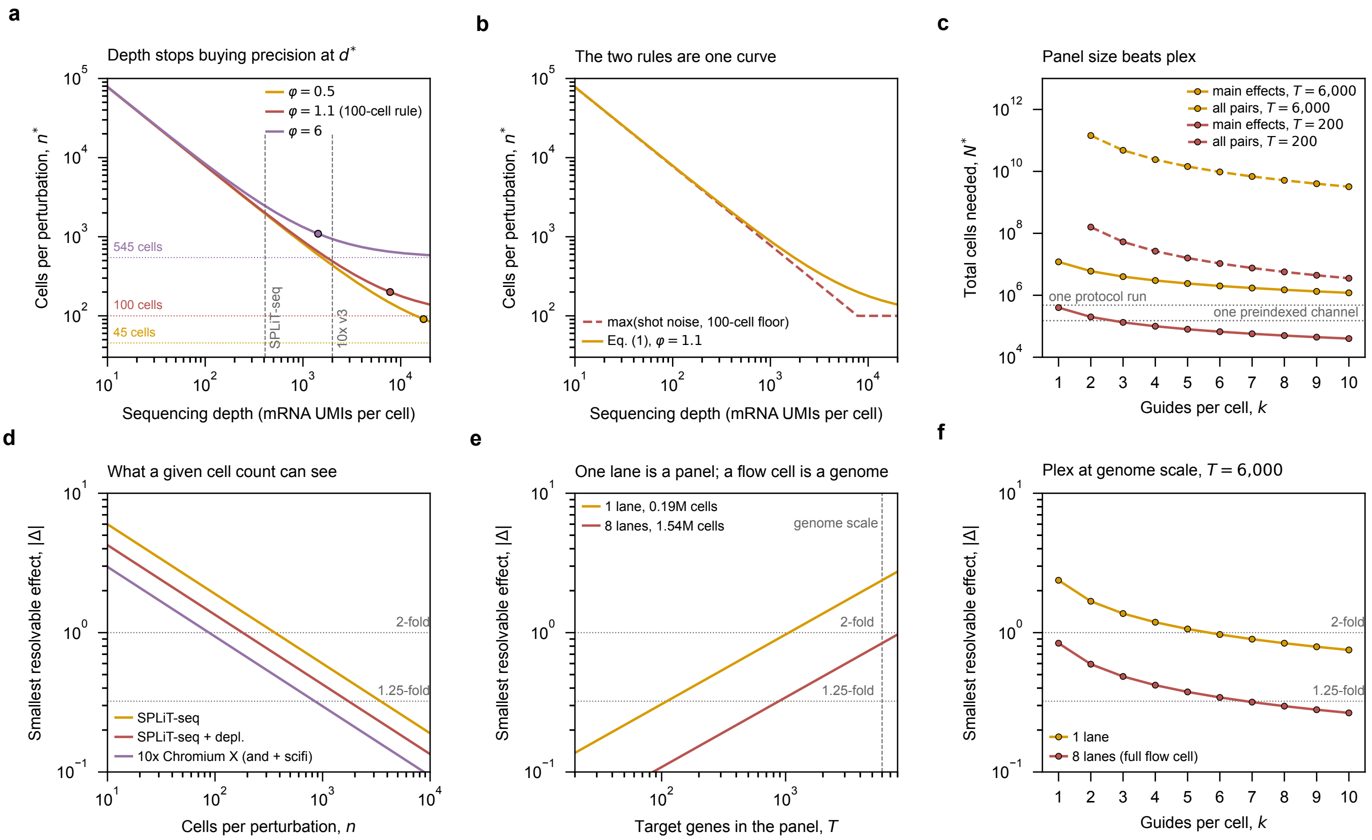

Figure 13: One equation reproduces both design rules, and turns a lane of sequencing into a statement about the smallest effect a screen can see. Top row, the structure of eq. (5). (a) Cells per perturbation against depth, swept over biological overdispersion φ. Every φ implies a cell floor, $n^* \to A\varphi$ as depth grows without bound, and those are the dotted horizontals – each labeled with the cell count it corresponds to, so the swept parameter can be read directly as the quantity a screen is designed in. The marked point on each curve is the depth-sufficiency threshold $d^* = 1/(\varphi p_j)$, where shot noise and biology contribute equally. Past d^* the requirement is within a factor of two of its floor, so further depth is close to wasted. Curves are evaluated at the *measured* effect size of section 4.5, $|\Delta| = 0.42$, not at a nominal two-fold; the middle curve is the $\varphi = 1.1$ that the field's 100-cell rule implies at that effect size, and its floor is accordingly 100 cells. At $\varphi = 1.1$, $d^* \approx 7{,}800$ UMIs per cell – roughly four times deeper than the best platform on offer – so *both* candidate chemistries are firmly shot-noise limited, and depth still buys precision for both. That is the argument for rRNA depletion in one number: it is the cheapest available way to move right along this axis. (b) The same curve against the piecewise rule the earlier subsections use. They agree exactly in both limits and differ near the knee, where the piecewise form *under*-estimates the requirement – the regime both candidate platforms occupy. (c) eq. (6) against plex, colored by panel size and dashed for pairwise. Raising k moves every curve down by a similar factor; shrinking the panel from 6,000 targets to 200 moves the pairwise curve by $\sim$900×. Restricting the panel, not multiplexing harder, is what brings interactions into reach. The two dotted horizontals are the batch units the two platform families come in – one split-pool protocol run at $\sim$480,000 cells, one preindexed Chromium channel at 151,788 – so a curve above them is a feasibility question rather than a budget one. They are a factor of three apart, not the order of magnitude the un-preindexed 20,000-cell channel would put there. Bottom row, eq. (5) inverted – the smallest resolvable $|\Delta|$ – for the two sequencing configurations the core actually sells: one NovaSeq X 25B lane (3.2 billion read pairs, \$3,380) and a full eight-lane flow cell (25.6 billion, \$25,440 at the volume rate). (d) Resolution against cells per perturbation, by platform. **There is no fourth curve for the preindexed droplet, and its absence is the finding rather than an omission**: preindexing changes cells per priced channel and not UMIs per cell, so it lies exactly on the 10x curve here and on every other statistical panel of this figure. What it changes is the cost of reaching a given n, which is fig. 15. (e) Against panel size, at split-pool with rRNA depletion. This is the panel to design from: one lane yields $\sim$192,000 usable cells and resolves two-fold across a few hundred targets, whereas a full flow cell yields $\sim$1.5 million and reaches two-fold across all 6,000 genes. One lane is a focused panel; a flow cell is a genome. (f) Against plex at genome scale – multiplexing to $k \approx 8$ buys a single lane roughly what eight lanes deliver unmultiplexed, which is the cheapest lever in the document. Note that the volume discount itself is minor (\$0.99 against \$1.06 per million read pairs, $\sim$6%); the reason to fill a flow cell is what it buys, not what it saves. Generated by `experiments/024-perturb-seq-costing/scripts/design_equation.py`.

are still assumed rather than measured. Three were, and the list below is kept at three so the one that has since been pinned can be read against the two that have not:

1. **φ_j, gene-level biological overdispersion in yeast.** The critical one: it alone sets the floor, and the whole 100-cell convention rests on it. Estimable from any existing yeast scRNA-seq count matrix, and it is the reason the figure sweeps φ instead of fixing it.
2. **The distribution of Δ – now measured.** Median $|\Delta| = 0.42$ among responding genes, a 1.34× change (section 4.5, table 6). $A(\Delta)$ goes as Δ^{-2}, so this was the most violent term in the equation; pinning it multiplied every cell requirement by ∼5.6 and resolved the φ puzzle above. Table 9 now carries it as measured, and $A(\Delta)$ with it as derived rather than assumed.
3. **q, per-guide detection probability.** Enters as q^r, so it is forgiving for main effects and punishing for interactions. It cannot be measured until the guide-barcode construct of section 3.6 exists.

Of the three, one is now measured and two remain. The curves here therefore rest on one measured parameter and one swept one, which is why φ is drawn as a family rather than a line, and why no number from this subsection is yet used in the budgets of section 5. Those budgets are built on the 100, 250 and 500 cells-per-gene tiers, which come from the biological floor rather than from a fold-change target, so they are NOT understated by the 5.6× that section 4.5 applies to a depth-derived requirement. The 5.6× bites only where a number was derived from the two-fold assumption, and section 4.9 shows the convention itself survives the measured effect size intact.

Two limitations of the derivation itself, for completeness. The delta-method step behind eq. (4) degrades when counts are small, and $A \approx 16$ counts is small – near the shot-noise limit eq. (5) should be read as an order-of-magnitude statement, with an exact negative-binomial test the right tool in that corner. And eq. (5) assumes equal per-cell depth; real libraries vary and are handled with size factors, so d should be read as a mean depth, correct to first order. Neither affects the structural conclusions, both of which concern the φ term. The published treatment of the cells-versus-depth optimum that panel (a) rediscovers is Zhang et al. [32], and the case for pseudobulk as the estimand is Squair et al. [31].

> **One equation contains both design rules, and they were never independent.** Setting biological overdispersion to zero recovers the shot-noise requirement exactly; letting depth run to infinity recovers the biological floor. The apparent tension in the 100-cell convention dissolves once the effect size is measured rather than assumed: at the measured 1.34-fold it corresponds to an entirely ordinary yeast overdispersion, not an implausibly bursty one. The convention was right and the two-fold assumption was wrong.

5 Application: Genome-Scale Screen Design at UIUC ✗

Applying Sec. 4 to our actual instruments and rates. Design target: 6,000 genes, six guides per gene in the MAGIC style, one environment.

5.1 Start-Up Cost Composition ✗

Brettner et al. quote a start-up cost of ∼\$10,000 and a per-protocol cost of \$1,300–1,700. The composition matters more than the totals, because the two scale completely differently.

Table 10: Brettner 2024 SPLiT-seq line items (`paper.md:183`). The start-up cost is one purchase order – three plates of custom IDT oligos, good for 215 runs.

Item	Cost	Scaling
IDT barcode plates (RT r1 + ligation r2 + r3)	\$7,699.40	**one-time, 215 runs**
Standalone oligos (lyophilized)	\$270.60	one time
Maxima H Minus Reverse Transcriptase (ThermoFisher EP0753)	\$840.00	**per run**
T4 DNA Ligase (NEB M0202M)	\$270.00	**per run**
RNase inhibitors (SUPERase-In + Enzymatics)	\$127.00	**per run**
Barcoding oligos drawn from the plates	\$36.00	**per run**
Dynabeads MyOne Streptavidin C1	\$14.00	per sublibrary
WGS fragmentation + ligation library prep (Enzymatics)	\$39.00	per sublibrary

The start-up cost is a single purchase order: three 96-well plates of custom barcoded oligos from IDT, \$7,699.40, which is 77% of it.

What "215 uses" actually means. The plates are *reagent stocks*, not devices that get reused. IDT ships them pre-diluted – 300 µL of 100 µM oligo in each of 96 wells – and a single pass through the protocol withdraws roughly one microlitre from every well. Nothing is recovered afterwards and no barcode is ever read twice; the plate simply has enough volume in it for about 215 withdrawals. Brettner et al. state the capacity per plate directly: "enough volume to complete 300 RT, 250 r2, and 215 r3 ligations with careful pipetting." The round-3 plate is the binding constraint at 215, so 215 is the honest number rather than the 300 the RT plate would allow.

Two practical points follow from the pre-diluted format, and they answer the obvious worry about committing a whole plate at once. Because the oligos arrive already in solution, there is no rehydration event – you are drawing from a reservoir, not reconstituting a plate that must then be consumed. What does degrade them is freeze–thaw, which is why the working practice is to aliquot each source plate into several *working plates* and cycle only those. Gaisser et al.'s troubleshooting guide treats "remake fresh working plates" as the standard response to a barcoding failure, which tells you the source plates are expected to outlive many working copies.

And what counts as one "run". One run is one complete pass through the barcoding workflow of Sec. 3.1: fix and permeabilize a batch of cells, distribute them across the round-1 plate, two ligation rounds with pooling and re-splitting between, then split into sublibraries and prepare libraries. It consumes one withdrawal from each barcode plate, one tube of reverse transcriptase, one tube of ligase, and about a fifth of the RNase inhibitors – and it can carry up to 96 samples and, on a full round-1 plate at Brettner et al.'s stated density, ∼480,000 cells. Note that this is the plate's capacity, not what their published run did: they loaded 48 of the 96 wells, so the run described in section 3.1 put ∼240,000 cells through the chemistry. Every budget here assumes the full plate. It is a unit of *bench work*, not a unit of sequencing: the sublibraries a run produces are sequenced separately and are priced separately.

So the plate set is a capital asset. At a realistic five to ten runs per year, 215 withdrawals is a twenty-year supply, and a plate on the bench for twenty years will not survive that long. Two consequences: we are over-buying by roughly an order of magnitude and should ask IDT for a smaller synthesis scale or volume, and one plate set can serve an entire department. The genuinely recurring items are the reverse transcriptase ($840) and the ligase ($270), repurchased every run. Whether the degradation actually costs capacity is the next question, and the answer turns out to be no.

What one plate set buys, and whether freeze–thaw invalidates the amortization. The start-up cost is amortized over those 215 runs, and the objection is fair: if a plate degrades before it is drawn down, the amortization is fiction and the real per-run cost is higher than quoted. The two failure modes are separable, and separating them shows the objection is answerable by aliquoting rather than by a shorter plate life.

Depletion is a volume budget and is a hard count. At a full round-1 plate the 215 runs barcode **103 million cells**, and the depleted split-pool row of table 16 needs 13 runs for a genome-scale screen at 250 cells per target gene, so **one plate set is about 16 genome-scale screens** – or the same screening capacity spread over environments, which is where section 7.3 argues it should go.

Freeze–thaw is a cycle budget, and it applies to whichever plate is being pipetted from rather than to the oligo stock as such. Split a source plate into w working plates in a single thaw and the source sees *one* cycle whatever follows, while each working plate serves $215/w$ runs and therefore sees $215/w$ cycles. Keeping every plate under a tolerance of f cycles needs $\lceil 215/f \rceil$ working plates: 22 at $f = 10$, 9 at $f = 25$, 3 at $f = 100$. Empty plates and one thaw are close to free against $7,699, so the freeze–thaw limit converts into a plate-count decision and not into lost capacity. Two qualifications keep this honest. The tolerance f is not measured anywhere in the mirror and the values swept above are illustrative, so what survives is the shape of the answer rather than the plate count. And the working-plate practice quoted above is itself evidence that f is small: a protocol whose standard response to a barcoding failure is to remake the working plates is not one that expects a plate to survive 215 cycles.

Independently replicated. Gaisser et al. [7] costed the same chemistry for bacteria and put the identical item at $7,844.52 against Brettner et al.'s $7,699.40 – two labs, two organisms, two years apart, 1.9% apart. Their barcode cost per run is flat regardless of sample count, which is the split-pool economics in one number.

> **The split-pool start-up cost is a capital asset, and we are over-buying it by an order of magnitude.** 77% of the $10,000 is three plates of custom barcoded oligo, and their capacity is 215 protocol runs, which is 103 million cells or about 16 genome-scale screens. At five to ten runs a year that is a twenty-year supply, so ask IDT for a smaller synthesis scale and expect one plate set to serve a department. Freeze–thaw does not eat into that capacity, it just decides how many working plates the aliquoting has to make, since w working plates each see $215/w$ cycles. The genuinely recurring items are the reverse transcriptase and the ligase, about $1,100 per run.

5.2 Loaded Cost per Usable Cell ✗

The figure the literature quotes is library preparation only.

Table 11: Cost per cell, *library preparation only.* "Quoted" means the source states a per-cell rate; "derived" means a total divided by a cell count the same source supplies. Every row excludes sequencing.

Method	Organism	Isolation	$ / cell	Basis	Source
MATQ-seq	*bacteria*	plate	$50.00	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2
Plate FACS + STRT-seq (Nadal-Ribelles 2019)	*S. cerevisiae*	plate	$4.15	quoted	Jariani 2020, eLife 9:e55320
10x Chromium 3' v2 (Jariani 2020)	*S. cerevisiae*	droplet	$0.3000	quoted	Jariani 2020, eLife 9:e55320
ProBac-seq	*bacteria*	droplet	$0.1649	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2
PETRI-seq	*bacteria*	split-pool	$0.0367	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2
microSPLiT (1 sublibrary)	*bacteria*	split-pool	$0.0172	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2
SPLiT-seq (Brettner 2024)	*S. cerevisiae*	split-pool	$0.00500	derived	Brettner 2024, Yeast 41:242
microSPLiT (full scale)	*bacteria*	split-pool	$0.00251	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2

That spread – about four orders of magnitude – is real but misleading, because every row excludes sequencing and every row counts cells *processed* rather than cells that survive QC and carry an identified guide. Costing the same platforms end to end:

Table 12: Cost per cell end to end: 6,000 genes at 250 cells/gene, per cell that survives QC *and* carries an identified guide. ∗ projected rather than measured: the scifi row changes one parameter of the 10x row, cells recovered per channel, and is unmeasured in yeast (section 3.3).

Platform	Reagents $/cell proc.	Loaded $/usable cell	Hidden mult.	Seq. share
SPLiT-seq, as published	$0.0049	**$0.1044**	21.3×	81.2%
SPLiT-seq + rRNA depl.	$0.0053	**$0.0383**	7.2×	44.2%
10x Chromium X	$0.1076	**$0.2424**	2.3×	19.2%
10x + scifi preindex.∗	$0.0141	**$0.0724**	5.1×	64.5%

The headline inverts, and then the ordering does. On reagents alone, split-pool beats droplet by ∼22×. End to end it is 2.3×: the gap closes because 94% of split-pool reads are ribosomal and ∼75% of its sequenced cells are thrown away. rRNA depletion reopens it to 6.3×, which is what makes depletion the highest-leverage change available to the split-pool route, worth 2.7× on cost per usable cell. The droplet column has a lever that is slightly larger. Preindexing puts 7.6 times as many cells through a priced channel and is worth 3.3×, from $0.2424 to $0.0724, which undercuts split-pool *as published* ($0.1044) and still leaves depleted split-pool cheapest at $0.0383. So each route has one change worth about a factor of three, the two are independent, and neither has been measured in yeast.

5.3 UIUC Core Sequencing Capacity ✗

Rates from the UIUC Roy J. Carver Biotechnology Center DNA Services, effective 2026-08-01, retrieved 2026-08-18. Listed rates are the on-campus rate; external users pay +30.8%.

Table 13: UIUC Carver Biotechnology Center sequencing rates, effective 2026-08-01, on-campus. External users pay +30.8%. The core has no NextSeq and no NovaSeq 6000.

Option	Instrument	$ / lane	Read pairs / lane	$ per M pairs
Nova X 25B PE150, 8+ lanes	NovaSeq X Plus	$3,180	3,200M	$0.99
Nova X 25B PE150, per lane	NovaSeq X Plus	$3,380	3,200M	$1.06
Nova X 10B PE150, 8+ lanes	NovaSeq X Plus	$1,710	1,200M	$1.43
Nova X 10B PE150, per lane	NovaSeq X Plus	$2,000	1,200M	$1.67
Nova X 1.5B SR100, per lane	NovaSeq X Plus	$1,360	800M	$1.70
Nova X 10B PE150, SPLIT LANE	NovaSeq X Plus	$1,160	600M	$1.93
Nova X 1.5B PE150, per lane	NovaSeq X Plus	$1,580	800M	$1.98
i100 25M SR100nt	MiSeq i100	$986	25M	$39.44
i100 25M 2x150nt	MiSeq i100	$1,450	25M	$58.00
i100 titration run (26nt + indexes only)	MiSeq i100	$398	5M	$79.60
i100 5M 2x150nt	MiSeq i100	$661	5M	$132.20

The available instruments. The core runs the NovaSeq X Plus and the MiSeq i100, plus PacBio Revio and Nanopore GridION. It has no NextSeq and no NovaSeq 6000. The microSPLiT protocol is written against a NextSeq 2000 P2, so that configuration is not available here; the choice is a 25 to 50 M-read MiSeq run or a ≥800 M-read NovaSeq X lane, with a 30-fold gap between them (fig. 14a). The NovaSeq X 25B is the highest-throughput configuration on campus at 3.2 billion read pairs per lane and 25.6 billion per flow cell, and at the 8+ lane rate it is also the cheapest per read, which is why every sequencing figure in this document is priced on it.

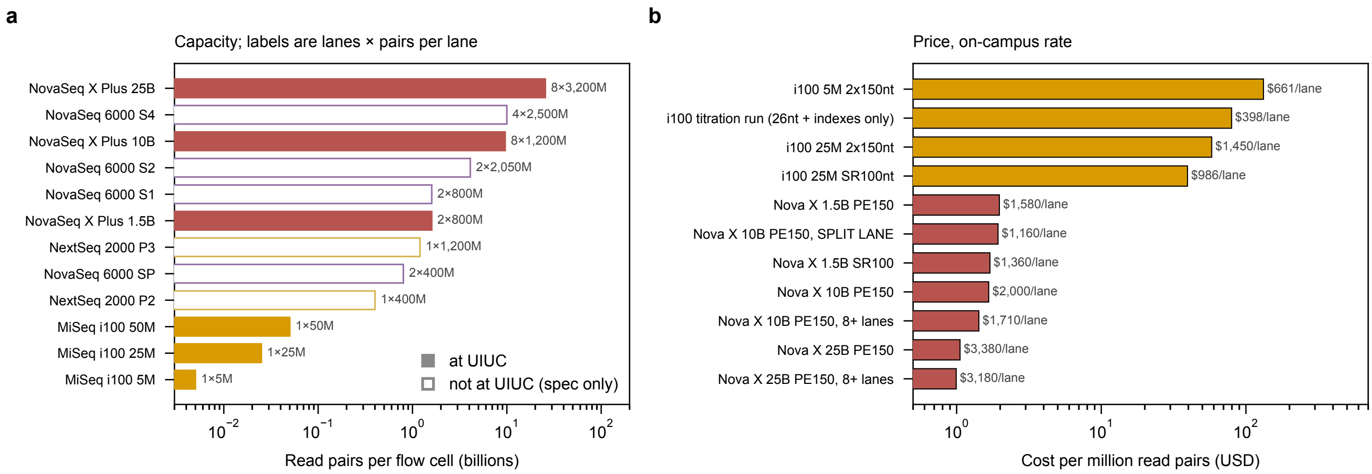

Figure 14: The instruments on campus leave a 30-fold hole exactly where a pilot screen wants to sit, and the largest flow cell is also the cheapest per read. (a) Output per flow cell, with lanes and reads-per-lane drawn separately, since a platform gets large by either route and the two are not interchangeable for a screen: lanes partition a library, reads-per-lane deepen it. Filled bars are available at the UIUC Roy J. Carver Biotechnology Center; open bars are manufacturer specifications for instruments the core does not have, shown for comparison and therefore unpriced. The gap between the MiSeq i100 at 50 M read pairs and the NovaSeq X 1.5B at 1.6 billion is the practical constraint – a pilot sized at a few hundred million reads has to either waste most of a NovaSeq lane or be redesigned down to a MiSeq, and the microSPLiT protocol's own NextSeq 2000 P2 configuration falls squarely in the hole. (b) Price per million read pairs, on-campus rate, for the configurations the core sells. Cost per read falls monotonically with flow-cell size, so the NovaSeq X 25B at 8+ lanes is simultaneously the largest and the cheapest option ($0.99 per million read pairs); there is no economy-of-scale penalty to buying the biggest flow cell, which is why every sequencing figure in this document is priced on it. A consequence worth noting for submission: the 25B is paired-end only, so the paired-end reads both method families require (Table 14) cost nothing extra here – a conclusion that would reverse at a core whose cheapest configuration is single-read. Generated by `experiments/024-perturb-seq-costing/scripts/plot_sequencers.py`.

Both methods need paired-end reads, and it costs nothing. Neither family can use a single-read flow cell – with one read you get barcodes and no transcript, or transcript and no cell identity. They put the barcode on *opposite* reads, which is the thing to get right at submission:

Table 14: Read configurations. Both families require paired-end runs but put the cell barcode on *opposite* reads. For SPLiT-seq the index is barcode round 4, not merely a sample tag, so unique dual indices are mandatory.

Method	Read 1	Read 2	Index	Cycles
SPLiT-seq	74 nt cDNA	86 nt UMI+BC3+BC2+BC1	6 or dual 8 nt (= BC round 4)	160
10x Chromium 3'	28 nt barcode + UMI	90 nt cDNA	dual 10 nt	118
Drop-seq	20 nt barcode + UMI	100 nt cDNA	single 8 nt	120

Paired-end is free here because the largest flow cell is paired-end only and still wins on price per read: 25B PE150 at 8+ lanes is $0.99 per million read pairs, against $1.70 for the only single-read configuration the core sells, the 1.5B SR100. Choosing paired-end is not a compromise here, it is the cheaper option. That would not hold at a core whose cheapest configuration were single-read.

Split-pool also spends cycles on nothing, and it is measured. Two thirds of the cycles SPLiT-seq uses to read its composite cell barcode are **uninformative**: they read ligation overhangs, primer binding sites and Tn5 mosaic ends rather than barcode. Datlinger et al. measured this across four protocols and it is a property of the read structure rather than of any one run – 67% for SPLiT-seq, 42% for sci-RNA-seq v1, 13% for sci-RNA-seq v3 and sci-Plex, and none for a two-round design where the barcode is placed once by the RT primer and once by ligation [6].

The figure reconciles against table 14 to within a point, which is worth checking because it is the only place this document can audit an outside claim against its own numbers. Read 2 is 86 nt of UMI plus BC3, BC2 and BC1. Take off the 10 nt UMI and 76 cycles remain for the composite barcode, of which three 8 nt barcodes are 24; the other 52 are linker, and $52/76 = 68\%$.

The size of the charge then depends on the kit, and here it is small, which is worth saying because 67% invites a larger conclusion than it supports. The waste falls on barcode cycles, not on the whole read, and split-pool uses 160 of the 300 cycles a 2×150 kit delivers. The 52 wasted cycles come out of 140 spare ones, so recovering them buys transcript length rather than money. They become a cost only on a cycle-limited kit, which is the NovaSeq X 100-cycle configuration and not the PE150 everything here is priced on. So the 67% is a real inefficiency of the chemistry that this particular rate card absorbs, and it is an order of magnitude smaller than the 94% rRNA fraction of Sec. 3.5, which the same rate card does not absorb at all.

Three split-pool-specific hazards. PhiX at ≥10% is a protocol requirement, not hygiene – read 2 carries fixed linker sequence at defined cycles, so base diversity collapses and patterned flow cells mis-register clusters; combined with 70–90% valid-barcode rates only ∼72% of purchased reads become usable data, which is in the model. Unique dual indices are mandatory, because the sublibrary index *is* barcode round 4, so an index hop moves a cell into another sublibrary's namespace and manufactures a barcode collision rather than mere sample bleed. And a 2×150 kit gives 300 cycles when only 160 are needed, so read 1 should be set to 150 nt – free, and it helps with the whole-genome-duplication paralogs that forced Brettner et al. to keep multi-mapping reads.

The largest flow cell is also the cheapest per read, and there is nothing between it and a MiSeq. NovaSeq X 25B at 8+ lanes is $0.99 per million read pairs, so there is no economy-of-scale penalty for buying the biggest option, and paired-end costs nothing because the largest flow cell is paired-end only. The problem is the hole underneath it: a pilot sized at a few hundred million reads either wastes most of a NovaSeq lane or gets redesigned down to a MiSeq.

5.4 Barcode Space and Collision Rates ✗

Table 15: Barcode collision rate, $100(1-((B-1)/B)^{C-1})$. For split-pool $B = 96^R \times S$; the sublibrary index S is a real barcode dimension, because cells are split into sublibraries *after* the last in-cell ligation. The final row is the preindexed-droplet scheme of section 3.3, where B is the 384 round1 wells times the 737,280 round2 barcodes the Chromium ATAC reagents supply. **That row is an upper bound on its own performance.** The formula assumes cells are spread uniformly over the space, which is true of plate wells and not of droplets; Datlinger et al. replace the uniform assumption with a zero-inflated Poisson fit for exactly this reason, so read the row as "the space is not the binding constraint" rather than as a predicted collision rate.

Scheme	Sublib.	Barcode space	@45k	@480k	@3M	@12M cells
split-pool, 3 rounds	1	884,736	4.96%	41.87%	96.63%	100.00%
split-pool, 3 rounds	24	21,233,664	0.21%	2.24%	13.18%	43.17%
split-pool, 3 rounds	96	84,934,656	0.05%	0.56%	3.47%	13.18%
split-pool, 4 rounds	1	84,934,656	0.05%	0.56%	3.47%	13.18%
split-pool, 4 rounds	**24**	**2,038,431,744**	0.00%	0.02%	0.15%	0.59%
split-pool, 4 rounds	96	8,153,726,976	0.00%	0.01%	0.04%	0.15%
scifi, 384×droplet	–	283,115,520	0.02%	0.17%	1.05%	4.15%

What a fourth round buys, concretely. Each barcoding round multiplies the space by the number of wells, so three rounds of 96 give $96^3 = 8.8 \times 10^5$ trajectories and four give $96^4 = 8.5 \times 10^7$ – a **96-fold** increase, two orders of magnitude. Multiplied by 24 sublibrary indices that is 2.0×10^9 distinct labels against the ∼12 M cells a genome-scale screen needs.

The practical consequence is the sublibrary count. At 480,000 cells per run and a 1% collision target, three rounds need 54 sublibraries at $55 each in beads and library prep ($2,970 per run); four rounds clear 1% at a single sublibrary, so the count falls to whatever PCR complexity dictates – about 11 at Gaisser et al.'s 45,000-cells-per-sublibrary working limit, or $605 per run. A fourth plate costs $2,566 once, being the three-plate purchase divided evenly, and $147 per run. That per-run figure is worth unpacking, because a fourth BARCODE round is a third LIGATION round: $135 of it is the ligase line growing by half, and only $12 is the extra plate being drawn down. Either way it repays itself inside the first run.

Against that sits the one real argument: every split-pool ligation loses material, and the yield cost of a fourth round must be measured in the pilot before it is committed to.

The preindexed scheme buys the same space with one plate. A round1 plate of 384 wells against the 737,280 round2 barcodes in the Chromium ATAC reagents is 2.8×10^8 labels from a single in-cell reaction (table 15) – more than three rounds of 384 would give, and between what four rounds of 96 deliver at one and at 24 sublibraries. The barcode space is not what distinguishes the two routes. What distinguishes them is where the space comes from: split-pool builds it from repeated ligations, each of which costs yield, and the preindexed scheme buys most of it from a bead the vendor already sells. That is the same argument as the fourth-round one above with the ligation-yield term removed, and it is why the yield question the pilot has to answer is different for the two routes rather than merely smaller.

A fourth barcode round pays for itself inside one run. At 480,000 cells and a 1% collision target, three rounds force 54 sublibraries at $2,970 per run. Four rounds clear 1% in a single sublibrary, so the count falls to whatever PCR complexity dictates, about 11, or $605. The plate costs $2,566 once and $147 per run. The one cost that is not on this ledger is molecular yield through an extra ligation, and the pilot has to measure it.

5.5 Genome-Scale Budget ✗

Table 16: Genome-scale CRISPRi Perturb-seq budget: 6,000 target genes, six guides per gene, one environment, UIUC rates effective 2026-08-01. **Cells per target gene** = cells carrying a guide against that gene, pooled over all six of its guides – not cells per guide and not cells per plasmid. **Analyzed** = cells surviving QC *and* assigned to a guide; **sequenced** = cells whose reads were paid for, which is larger because ~75% of split-pool cells are discarded after sequencing. **Protocol runs** is a count of wet-lab repetitions of the barcoding workflow (split-pool) or of 10x channels, *not* a cost. All sequencing is priced on NovaSeq X 25B PE150 at the 8+ lane rate, the cheapest per-read option the core offers. Excludes labor, oligo pool synthesis, strain construction, and the ~$10,000 one-time split-pool start-up. ∗ **The scifi row is a projection, not a measurement.** It holds every 10x parameter fixed and changes one: cells recovered per channel becomes Datlinger et al.'s measured 151,788 in place of the UIUC baseline of 20,000. So it answers what the droplet budget looks like if a channel goes 7.6 times further and nothing else changes. Per-cell depth in yeast is unmeasured for this chemistry and is held optimistically at the 10x value; no scifi run in any microbe has been published (section 3.3).

Cells per target gene	Platform	Cells analyzed	Cells sequenced	Protocol runs	Read pairs (billions)	Reagents	Sequencing	Total
100	SPLiT-seq	600,000	2,400,000	5	50.0	$11,470	$50,880	**$62,350**
100	SPLiT-seq + depl.	600,000	2,400,000	5	10.0	$12,470	$12,720	**$25,190**
100	10x Chromium X	600,000	1,090,910	55	27.3	$117,865	$28,620	**$146,485**
100	10x + scifi preindex.*	600,000	1,090,910	8	27.3	$17,144	$28,620	**$45,764**
250	SPLiT-seq	1,500,000	6,000,000	13	125.0	$29,470	$127,200	**$156,670**
250	SPLiT-seq + depl.	1,500,000	6,000,000	13	25.0	$32,070	$25,440	**$57,510**
250	10x Chromium X	1,500,000	2,727,273	137	68.2	$293,591	$69,960	**$363,551**
250	10x + scifi preindex.*	1,500,000	2,727,273	18	68.2	$38,574	$69,960	**$108,534**
500	SPLiT-seq	3,000,000	12,000,000	25	250.0	$57,185	$251,220	**$308,405**
500	SPLiT-seq + depl.	3,000,000	12,000,000	25	50.0	$62,185	$50,880	**$113,065**
500	10x Chromium X	3,000,000	5,454,546	273	136.4	$585,039	$136,740	**$721,779**
500	10x + scifi preindex.*	3,000,000	5,454,546	36	136.4	$77,148	$136,740	**$213,888**

The two platforms are not equally worth measuring. Panel (e) of Fig. 15 puts both on one axis, cells per batch, but the curves say different things about it. Droplet cost is reagents and reagents are per batch, so its curve falls steeply and keeps falling; split-pool cost is sequencing, which does not depend on this number at all, so its curve flattens almost at once and learning that yeast tolerates a denser RT well buys little past a few hundred thousand cells per run.

Multiplexing alone does not open up epistasis. Panel (f) of Fig. 15 prices a 200-gene panel. Run the same calculation over all 6,000 genes and coverage per pair falls by $(200/6000)^2 \approx 1/900$, to well under one cell per pair at any k, because the pairs to be covered grow quadratically while the pairs a cell reports grow only as k^2. Restricting the panel is what makes pairs measurable and multiplexing is what then makes it affordable; neither substitutes for the other.

The scifi column of Fig. 15 is a projection throughout. It changes exactly one parameter of the 10x column – cells recovered per channel becomes Datlinger et al.'s measured 151,788 rather than the UIUC baseline of 20,000 – and per-cell depth in yeast is unmeasured for that chemistry and held at the 10x value, the optimistic end. No scifi-RNA-seq run in any microbe has been published (section 3.3).

Reading the table: the dominant term *moves*. For split-pool as published, sequencing is 81% of spend; for 10x it is 19% and library preparation is 81%, because a genome-scale screen needs 137 channels at $2,143 each. These are different businesses and the optimizations do not transfer between them. rRNA depletion is worth ~$99,000 at the 250-cells-per-gene tier. And the 13 protocol runs behind the split-pool row are ~13 weeks of skilled bench work with 13 chances to fail, against 137 channels run by the core – a real argument for droplet that the dollar column conceals.

What preindexing does to that reading. The 137 channels are what makes the droplet row expensive, and they are a consequence of one number: 20,000 cells recovered per channel. Replace it with the 151,788 Datlinger et al. measured and hold everything else fixed, and the same screen is 18 channels and $109 k against $364 k. Three things follow.

The dominant term moves again, and it moves onto the axis this document has been optimizing all along. At 18 channels, reagents are 36% of the preindexed droplet column and sequencing is 64%, so a preindexed droplet screen is a sequencing-limited experiment in the same way split-pool already is. It does not inherit split-pool's fix: an oligo-dT-primed droplet library is already ~85% mRNA, so there is no rRNA to deplete and the lever there is reads per cell, not library chemistry.

The labor argument reverses direction. Eighteen core-run channels against 13 protocol runs is not the 137-against-13 comparison that made droplet attractive on bench time, and the preindexing plate is itself a wet-lab step, so the two routes converge on effort as well as on price.

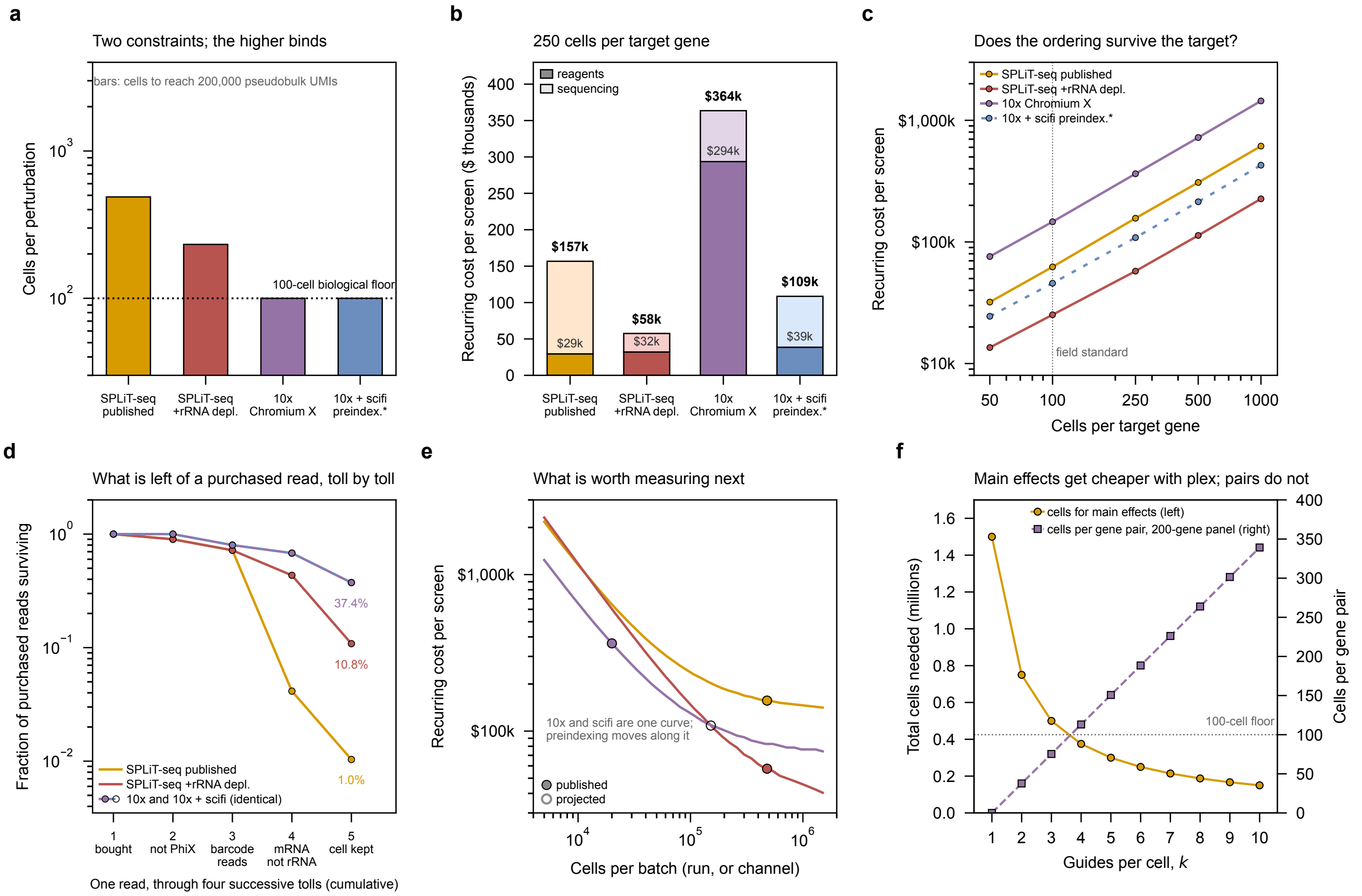

Figure 15: Six views of one decision: which platform to run. Color is the platform in every panel and only ever the platform, so the key is learned once; where a panel needs a second dimension it uses lightness within the platform's own color, or hatching against white. The asterisk marks the projected row, drawn inside the panels because a screenshot loses a caption. All four columns are 6,000 target genes in one environment at UIUC rates; panels (a), (b), (d) and (e) are priced at 250 cells per target gene. **(a) Cells per perturbation, under two independent constraints** (section 4.4): the depth requirement, i.e. how many cells it takes to accumulate 200,000 pseudobulk UMIs at each platform's per-cell yield, against the 100-cell biological floor below which cell-cycle and metabolic state are not averaged out. The taller bar binds, and the floor binds for every deeper platform, so past a few hundred UMIs per cell the design is limited by how many cells must be observed rather than by how deeply each is read. The preindexed column is identical to the 10x column here, which is the point of showing it: preindexing buys cells per dollar, not UMIs per cell, and is invisible to every statistical panel in this document (fig. 13d). **(b) Where the recurring money goes**, split into the two terms that carry the message: the platform's own color for reagents – barcoding, library preparation and, for the split-pool columns, sublibrary preparation – and its pale companion for sequencing. Sublibrary prep sits inside the reagent term rather than being drawn separately, which is how table 16 groups the same money and is also the only legible choice: at $2,970 of a $156,670 screen it is under 2% of a bar, and a band four points tall cannot carry a pattern. The dominant term *moves*: sequencing is 81% of split-pool spend, while for 10x it is 19% and library preparation 81%, because a genome-scale screen consumes 137 channels at $2,143 each. These are different cost structures, and an optimization that helps one (rRNA depletion, worth ∼$99,000 here) does nothing for the other. Preindexing takes the 137 channels to 18 and the column to $109 k, and in doing so moves that platform's dominant term onto sequencing as well – reagents 36%, sequencing 64% – but with no rRNA left to deplete. **(c) Whether the ordering survives a change of target.** Panel (b) prices one design; this sweeps 50 to 1,000 cells per target gene. The ordering is stable across the whole range, so the platform choice is not an artifact of the 250-cell point, and the vertical rule marks the 100-cell field standard the recommendation is actually made at. **(d) What happens to a purchased read**, stage by stage, and this is where the reagent comparison and the cost comparison part company. Split-pool reagents are ∼22× cheaper per cell than droplet and it still loses end to end; the reversal is entirely here. Of 100 reads bought for a split-pool run, 28 are lost to the PhiX spike (10%) and to barcodes that do not resolve (a further 20% of what remains), 94 of the remainder are ribosomal, and three quarters of the cells they came from are then discarded – 1.0 read in 100 survives all four tolls. Depletion takes that to 10.8, and both droplet columns reach 37.4. **The two droplet lines coincide exactly at every stage**, which is worth seeing: preindexing touches no term in this panel. **(e) What is worth measuring next.** Every platform here has exactly one number that is both uncertain and load-bearing, and for all of them it is the same field – cells barcoded per protocol run, or cells recovered per channel – so they share an axis. 10x and scifi are one curve, and preindexing is a move along it rather than a move to a different one. **(f) What multiplexing buys as guides per cell k rises.** Left axis: total cells needed to hold main-effect precision across 6,000 genes falls as $1/k$, from 1.5 M cells at $k = 1$ to 0.15 M at $k = 10$, because every cell reports on k genes at once. Right axis: what that same cell budget delivers for one *named* gene pair if the library is restricted to a 200-gene panel, which grows roughly linearly in k and crosses the 100-cell floor near $k = 4$. Generated by `experiments/024-perturb-seq-costing/scripts/plot_economics.py`, which fails rather than writing the figure if any two labels in it collide.

And the ordering does not change. Depleted split-pool stays cheapest at \$57.5 k against \$109 k. What changes is the size of the gap, from 6.3× to 1.9×, and a factor of two is small enough to be settled by considerations the dollar column does not hold: instrument access, whether the guide readout works, and which of the two unmeasured assumptions turns out to be true.

Recommended entry point, unchanged. 100 cells per gene, split-pool with rRNA depletion: ∼\$25 k recurring plus ∼\$10 k one-time. That is a genuine genome-scale yeast Perturb-seq for under \$40 k of consumables and sequencing, and 100 cells per perturbation is the published field standard.
What preindexing changes is the fallback, and section 5.6 promotes it further. The same screen on preindexed droplet is ∼\$46 k, so if the split-pool guide readout or the cells-per-RT-well number fails in the pilot, the droplet route is no longer nearly six times the price – it is under twice, on a platform the core already runs. That is worth one measurement in the pilot rather than a change of direction now: both figures rest on assumptions that have never been tested in yeast, and split-pool's rest on fewer of them.

5.6 What the Core Said, and What It Changes ✗

A meeting with the Carver Biotechnology Center on 2026-08-18 (Alvaro Hernandez and Chris Wright) settled several open items in section 5.3 and opened one new option. The claims are recorded with the status of each in `uiuc_core_data.py`, because a verbal source is not a rate card and three of these numbers disagree with the published page. None is used in a budget here.

What is settled. Three things stop being open questions. The core runs the GEM-X Universal 5′ Gene Expression kit, a 10x Genomics product. Roughly one microlitre of zymolyase can go into the 10x reaction, which is the Jariani et al. in-droplet spheroplasting change of section 3.2, so the yeast adaptation that section describes is one the core already accepts rather than one we would have to argue for. And quality control is theirs: they asked to be relied on for it, which is consistent with a page that charges nothing for 10x test counts.

Cell counting, and where it matters less than it sounds. Their strongest advice was that getting the cell count right is the single most critical step. That is exactly right for ordinary droplet loading, where the count sets the Poisson loading rate and therefore the doublet rate, and a count that is wrong by a factor of two moves the doublet rate by roughly the same factor. It matters much less under the preindexed regime of section 3.3: at 100-fold overloading the droplets are 95.5% occupied, the operating point is deliberately saturated, and cell identity comes from the round1–round2 pair rather than from one-cell-per-droplet. Counting accuracy is a first-order requirement for the plain droplet route and a second-order one for the preindexed route, which is a point in the latter's favor that the cost tables do not show.

Which counter, since the advice was to buy one. Yeast is the awkward case for automated counting: *S. cerevisiae* runs about 3.5 to 5 µm▲ [external: vendor sizing notes; no primary in the mirror] across, at or below the stated floor of most brightfield instruments, and 10x's own guidance is that cells under 10 µm "may be difficult to distinguish from debris" and that a fluorescent viability stain rather than trypan blue is "strongly recommended"▲ [external: 10x CG00053]. Two instruments are validated on this organism by their makers: the Logos LUNA-FX7, which has a *S. cerevisiae* application note reporting $R^2 > 0.99$ against a hemocytometer on both concentration and viability, and the ChemoMetec NC-203▲ [external: Logos and ChemoMetec application notes]. Both appear on 10x's evaluated list. Worth knowing before buying anything: the Bio-Rad TC20 specifies a 6 µm floor▲ [external: Bio-Rad TC20 brochure], which excludes yeast, and it is the counter Jariani et al. used. No published yeast single-cell RNA-seq run has loaded inside 10x's recommended 700–1,200 cells per microlitre window▲ [external: 10x CG000127], which is a small, real gap our protocol could close.

The Illumina kit, and a resolved mishearing. "Preserve Seek" is *Perturb-seq*: Illumina has a page of that name, which is where a search of their site lands. There is no Illumina product called PreserveSeq and no branded preservation reagent. What Illumina does sell is the Single Cell 3′ RNA Prep, the PIPseq chemistry acquired from Fluent BioSciences, which partitions cells on templated particles with a vortexer instead of a microfluidic chip. It is genuinely cheaper per cell at scale, and its guide-enrichment kit ships today. It is also unusable here without method development: it has never been run on a walled organism, and Illumina state their own nuclei-isolation reagents are "not recommended for use with non-mammalian sample types"▲ [external: Illumina support]. It belongs on the watch list, not in the plan.

Three numbers that do not agree with the published page. A lane was quoted as 3.82 billion; the instruments page gives 3.2 billion read pairs per NovaSeq X 25B lane, which is what every figure in this document is priced on and what the flow-cell naming implies. Test counts were quoted at about \$153 with a few not charged; the page says there is no charge, without qualification. The Ultima rate was given as roughly \$1,990 per lane alongside a figure of \$3,310 for three lanes, and those cannot both be per-lane rates. The conservative reading is retained in each case, and all three need confirming in writing before anything is built on them.

Which way this pushes the recommendation. section 5.5 recommends split-pool with rRNA depletion on price, and nothing in the meeting changes the arithmetic. What it changes is the weight on the things the dollar column does not hold, and all of them moved the same way. The core runs the droplet platform as a service and will own quality control on it, so 137 or 18 channels are their bench time rather than ours, against the 13 protocol runs of skilled in-house work the split-pool row implies. The zymolyase modification is one they already accept. The kit they run is the 5′ one, which is the assay that might read our guides without a new construct (section 3.6). And counting accuracy, their strongest warning, is a much weaker constraint under preindexed overloading than under ordinary droplet loading.

None of that is a costing argument and it should not be presented as one: depleted split-pool is still cheapest, at \$57.5 k against \$109 k for the preindexed droplet route. But the gap is a factor of 1.9, and the considerations above are exactly the ones section 5.5 said a factor of two is small enough to be settled by. Preindexed droplet has moved from fallback to co-equal candidate, and the pilot should be designed to discriminate between them rather than to confirm the cheaper one.

> **Preindexed droplet has moved from fallback to co-equal candidate.** Depleted split-pool is still cheapest, \$57.5 k against \$109 k, and the arithmetic has not changed. What changed is everything the dollar column does not hold, and it all moved the same way: the core runs the droplet platform as a service and will own quality control on it, they already accept the zymolyase modification, the kit they run is the 5′ one that might read our guides unmodified, and counting accuracy binds far less under preindexed overloading. Design the pilot to discriminate between the two rather than to confirm the cheaper one.

5.7 Ultima Sequencing, and Whether Genotypes Survive It ✗

The core also offers Ultima Genomics “wafer” sequencing, which is not on the public rate page. At the quoted rate it is several times cheaper per read than the NovaSeq X lane that section 5.5 prices everything on, and since sequencing is 81% of the split-pool budget, a cheap read is the largest imaginable lever. The question is whether the assay still works, and for the perturbation readout specifically the published answer is no.

What the instrument actually is. The UG 100 is single-end. It emits one read of roughly 200 to 300 bases, and paired-end is announced for the successor series rather than available now [34]. The core’s remark that it has no 2×150 configuration is therefore a property of the platform, not of their menu. The “20 nt read 1, 90 nt read 2” heard in the meeting is not a read-length limit either: the pipeline splits the single long read into a synthetic read 1 carrying the cell barcode and UMI and a read 2 carrying the cDNA, then trims the cDNA to 90 bases. So cDNA length is not the constraint.

Its chemistry has no terminators, so homopolymer run length is inferred from signal intensity, and indels in homopolymers are its characteristic error mode. That is the reason the core described it as suited to work with a well-assembled reference: counting reads against a known genome tolerates the error mode, and calling variants does not.

Cell barcodes survive; UMIs are shortened. On matched libraries the two platforms called almost the same cells, 7,916 against 7,926, with expression correlated at $r = 0.98$ [34]. The 16 nt cell barcode is an error-correctable whitelist, which absorbs indel errors well. The UMI does not fare as well on a 3′ assay: the poly(T) stretch adjacent to it degrades the bases flanking it, and the standard mitigation is to mask the last three, leaving an effective 9 nt UMI. That is still $\sim 2.6 \times 10^5$ labels, comfortably above the few thousand molecules of any one gene in a yeast cell (section 4.2), so UMI collision remains negligible. A 5′ assay avoids the problem, which is a second reason the core’s 5′ kit is the relevant one.

> **The guide readout is the part that does not survive, and this is published.** The paper that validates Ultima for single-cell RNA-seq ran a Perturb-seq case study and put the transcriptome on Ultima while sequencing the guide-detection libraries on Illumina, “because the read length was not sufficient for guide detection” [34]. Guide-to-cell assignment matched Illumina in that study because it *came from* Illumina. No published workflow reads CRISPR guides on this platform, and guide capture does not appear among the 10x library types verified as Ultima-compatible. So the answer to whether genotypes can be resolved on cheap sequencing is: not on this platform alone, on present evidence.

The hybrid, and why it is still interesting. Nothing above rules out running the transcriptome on Ultima and the guide library on a MiSeq or a fraction of a NovaSeq lane. The guide library is small, since it carries one short amplicon per cell rather than a whole transcriptome, so its sequencing cost is a minor term while the transcriptome is the term Ultima would attack. Against that: 10x libraries require a PCR adapter conversion before they can run on Ultima, so the hybrid is two library-preparation paths and two submissions. That is a real complication for a first experiment and a reasonable trade later.

What would have to be measured. Ultima enters this document as an option with one published negative result attached, so the honest ranking puts it behind the items in section 6. **Hypothesis (untested):** a polyadenylated guide-barcode transcript of the kind section 3.6 proposes would read on Ultima, because it is an ordinary mRNA-like molecule rather than the short targeted amplicon that defeated guide detection in the published study. If that held, the whole readout could run on one cheap platform. It has not been tried, and the construct does not exist yet.

6 Bottlenecks and Development Priorities ✗

What has to be true for 8–10 simultaneous perturbations to be measurable, and what to do first given that we will start with single-guide CRISPRi and move to two.

6.1 Priority Ranking ✗

1. Guide capture (blocking). Nothing else matters until guide identity is recoverable from the RNA (Sec. 3.6). It has been treated here as a construction problem rather than a cost problem, with a published yeast template, and it may turn out not to be one: the 5′ assay captures guides with a primer against the sgRNA scaffold rather than with an engineered capture sequence, so an unmodified library might already be readable. Test that first, because it costs one channel and it decides whether anything has to be built at all. Let q be the *per-guide detection rate* – the probability that one guide present in a cell is actually seen in that cell's reads. A k-plex cell is only usable if all k of its guides are detected, so the usable fraction is q^k, and the whole feasibility of multiplexing turns on which q we inherit. At Brandner et al.'s bacterial $q = 0.25$ a 3-plex yields $0.25^3 \approx 1.6\%$ of cells and an 8-plex is arithmetically hopeless. At the $q > 0.71$ Nadal-Ribelles et al. achieve in yeast, a 3-plex yields 36% and even an 8-plex yields 6% [5]. That is the difference between multiplexing being impossible and merely expensive, and it is one measurement. Raising q is worth more than any other single change, because it enters the multiplex ceiling exponentially while everything else enters linearly – and the pilot's first job is to find out which of those two regimes a plasmid-borne guide barcode lands in.

2. mRNA fraction. At 5.75% mRNA, 94 of every 100 reads buy ribosomal RNA and the sequencing bill is inflated by roughly the reciprocal. Two independent levers: shift the random-hexamer:oligo-dT ratio, since the 5′ coverage hexamers buy is worthless for a 3′-counting screen; and add Cas9-guided rRNA depletion after cDNA synthesis. Brandner et al.'s depletion took *E. coli* from 4.6% to 60.2% mRNA, but only 2.2% to 10.4% in *P. putida*, so the transfer to yeast must be measured, not assumed. Worth ∼\$99 k at genome scale (Sec. 5.5).

3. Cells per RT well. Brettner et al. verified 5,000 yeast cells per well; microSPLiT runs 40,000–100,000 for bacteria. This number sets how many cells one protocol run can barcode, and therefore how many runs a screen needs. The 250-cells-per-gene split-pool row of table 16 requires 13 protocol runs at Brettner et al.'s loading; at 20,000 cells per well the same screen is four runs, because a run would barcode ∼1.9 M cells instead of ∼480,000. That is ∼13 weeks of skilled bench work reduced to ∼4, which is most of the labor argument for droplet (Sec. 5.5). It is also the cheapest experiment on this list. The comparison it is measured against has moved: the droplet alternative is 18 preindexed channels rather than 137 plain ones, so the bench-time gap this number closes is against 18, not 137.

4. Barcode space. A fourth ligation round is necessary at genome scale and pays for itself within one run (Sec. 5.4). Its only cost is molecular yield through an extra ligation, which the pilot must measure. Under multiplexing the requirement tightens further, since a collision fabricates a gene pair rather than merely diluting a main effect.

5. Transformation efficiency. The ceiling on multiplexing is library construction, not measurement (Sec. 4.8). Combinatorial diversity generated by co-transforming several single-guide libraries, rather than by constructing combinations, is the route worth costing – it converts a combinatorial cloning problem into a Poisson loading problem.

6. Effect size, now measured, and the two that took its place. This item used to read that section 4.5 was unresolved and that everything in section 4.4 inherited its uncertainty. It is resolved: the median responding gene moves 1.34-fold, which multiplied every depth-derived cell requirement by ∼5.6 and is the largest single correction in the document. What it did not resolve is the other two inputs to eq. (5), and both are now the thing to measure. φ_j, gene-level overdispersion, sets the floor by itself and is estimable from any yeast count matrix. p_j, the target gene's transcriptome share, is currently assembled from two sources that do not belong together and is worth a five-fold spread in cells per perturbation (section 4.2). Both come out of the first pilot's count matrix at no extra cost, which is what makes them cheaper than anything above them on this list and is why they are ranked here rather than as an experiment.

7. Preindexed droplet, as a hedge rather than a plan. The item that would most change the ranking above is the one nothing on this list depends on. Preindexing takes the droplet route from 6.3× the cost of depleted split-pool to 1.9× (Sec. 5.5), which converts droplet from a platform ruled out on price at *genome scale* into a live fallback there if items 1 or 3 fail. The qualifier matters, because the first two experiments in section 6.3 are both droplet channels and neither contradicts this: a single channel is cheap however it prices per cell at six thousand genes, the core runs the platform as a service, and one of the two exists precisely to test their 5′ kit. Price rules droplet out as a way to buy twelve million cells, not as a way to buy twenty thousand. It ranks last because it is the least evidenced: items 1–6 are quantities to be measured on a chemistry that has been run in yeast, whereas this is a chemistry that has never been run in any microbe, and the same in-cell reverse transcription it needs is what item 3 is already testing. That ordering is worth stating rather than leaving implicit, because the cost figure is attractive enough to invite reordering it, and the figure and the evidence run opposite ways.

6.2 The Host Organism, and Why the First Experiment Carries No Perturbations ✗

Everything above is written for *S. cerevisiae*. The intended production host is *Pichia kudriavzevii*, the same species as *Issatchenkia orientalis* and as the clinical isolate *Candida krusei*, whose genomes are "collinear ... 99.6% identical in DNA sequence" [35]. Moving there changes the risk profile sharply, and in a direction that decides what to run first.

What exists. A complete reference genome, five chromosomes and 10.8 Mb with 5,139 protein-coding genes annotated, so read alignment and guide design are not obstacles. Catalytic Cas9 works well: an *ADE2* disruption efficiency of 97.0% using an *RPR1′*-tRNA$^{\mathrm{Leu}}$ promoter for the sgRNA, and multiplexed deletions at 72.8% to 89.9% for pairs and 46.7% for a triple [36]. A landing-pad system reaches greater than 80% integration efficiency at 27 characterized loci and supports simultaneous integration of two to five genes [37].

What is not published, which is not the same as what does not exist. One of the gaps below is closed in house and the distinction matters, because a capability that exists but is unpublished is a validation task rather than a research project. There is no published dCas9 CRISPRi or CRISPRa system in this organism – every CRISPR result above is catalytic knockout or integration – but a working one is held here▲ [external: in-house, unpublished], as is a JGI assembly of the strain actually in use▲ [external: in-house, unpublished]. The assembly is not what the published reference genome above already provides: read alignment and guide design were never the obstacle, and a strain-level assembly matters because the production strain is not the reference strain and is diploid. Read against that, the missing piece is not the perturbation tool; it is everything downstream of it.

What remains genuinely open is delivery and scale. Episomal plasmids are mitotically unstable, with about 55% of cells still expressing a plasmid-borne reporter after five days [36], which is a problem for the delivery route of section 7.1 specifically; a centromere-like sequence found in the same organism improves plasmid stability and is the obvious thing to build on [38]. And the only transformation-efficiency anchor available, roughly 10^3 colony-forming units per microgram, is three to five orders of magnitude below what *S. cerevisiae* achieves. Section 4.8 showed that library scale is set by transformation; if that figure is representative, pooled library-scale screening in this host is not currently reachable and the gap is not one a budget closes. The published strain used by the bioproduction community is also diploid, which changes what a single guide does and what a knockout means.

The first experiment carries no perturbations, and only part of that is forced. A pooled library-scale screen in this host waits on transformation efficiency whatever anyone decides. What is chosen, and is the reason to run now rather than wait, is spending the first run on learning the assay: nobody on this project has run single-cell RNA-seq on a walled microbe, so it answers whether the cell wall yields to the in-droplet zymolyase of section 3.2 in this species, how many mRNA UMIs per cell are recovered, and what fraction of reads is ribosomal. That is the collaborator's recommendation and it is also what the arithmetic supports, since the third of those is the largest cost lever in the document (section 3.5) and is a proportion, answerable at any sequencing depth. None of the three needs a library, so none of them waits on the delivery and scale questions above. If a run with no perturbations looks like a thin return for a channel, the way to thicken it is to spread the same run across several environments (section 7.3), which costs barcode capacity that is otherwise idle rather than costing genetics that do not exist. Running this first also frees the guide-capture question of section 3.6 to be settled in *S. cerevisiae*, where the library already exists, in parallel rather than in sequence.

Two runs, not one: a MiSeq to check, a NovaSeq lane to profile. The core's own advice is to use the MiSeq for quality control rather than for production, and that maps cleanly onto this experiment because the two things being asked are different questions.

The check is whether the chemistry worked at all: did the wall yield, did cells barcode, what fraction of reads is ribosomal. All three are proportions, so they are answerable at almost any depth. A MiSeq i100 25 M paired-end run ($1,450) carries ~1,250 cells at 10x's recommended 20,000 read pairs per cell, and the $398 titration run answers the barcode question alone. Either is cheap enough to repeat if the first attempt fails, which is the property that matters at this stage: section 6.3 notes that cDNA traces and library concentration do not predict barcoding success, so there is no earlier checkpoint than sequencing something.

The profiling run is a different job. A first transcriptome atlas of this organism is a dataset we would want to keep and analyze, and 1,250 cells is not one. Section 5.3 showed that the core's gap sits exactly here, but the split-lane row is the way through it: a NovaSeq X 10B split lane gives 600 M read pairs for $1,160, which is ∼30,000 cells at full depth, for less than the MiSeq run costs. So the sequencing is not what constrains this experiment; the 10x channel is, at 20,000 cells recovered.

Sequence the pilot twice, and the second time is cheaper than the first. Run the MiSeq to confirm the chemistry, then a NovaSeq X 10B split lane ($1,160 for 600 M read pairs) for the actual profile. The split lane costs less than the 25 M MiSeq run and delivers 24 times the reads, so the only reason to sequence shallowly here is turnaround, not money. Depth is not the binding constraint on this experiment and neither is cost.

6.3 Staged Execution Plan ✗

1. **Unperturbed *P. kudriavzevii*** (section 6.2). No library and no guides, so it depends on none of the delivery and scale questions still open in this host, and it is where the assay itself gets learned. It returns the wall-digestion verdict, the mRNA UMIs per cell, and the ribosomal fraction, which is the largest cost lever in the document. Sequence it twice: a MiSeq run to confirm the chemistry, then a NovaSeq X 10B split lane for a profile worth keeping. If the channel looks under-used carrying one condition, run it across several environments instead of adding guides. This is the current first experiment.
2. **Test 5′ guide capture on the existing *S. cerevisiae* library**, in parallel with step 1 rather than after it, because it uses a different organism and a library that already exists. One channel on the core's GEM-X 5′ kit answers whether the scaffold primer reverse transcribes an unmodified Pol III guide (section 3.6). If it does, the construction project below is not needed.
3. **Construct and validate the poly(A) guide barcode**, *only if step 2 fails*, in an arrayed set of ∼10 strains with known knockdown phenotypes, scored against bulk RNA-seq from the same strains. Ground truth is only available while the design is arrayed; a pool cannot tell you what was in a given cell.
4. **A split-pool run in *S. cerevisiae***, which is the first step on this list to use that chemistry: steps 1 and 2 are droplet channels at the core, and everything the split-pool route needs measured is measured here. Establish the chemistry numbers the model is most sensitive to: mRNA fraction after depletion, cells per RT well, guide-assignment rate q, the read-depth saturation curve, and – from the same data, at no extra cost – the transcriptome share p_j of a few representative target genes, which section 4.2 shows is currently worth a five-fold spread in cells per perturbation and is the largest unforced uncertainty in the budget. One $398 MiSeq titration run protects a $3,180 NovaSeq lane, and there is no earlier checkpoint – Brettner et al. is explicit that cDNA traces and library concentration do not predict barcoding success.
5. **Answer the preindexing question from data step 4 already produces**, at no additional cost. Whether a fixed, wall-digested yeast cell reverse transcribes in situ and then survives handling is measured by the round-1 step of step 4's split-pool run; what step 4 does not answer is whether such a cell survives the pressures of a microfluidic chip, which is one channel to test and is the only part of Sec. 3.3 with no yeast precedent at all. Worth running only if step 1 or step 3 goes badly, because it buys a fallback rather than advancing the main route.
6. **Single-guide pilot** on a focused sub-library, at the 100-cell floor.
7. **Two-guide screen** on the focused metabolic sub-library, where named pairs are reachable, to measure how guide detection and collisions actually degrade at $k = 2$ before trusting any extrapolation to $k = 8$.
8. **Genome-scale single-guide screen**, four barcoding rounds, one environment.

6.4 Outlook ✗

Two extensions change what the platform is for, and both are worth designing toward now rather than retrofitting.

Splitting the sample – and why the obvious version does not work. Pairing a transcriptome with mass spectrometry on the *same* perturbation is the extension that would matter most for metabolic engineering, since flux is what we actually want and it cannot be inferred from expression alone. Fulcher et al. [39] is the precedent for parallel transcriptome and proteome from the same single cells. But the naive version – take half the contents of a compartment – is not available in either format, and for different reasons worth separating.

A semipermeable capsule cannot be aliquoted. It is a hydrogel shell with a pore cutoff around 30 nm: enzymes up to ∼160 kDa diffuse in, and that permeability is the entire point, since it is what lets reagents reach the cells without breaking the compartment [40]. There is no valve; the pores are a property of the material, not a gate that can be closed.

And the same cutoff is fatal to the specific application: *small molecules leak.* Metabolites are precisely the species that pass a 30 nm pore freely, so an SPC cannot hold a metabolome in the first place, let alone divide one. Proteins are the more plausible target, since much of a proteome sits above the cutoff – but a metabolomic readout from SPCs is not a matter of engineering the split, it is excluded by the format.

Two routes remain, and they split the population rather than the compartment. Capsules are FACS-sortable and their cells can be released alive by brief protease treatment, so whole capsules can be *routed* – one fraction sequenced, another pooled for bulk mass spectrometry – which gives paired measurements over matched sub-populations rather than over identical cells. Alternatively, because each capsule holds a *clonal* population expanded from one founder, releasing that clone and dividing the cells does give two aliquots of the same genotype; what is lost is the capsule identity, which then has to be re-established by a barcode rather than by physical containment.

The unsolved part. Either route needs the mass-spectrometry channel to carry perturbation identity, and MS has no equivalent of a sequencing barcode. The plausible designs are chemical isobaric labeling of routed fractions, or physically arraying sorted capsules into wells so that position encodes identity – which trades throughput for it, and is the obvious tension to think about before this becomes a plan. Isotope labeling for flux sits downstream of solving that.

Environments as a second axis. Treated in full in section 7.3, which is where the arithmetic sits: environments multiply cell demand linearly where named gene combinations grow quadratically, and a round-1 plate already carries the labels they need, 96 of them in split-pool and 384 in preindexing.

7 Scaling the Screen ✗

The screen priced in section 5 perturbs one gene per cell in one environment. Neither restriction is where the value is. A metabolic engineering campaign typically arrives at a final strain through fewer than ten edits, and the aim of running a screen at all is to reach that endpoint in one pass, or to collect enough data that a model can propose the edits. Both goals need the screen to move along two axes: more perturbations per cell, and more environments per experiment. They behave completely differently, and the difference is the subject of this section.

7.1 Delivering More Than One Guide per Cell ✗

Two routes put k guides in one cell, and the yeast literature has settled firmly on the first.

Route A, an array on one plasmid. Several guide cassettes are cloned into a single vector, so the combination is designed and known by construction. This is what our own library's lineage does: sMAGIC built its pairwise library by piecing "two MAGIC plasmid libraries together in the same vector using Golden-Gate Assembly" [2]. The ceiling is set by repeated sequence, not by biology. Independent identical Pol III cassettes degrade sharply: targeting two, three and four genes gives 89.2%, 84.1% and 50.3% efficiency, and "targeting of the 5th gene resulted in a drastically lower efficiency of only 1.96%" [41]. Spacing the guides with tRNAs reaches eight, and the authors say plainly why they stopped there: "because of limited plasmid construction tools available for assembly of multiple fragments with repetitive sequences" [42]. Twenty-four is possible with Csy4 processing, but only genome-integrated and only while the array is switched off, since such arrays "can recombine over multiple cell passages while induced" [43].

For a *pooled* library the repeats bite harder than these efficiency numbers suggest. Repeated scaffold sequence means an oligo pool cannot span the array, so the library needs multi-step assembly, and every assembly step is an opportunity for guide pairs to scramble. Treat two to four guides as the library-safe ceiling on this route.

That ceiling matters for how the rest of the document reads. section 4.6 calls a 200-gene panel at $k = 8$ "the single most useful number in this document" and section 6 opens by asking what has to be true for 8 to 10 simultaneous perturbations. Neither is reachable by route A. They are reachable only if guides arrive on separate plasmids, which is why route B is worth the rest of this subsection rather than being a curiosity.

Route B, more plasmids per cell. The alternative keeps the single-guide library and raises the number of plasmid molecules a cell carries, so the combination is random rather than designed. The attraction is that nothing has to be cloned: the same library, moved to a higher-copy vector or a weaker selection marker, produces combinations for free.

Two facts have to be separated before this can be reasoned about, and conflating them is the trap. **Copy number is not guide diversity.** A cell that takes up one plasmid molecule and then amplifies it to fifty copies carries fifty copies of *one* guide. Diversity comes only from the number of independent uptake events at transformation. Raising average copy number by attenuating the marker, which is well established and finely tunable in yeast over "5–100 copies per cell" with a dose response to antibiotic concentration [44], therefore does not by itself put two guides in a cell. It changes how many copies of whatever arrived are maintained.

What does produce multiple distinct guides is multiple uptake events, and that is both real and dose-tunable. Multi-vector transformants are a documented and usually unwanted feature of yeast library construction: "up to 90% of clones in yeast homologous recombination libraries may be multiple vector transformants," carrying a median of 3.9 to 4.9 unique plasmids, with the population proportion moving from 0.35 to 0.88 as vector and insert mass are raised [45].

The arithmetic of route B. This is the third of the three appearances of Poisson sampling introduced in section 4.1, and fig. 8 is drawn on exactly this problem: the protocol sets an average uptake rate and the cells arrange themselves around it, so the property to carry here is the middle panel, that selection kills the cells which took up nothing and removing the smallest value from a distribution raises the average of what is left.

Selection for the marker conditions on carrying at least one plasmid, so the number per surviving cell is a zero-truncated Poisson. Setting the post-selection mean to $\bar{m}$ requires λ solving $\bar{m} = \lambda/(1 - e^{-\lambda})$, and the fraction of surviving cells carrying two or more guides follows. Two plasmids can also carry the same guide, which costs almost nothing at library scale: drawing from 6,000 guides at $\bar{m} = 8$, the expected number of *distinct* guides is 7.995. Table 17 gives the operating points.

Table 17: Delivering several guides per cell by multiple plasmids rather than by an array (route B of section 7.1). Selection for the marker conditions on carrying at least one plasmid, so the count per surviving cell is a zero-truncated Poisson with mean $\bar{m} = \lambda/(1 - e^{-\lambda})$. **Plasmids per cell** is the average number of plasmid molecules carried by a cell that survived selection, written $\bar{m}$ in the text. It is a copy number, not a count of genes or of library members, and it is the quantity a marker attenuation actually sets. λ is the underlying uptake rate before selection, which is lower because selection discards the cells that took up nothing. **Distinct guides** is lower than $\bar{m}$ because two plasmids can carry the same guide, and the gap is negligible at this library size (6,000 guides). The last two columns are the reason k cannot be treated as a constant: even at $\bar{m} = 3$ nearly a fifth of cells carry a single guide. Generated from `experiments/024-perturb-seq-costing/scripts/scaling\_analysis.py`.

Plasmids per cell (target average)	Poisson λ	Distinct guides per cell	Cells with 1 guide	Cells with ≥ 2 guides
2	1.59	2.000	40.6%	59.4%
3	2.82	2.999	17.9%	82.1%
4	3.92	3.999	7.9%	92.1%
5	4.97	4.998	3.5%	96.5%
8	8.00	7.995	0.3%	99.7%

Three properties of this route decide whether it is usable.

1. **k is a distribution, not a number.** At $\bar{m} = 2$ only 59.4% of selected cells carry two or more guides; the rest are effectively a singles screen running alongside. That is not waste, since section 4.6 shows singles inform main effects perfectly well, but every per-cell k used in a design equation has to be read as an average over this spread.

2. **It does not buy transformants.** section 4.8 showed that cloned pairs are limited by yeast transformation at 10^6 to 10^7, which covers "∼0.01% to ∼0.1% of the total combinations" of the sMAGIC space [2]. Route B removes the cloning bottleneck and leaves the transformation bottleneck exactly where it was. It changes what each transformant contains, not how many there are.

3. **The multi-plasmid state is transient.** Multi-vector transformants segregate, persisting through roughly a day of liquid outgrowth rather than indefinitely [45], and 2μ plasmids are retained in only about 60% of cells even in a lab strain [46]. Any protocol on this route has to fix and capture the cells while the combination is still present, which couples library delivery to experiment timing in a way route A does not.

Route B is unprecedented in yeast, and the missing measurement is small. The 2025 review of CRISPR screens in yeast states that guide libraries are cloned into "typically, low-copy number vectors to prevent multiple guide plasmids in a single cell" [47]: the field treats multiple plasmids per cell as an artifact to suppress, which is precisely the regime this route wants. The mammalian field already runs the equivalent at MOI 2.5–10 with direct guide capture▲ [external: Nat. Methods 2026; not in the mirror], so the concept is proven, just not here. What is missing is one number that nobody has published: the per-cell distribution of distinct guides under tuned delivery. It is measurable in a single experiment once guide capture works (section 3.6), and it is the natural second pilot.

7.2 Do You Have to See a Combination Twice? ✗

This is the question that decides whether high-plex screening is affordable, and it has two opposite answers depending on which quantity is wanted.

For main effects, no, and never. Under the additive model of section 4.6 a cell carrying k guides contributes to k different main effects. No combination has to recur, and no combination even has to be identified as such. Table 18 is evaluated at the 100-cell biological floor of section 4.4, where table 7 in section 4.6 used the 250-cells-per-gene precision tier; that is the only reason their main-effect columns differ, and their pair columns are identical because Eq. 3 does not depend on the choice. The requirement is Nk/T cells per gene, so multiplexing divides the cell count by k: a 6,000-gene library at the 100-cell floor needs 600,000 cells at $k = 1$ and 75,000 at $k = 8$.

For the joint transcriptome of a named combination, yes, and it is expensive. Estimating the cell state under a *specific* double perturbation requires cells carrying exactly that pair. The probability that a random k-subset of T targets contains a given pair is $k(k-1)/(T(T-1))$, and this is where the combinatorial space asserts itself. Table 18 evaluates both requirements together.

Table 18: Main effects never require a combination to recur; a named combination's joint transcriptome does. Both requirements evaluated at the 100-cell first-order floor and Yao et al.'s 400 cells per pair. k **is guides per cell**, each against a different gene, so k is also how many genes are knocked down in one cell. It is not the six guides per gene the library carries, and it is not a count of library members. T is how many genes the panel targets. **Cells for main effects** is $100\,T/k$, since a cell carrying k guides informs k genes. **Cells for all pairs** is Eq. 3. **Times a given pair is seen** is the last column of the first divided over the pair space, that is, how often one specific gene pair appears while running only the main-effect budget: below 1 means most pairs never appear at all. It falls by a factor of about 30 between the 200-gene panel and the genome, because the pair space grows quadratically in T while the cell budget grows linearly. Generated from `experiments/024-perturb-seq-costing/scripts/scaling\_analysis.py`.

Genes in panel, T	Guides per cell, k	Cells for main effects	Cells for all pairs	Times a given pair is seen
200	2	10,000	7,960,000	0.50
200	3	6,667	2,653,333	1.01
200	5	4,000	796,000	2.01
200	8	2,500	284,286	3.52
6,000	2	300,000	7,198,800,000	0.02
6,000	3	200,000	2,399,600,000	0.03
6,000	5	120,000	719,880,000	0.07
6,000	8	75,000	257,100,000	0.12

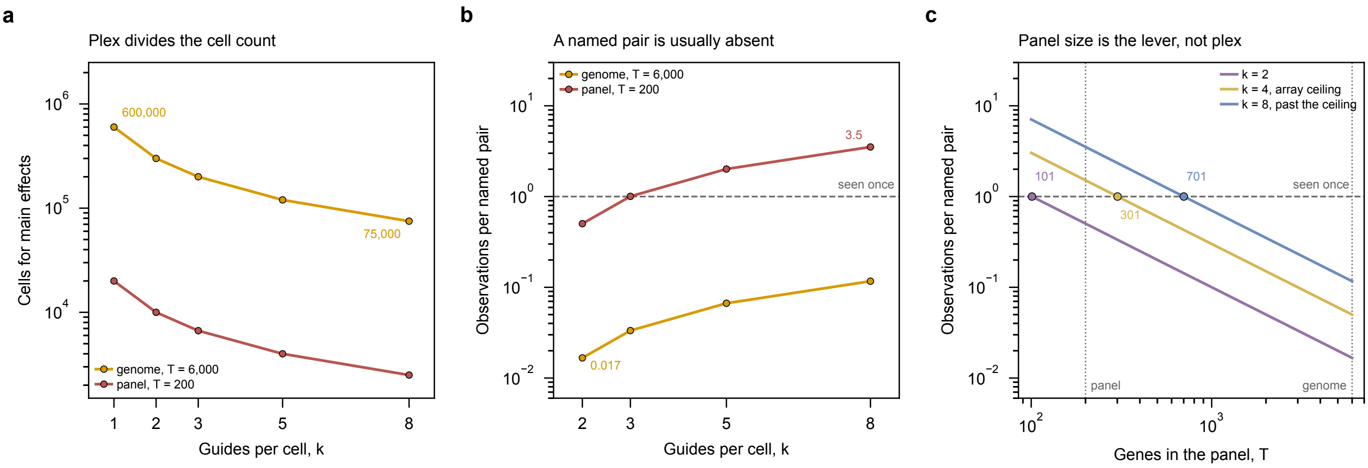

Figure 16: Multiplexing makes main effects cheaper and named pairs no more visible; only a smaller panel does that. (a) Cells needed to put every gene at the 100-cell floor of section 4.4. A cell carrying k guides reports on k genes, so the requirement is Nk/T and the count falls as $1/k$: 600,000 cells at $k = 1$ against 75,000 at $k = 8$ for the genome-scale library. **(b)** Expected observations of one *named* gene pair at that same budget. The probability a random k-subset of T targets contains a given pair is $k(k-1)/(T(T-1))$, so at genome scale the pair is seen 0.017 times at $k = 2$ and 0.117 at $k = 8$: fewer than one pair in eight is observed even once, and the joint transcriptome of a named pair is usually absent from the dataset rather than merely under-powered. The reference line marks one observation. **(c)** The same quantity against panel size at fixed plex. Both levers enter quadratically, so on the arithmetic alone they are interchangeable; what separates them is that k is capped at two to four by array construction (section 7.1) while T is a free choice spanning a factor of thirty. Over that range T carries each curve across the line and k barely moves the crossing.

The right-hand column is the direct answer. Running a genome-scale library at the budget that powers main effects, a given gene pair is seen 0.017 times at $k = 2$ and 0.117 times at $k = 8$. Fewer than one pair in eight is observed even once, so the joint transcriptome of a named pair is not merely under-powered, it is usually absent from the dataset. Restrict to a 200-gene panel and the same budget delivers about one observation per pair at $k = 3$ and 3.5 at $k = 8$. The transition is not gradual, because T enters quadratically: dropping from 6,000 targets to 200 shrinks the pair space by a factor of 900 while shrinking the library by a factor of 30.

Panel size, not plex, is what makes combinations recoverable. Both levers, the guides per cell k and the panel size T, enter Eq. 3 quadratically, so on the arithmetic alone they are interchangeable. What separates them is that one is capped and the other is free. Raising k runs into a construction ceiling of two to four guides on an array (section 7.1) and into q^k, where q is the fraction of cells whose guides are correctly read (section 4.6), so its useful range spans a factor of a few. Choosing T costs nothing to build and spans a factor of thirty here. Restricting the panel is what makes named combinations measurable at all; multiplexing is what then makes them affordable. This repeats section 4.6's conclusion from the delivery side, and the two arguments are independent.

7.3 Environments as the Other Axis ✗

Perturbation and environment are independent, so a screen is a choice of subset of $\mathcal{L} \times \mathcal{E}$, and the two axes cost very differently. Environments multiply cell demand *linearly*: at 600,000 cells per condition, 4 environments need 2.4 million cells and 12 need 7.2 million. Compare that to the quadratic growth of named pairs, and the environment axis is by far the cheaper way to enlarge a dataset.

Labeling a condition is free wherever round 1 is a plate. A cell can be assigned to a condition only if something in the protocol records which condition it came from. Split-pool records it for nothing: round 1 is a 96-well plate, its index is read out of every cell whether or not it has been given a meaning, and every cell from one well shares it. Growing 96 conditions in a deep-well plate and barcoding them into the same run therefore costs no barcode space at all. Two cells collide when they share a *full* barcode, so pinning conditions to round-1 wells partitions the 96^R space rather than shrinking it, and with comparable cells per condition it does not change how uniformly the space is used. What a single-condition screen wastes is the sample label, not the wells; all 96 are loaded either way.

The same is true of preindexed droplet, and its plate is four times larger. The scifi round-1 plate has 384 wells and round1 is shared by every cell from one well (section 3.3), so conditions preindexed into different wells can be pooled into shared channels and separated afterwards by the well index.

96 and 384 are plate formats, not platform properties. Each is what its protocol runs, and either platform could run the other: preindexing on a 96-well plate and split-pool on a 384-well one are both buildable. The formats are worth keeping apart anyway, because the round-1 well does different work on the two. In split-pool it is one of three or four barcode rounds multiplied by the sublibrary index, so its size barely moves the collision rate. In scifi it is the only thing separating cells that share a droplet, which is what permits the 100-fold overloading of section 3.3: preindexing on 96 wells would take the barcode space of table 15 from 2.8×10^8 to 7.1×10^7 and, at the 12 million cells a genome-scale screen needs, its collision estimate from 4.2% to 15.6%, on a formula that already understates collisions in droplets for the reason that table gives. So the plate is a free choice in split-pool and a loaded one in scifi.

Unmodified droplet is the one platform with no such choice. Nothing in a channel says which condition a cell came from, so each condition needs its own channels, and channels are what section 5.5 shows dominates that platform's cost.

Why this is the axis that matters for model training. The purpose stated at the top of this section is a model that proposes edits. A model learns a map from genotype and environment to cell state, and a dataset that varies only genotype in one environment can only ever constrain the genotype half of it. Environments are also where metabolic engineering objectives actually live, since a production condition is an environment rather than a genotype. **Hypothesis (untested):** for a fixed cell budget, spending it on environments rather than on higher plex yields a model that extrapolates better to unseen conditions. Nothing here measures that, and it is stated because it is the assumption the program would be built on, not because it has been checked.

Combining the axes. Perturbations and environments can be crossed in one run, and the barcode arithmetic permits it: round 1 encodes the environment, rounds 2 and 3 encode the cell. The cost is interpretive rather than financial. A cell carrying an unknown combination of guides, grown in one of many environments, contributes to a main effect only after both its guide set and its environment are correctly assigned, so the assignment failures of section 3.6 compound with a second labeling problem. Crossing the axes is worth doing once each axis works alone.

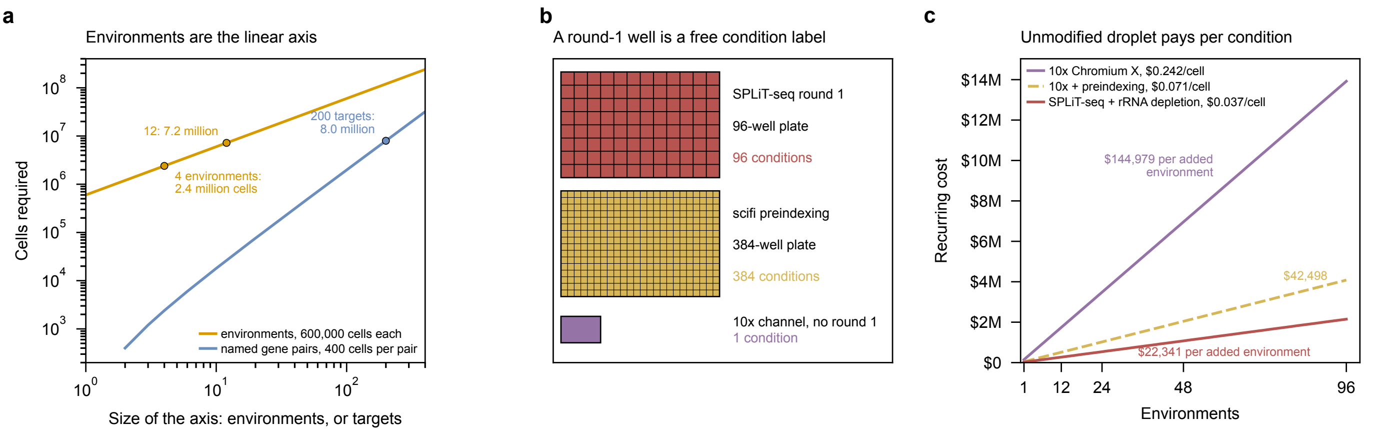

Figure 17: Environments enlarge a dataset linearly where named combinations do so quadratically, and any platform whose round 1 is a plate of wells labels them for nothing. (a) Cell demand against the size of each axis, on log axes so the exponent is the slope. Environments cost 600,000 cells each and rise with slope one: 4 environments need 2.4 million cells and 12 need 7.2 million. Covering every named pair of a T-target panel rises with slope two, so a 200-target panel already costs about as much as 12 environments and every further target costs more than the last. **(b)** Condition labels each platform already owns. The two plates are drawn on one footprint because a 96-well and a 384-well plate have the same one; every well is filled because a round-1 plate is loaded in full whatever the experiment is, round 1 being a barcode round before it is a sample round. What a single-condition screen leaves unused is the label the well index also carries. The two well counts are the formats these protocols run rather than properties that separate them, and either platform could run the other; what makes 384 worth keeping in scifi is that its round-1 well is also the only thing separating cells that share a droplet. An unmodified droplet channel carries no such index, so the only thing that can separate two conditions is running them in separate channels. **(c)** Recurring cost against environments, on the section 5.5 cost model at 600,000 usable cells per condition. All three are linear, so the claim is about slope. Unmodified droplet buys one set of channels per condition. The other two pool conditions through a round-1 well index, so what remains between them is cost per cell: an environment is the same 600,000 extra cells on every platform, and dividing each slope by that count returns the loaded per-cell figures of table 12. The key gives each of them, in dollars per usable cell. Split-pool keeps the environment axis at half the price of preindexed droplet for the reason it is cheaper per cell in section 5, not for a reason of its own.

Environments are the cheap axis, and both platforms with a round-1 plate label them for free. Cell demand grows linearly in environments and quadratically in named gene pairs, so for a fixed budget the environment axis enlarges a dataset far faster. Split-pool carries 96 condition labels in the round-1 plate it runs and scifi preindexing carries 384 in the one it runs, and in neither case does using them cost barcode space: the well index is read out of every cell already, and assigning conditions to wells partitions that space rather than shrinking it. Unmodified droplet has no such index, so every condition buys its own channels, which is the term that dominates that platform's cost. What is *not* free anywhere is the cells, and that is what still separates the two plate platforms: an added environment is 600,000 more cells whatever runs it, so its price is that count times what a cell costs, $0.037 on depleted split-pool against $0.071 on preindexed droplet, or $22,341 against $42,498 per environment. The environment ranking is the section 5 per-cell ranking restated, with nothing added.

7.4 What to Build First ✗

The ordering follows from the three subsections above and from the state of the host organism.

1. **Guide capture**, unchanged from section 6. Every statement in this section is conditional on reading k guides out of one cell, and q^k punishes failure exponentially.
2. **The per-cell guide-count distribution under route B.** One experiment, and it is the number the literature does not have. It also settles whether route B is a real alternative to cloned arrays or a curiosity.
3. **A 200-gene metabolic panel at $k \approx 3$**, which table 18 shows is the smallest design in which named pairs are observed at all.
4. **Environments**, added to whichever perturbation design works, since they are linear and a round-1 plate already carries the labels they need.

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Appendix: Provenance of Hand-Drawn Figure Numbers ✗

Table 19: Provenance for Fig. 1, the yeast scRNA-seq method taxonomy: every number drawn by hand on that canvas, and the sentence it comes from. A **note** marks a number that needed a judgment call or that was corrected in review; an entry with no citation key is not backed by the library mirror and should not be quoted. Generated from `experiments/024-perturb-seq-costing/scripts/figure_sources.py`.

Element	As drawn	Source
scrnaseq-method-taxonomy.drawio		
Microfluidic chip (C1) throughput	`up to 96 per chip`	“It uses an integrated fluidic circuit to isolate and process up to 96 cells at once.” [8] *Note:* Was drawn as ~160 cells, which is Gasch et al.’s 163-cell study total across four chip runs – a study total on a per-experiment axis. The capacity of one chip is the comparable quantity.
FACS into plates throughput	`~285 cells`	“we applied yscRNA-seq to 285 individual yeast cells” [17]
Microdissection throughput	`~2,000 cells`	“Microdissection coupled with imaging has been used to profile the transcriptomes of Schizosaccharomyces pombe across a variety of environmental stressors and S. cerevisiae during aging (Saint et al., 2019; Wang et al., 2022)” ▲*Note:* NOT SOURCED TO A NUMBER. The review names the studies but states no cell count, and neither primary is in the mirror. The ~2,000 is an order-of-magnitude placeholder and is the one number in this figure that should not be quoted.
Droplet throughput range	`6,000-100,000`	“applied it to over 100,000 single cells from three crosses [upper bound; lower bound is Jariani et al.’s 6,118 cells]” [21]
Microwell array throughput	`up to 1,061,865`	“we used a microwell-based platform for single-cell isolation … We profiled a total of 1.061.865 cells” [5] *Note:* Was ~75,000, taken from mDrop-seq – which is a droplet method, not a microwell one. Replaced with the microwell study that exists.
Combinatorial indexing throughput	`25,000-240,000 per run`	“Reverse transcription (in situ) was performed in 25 uL reactions in 48 wells of a 100 uL 96-well plate … 200,000 cells/mL [= ~5,000 cells x 48 wells = ~240,000 barcoded per run; lower bound is Kuchina et al.’s 25,214-cell combined dataset]” [3] *Note:* The upper bound was 1,000,000, which is Gaisser et al.’s protocol capacity claim ("enables … up to 1 million"), not a profiled count. Same correction as Fig. 4.
preindexed droplet throughput	`152,000 preindexed (human; Fig. 5)`	“we performed a large-scale scifi-RNA-seq experiment with 383,000 nuclei loaded into a single microfluidic channel of the Chromium system … This experiment resulted in 151,788 single-cell transcriptomes passing quality control” [6] `(md:57)` *Note:* Sits under the droplet column rather than in a seventh box, because preindexing is a modifier of droplet capture and not a separate way of keeping cells apart. Rounded to 152,000 on the canvas, where every other entry is rounded too. The organism caveat is IN the label rather than in a footnote: this is a figure of yeast methods and the one number in it that is not from a microbe has to say so even when the panel is screenshotted out of the document.

Table 21: Provenance for Fig. 2, split-pool barcoding: every number drawn by hand on that canvas, and the sentence it comes from. A **note** marks a number that needed a judgment call or that was corrected in review; an entry with no citation key is not backed by the library mirror and should not be quoted. Generated from `experiments/024-perturb-seq-costing/scripts/figure_sources.py`.

Element	As drawn	Source
splitseq-barcoding.drawio		
round-1 wells and cells per well	`48 wells, ~5,000 cells each`	“Reverse transcription (in situ) was performed in 25 uL reactions in 48 wells of a 100 uL 96-well plate with each well containing wellspecific, barcoded primers … 200,000 cells/mL” [3]
barcode space over three rounds	`963 = 884,736`	“Cells are then pooled and randomly split into a new 96-well plate, and a well-specific barcode is attached” [3] *Note:* Protocol capacity. The published run loaded 48 wells in round 1, so its realized space was 48x96x96 = 442,368 (Sec. 3.1).
sublibrary size	`5,000-20,000 cells`	“In experiment 3, the barcoded cells were evenly divided into 10 sublibraries. The two sequenced sublibraries returned ~5,500 and 10,000 barcoded cells that passed computational filtering.” [3]
rRNA fraction of recovered transcripts	`93.7%`	“On average, the transcript proportions we recover are 93.7% rRNA, 0.005% tRNA, 0.04% ncRNA, and 5.75% mRNA.” [3] `(md:70)`

Table 23: Provenance for Fig. 4, droplet barcoding: every number drawn by hand on that canvas, and the sentence it comes from. A **note** marks a number that needed a judgment call or that was corrected in review; an entry with no citation key is not backed by the library mirror and should not be quoted. Generated from `experiments/024-perturb-seq-costing/scripts/figure_sources.py`.

Element	As drawn	Source
droplet-10x-barcoding.drawio		
doublet rate at 20,000 cells	`~8% at 20,000 recovered`	“[10x Chromium specification for the GEM-X/v3.1 chemistry priced in Sec. 5; the protocol paper used 3’ v2]” [10] *Note:* MANUFACTURER SPECIFICATION, not a measurement from a mirrored paper. Verify against the current 10x user guide before quoting.

Table 23, continued

Element	As drawn	Source
digestion timing and cell-count drop	`6 min hold, 200-fold drop within 20 min at 53 C`	"cell counts hold on ice and through the 6 min of droplet generation, then drop 200-fold within 20 min at 53 C" [10]
bead barcode and UMI lengths	`16 nt CB, 12 nt UMI, 118 cycles`	"[10x 3' v3/GEM-X read layout; cycle count derived in experiments/024-perturb-seq-costing/scripts/read_structure.py]" [10] *Note:* Layout is the current GEM-X/v3.1 chemistry, not the v2 Jariani ran.

Table 25: Provenance for Fig. 5, combinatorial fluidic indexing: every number drawn by hand on that canvas, and the sentence it comes from. A **note** marks a number that needed a judgment call or that was corrected in review; an entry with no citation key is not backed by the library mirror and should not be quoted. Generated from **`experiments/024-perturb-seq-costing/scripts/figure_sources.py`**.

Element	As drawn	Source
scifi-fluidic-indexing.drawio		
maximum recommended load	`15,300 nuclei per channel`	"We first assessed the maximum recommended loading concentration (15,300 nuclei per microfluidic channel)." [6] `(md:26)`
occupancy at recommended load	`16.4% occupied; mean 0.2 nuclei per droplet`	"Counting 609 droplet images, we found that only 16.4% of droplets contained one or more nuclei (mean number of nuclei per droplet: 0.2)." [6] `(md:26)` *Note:* Measured by counting 609 droplet images, not inferred from Poisson.
overloaded condition	`100x overloaded, 1.53 million nuclei per channel`	"Remarkably, even 100-fold overloading (1.53 million nuclei per channel) resulted in a stable droplet emulsion and did not clog the microfluidic system" [6] `(md:26)`
occupancy when overloaded	`95.5% fill; mean 9.6 nuclei per droplet`	"up to a droplet fill rate of 95.5% and an average of 9.6 nuclei per droplet (1.53 million nuclei per channel), with highly consistent droplet diameter" [6] `(md:26)` *Note:* The droplet cartoons in panel a show three nuclei, not 9.6; they are schematic and the canvas says so. The number is the measured mean.
round1 plate format	`one 384-well plate`	"permeabilized cells or nuclei are first preindexed with barcoded oligo-dT primers by reverse transcription on multiwell plates (we use one 384-well plate with well-specific primers)" [6] `(md:34)`
round2 barcode space	`737,280 round2 barcodes`	"Using the 737,280 distinct microfluidic (round2) barcodes provided by the Chromium ATAC reagents" [6] `(md:30)`
resolvable transcriptomes at 96 round1 wells	`1 million from 96 round1 wells`	"our analyses indicated that scifi-RNA-seq can resolve 1 million single-cell transcriptomes already with 96 round1 indices" [6] `(md:30)` *Note:* MODELED, not observed: it comes from a zero-inflated Poisson model plus Monte Carlo simulation of barcode collisions. The canvas says 'a modeled bound, not a measurement' for exactly this reason.
large-run yield, left caption	`383,000 loaded; 151,788 recovered`	"we performed a large-scale scifi-RNA-seq experiment with 383,000 nuclei loaded into a single microfluidic channel of the Chromium system ... This experiment resulted in 151,788 single-cell transcriptomes passing quality control" [6] `(md:57)` *Note:* Deliberately separated from the occupancy numbers beside it. Those were counted under a microscope at a different loading concentration; this is a sequencing yield, and it is the only scifi number the cost model consumes.
channel saving carried into the budget	`7.6x more cells, so 137 channels become 18`	"[DERIVED HERE, no source sentence: 151,788 recovered per channel / 20,000 at the UIUC baseline = 7.6x; channel counts from cost_model.py, ScreenDesign(cells_per_gene=250)]" ▲*Note:* OURS, not the paper's. 151,788 recovered per channel over the UIUC baseline of 20,000 is 7.6x; the channel counts come straight from cost_model.py at 250 cells per target gene. Datlinger et al.'s own headline is 15-fold, against a ~10,000-cell standard Chromium run rather than against UIUC's more generous 20,000, so the figure drawn here is the conservative one. Recompute if the rate card's cells-per-channel changes.
round2 attaches by ligation, not RT	`round2 BC (ligated)`	"oligonucleotides carrying the microfluidic (round2) barcode (delivered via Chromium gel beads) are ligated to the cDNA via the 5'-phosphate group of the reverse transcription primer, directed by a complementary 3'-blocked bridge oligonucleotide" [6] `(md:34)`

Table 27: Provenance for Fig. 6, the end-to-end workflow: every number drawn by hand on that canvas, and the sentence it comes from. A **note** marks a number that needed a judgment call or that was corrected in review; an entry with no citation key is not backed by the library mirror and should not be quoted. Generated from **`experiments/024-perturb-seq-costing/scripts/figure_sources.py`**.

Element	As drawn	Source
perturb-seq-workflow.drawio		
cells recovered from those loaded	`25-50% recovered`	"The two sequenced sublibraries returned ~5,500 and 10,000 barcoded cells that passed computational filtering. Given we started the experiment with between 200,000 and 300,000 cells, this estimates a capturing efficiency between 25% and 50%." [3] `(md:68)` *Note:* A CELL-level efficiency, not a read-level one, and it is the authors' own back-calculation from a starting count rather than a directly measured recovery.
sequenced cells discarded at QC	`~75% of sequenced cells discarded`	"[DERIVED HERE: cost_per_cell_table.py, the ratio of cells passing QC with an identified guide to cells whose reads were paid for]" ▲*Note:* OURS, not a published figure. It is the gap between 'sequenced' and 'usable' that Sec. 5.2 is built on, and it is the reason a quoted cost per cell understates the real one by ~4x in split-pool.

Table 27, continued

Element	As drawn	Source
guide assignment rate q	`21-25% in bacteria, >71% in yeast`	"[Yeast end: genotype identity assigned for more than 71% of cells. Bacterial end: Brandner et al. raise correct assignment from 5.5% to 21-25% by adding an enrichment primer at cDNA amplification.]" [5] *Note:* TWO SOURCES IN ONE ROW, deliberately: the range is the span between two different organisms and two different constructs, not a measurement with an uncertainty. Sec. 4.5 shows it enters the multiplex ceiling as q^k, which is why the span matters so much.
reads surviving all four tolls	`~1 read in 100; ~11 with depletion`	"[DERIVED HERE: plot_economics.py panel d, composing the PhiX/barcode, rRNA and cell-QC fractions]" ▲*Note:* OURS. The rRNA term is Brettner et al.'s 93.7%; the cell-QC term is the row above. Composing them is this document's arithmetic, and it is the number that makes depletion the largest single lever.
reads lost to spike-in and barcode	`28 lost to the PhiX spike and to unresolved barcodes`	"[DERIVED HERE: cost_model.py, phix_fraction=0.10 and valid_barcode_fraction=0.80, i.e. usable_read_fraction=0.72]" ▲*Note:* OURS. The canvas used to say 10, which was the PhiX spike alone; the model applies a 10% spike AND a separate 80% valid-barcode rate, so usable_read_fraction is 0.72 and 28 reads in 100 are lost at this stage. The 1-in-100 survival figure was always computed from 0.72 and was right; only the drawn breakdown was wrong.
ribosomal share of remaining reads	`94 of the remainder are ribosomal`	"On average, the transcript proportions we recover are 93.7% rRNA, 0.005% tRNA, 0.04% ncRNA, and 5.75% mRNA." [3] `(md:70)` *Note:* Drawn rounded to 94 from the sourced 93.7%.
usable fragments, bulk vs single cell	`6 and 2 usable in bulk; 1 and 1 in single cell`	"[SCHEMATIC, not measured: the cut positions are drawn to illustrate that reads scale with length in bulk and do not in a 3' assay]" ▲*Note:* The only numbers on this canvas that are illustrative rather than sourced. The RATIO is the claim – six against two is the 3 kb against 1 kb case in Sec. 2.4 – and the cut count itself is not a measurement of anything.
value of rRNA depletion, drawn on canvas	`worth ~$99k (Fig. 7)`	"[DERIVED HERE: tables/t8-budgets.tex, the 250-cells-per-gene split-pool rows, $156,670 undepleted less $57,510 depleted = $99,160]" ▲*Note:* OURS. A difference of two budget rows rather than a figure any source states, and it is the number the whole rRNA argument turns on, so it is recorded here as well as in the text.

Table 29: Provenance for Fig. 7, ribosomal RNA depletion options: every number drawn by hand on that canvas, and the sentence it comes from. A **note** marks a number that needed a judgment call or that was corrected in review; an entry with no citation key is not backed by the library mirror and should not be quoted. Generated from `experiments/024-perturb-seq-costing/scripts/figure_sources.py`.

Element	As drawn	Source
rrna-depletion-options.drawio		
workflow step numbers	`fragment 134-139, ligate 152-154, index 156-157`	"The library then undergoes fragmentation and adapter ligation (Part 2, Steps 134-139 and 152-154). ... Illumina P5 and P7 sequence adapters and a final sub-library index are also appended at this final PCR step (Part 2, Step 157). e, A 0.5-0.7x double size selection then selects out unwanted fragments" [7] `(md:82)` *Note:* This is the correction. An earlier version of Sec. 3.5 placed depletion BEFORE fragmentation and indexing and argued it had to be there; the protocol says otherwise, and Brandner et al.'s stated input is the post-size-selection library. One precision caveat: the quote names step 157 for the index PCR, and 156 comes from the numbered protocol body rather than from this sentence.
Cas9 route, guide pool and conditions	`253 sgRNAs`	"We designed 253 sgRNA spacer sequences to target the 5S, 16S, and 23S rRNA sequences in E. coli and P. putida, weighted proportionally to the regions in the rRNA that appeared more frequently during NGS after microSPLiT. ... 100 pmoles of the transcribed, column-purified sgRNA molecules were pre-incubated with 10 pmoles spCas9 (NEB) ... then incubated at 37 C for 2 hours." [4] `(md:151)`
Cas9 route, measured outcome	`4.6% -> 60.2%; 2.2% -> 10.4%`	"With rRNA depletion, the mRNA fraction increased from 4.6% to 60.2% (Fig. 2B) and median mRNA transcripts per cell more than doubled, from 54 to 115 [E. coli]; ... the mRNA fraction increased from 2.2% to 10.4% [P. putida, md:119]" [4] `(md:71)` *Note:* Two organisms, one protocol, and the gap between them is the transfer risk the figure is drawing attention to. Both figures are proportions of ALIGNED transcripts computed before single-cell filtering, not of reads purchased.
RNase H chosen over Cas9	`75-80% rRNA left by the Cas9 arm`	"after testing two approaches for depleting ribosomal sequences from bulk libraries (Extended Data Fig. 3a-c), we chose an RNase H-based approach to complete our pipeline [and, md:92:] this is consistent with our trial runs using Cas9-based rRNA depletion (75-80% rRNA in the final library, Extended Data Fig. 3a)" [14] `(md:23)` *Note:* The head-to-head. An earlier version of Sec. 3.5 cited M3-seq as corroborating the Cas9 route; it tested both and chose the other one.
in-situ route, best published result	`~2.5-fold, ~7% of total RNA`	"we then tested polyadenylation with E. coli poly(A) polymerase I (PAP) ...; 5'-phosphate-dependent exonuclease ("Terminator", Epicentre); and RT with ribosomal RNA-specific probes followed by RNaseH mediated degradation ... We found that the treatment of fixed and permeabilized cells with PAP resulted in the highest (about 2.5-fold, or about 7% of total RNA) enrichment of mRNA reads" [9] `(md:36)`
why M3-seq avoids in-situ depletion	`in situ can decrease mRNA capture`	"we noted that depletion of rRNA in situ can decrease mRNA capture efficiency and thus focused on depleting rRNAs after amplification" [14] `(md:23)`
undepleted yeast transcript proportions	`93.7% rRNA, 5.75% mRNA`	"On average, the transcript proportions we recover are 93.7% rRNA, 0.005% tRNA, 0.04% ncRNA, and 5.75% mRNA." [3] `(md:70)` *Note:* Computed with multi-mapping reads RETAINED (STARsolo, 'the most basic uniform multi-mapping algorithm', md:195), which is the convention that inflates apparent rRNA. That makes it comparable to microSPLiT's numbers and NOT to any unique-only figure – see the reporting-convention paragraph in Sec. 3.5.
reads spent on rRNA, drawn on canvas	`roughly 94 of every 100 reads`	"On average, the transcript proportions we recover are 93.7% rRNA, 0.005% tRNA, 0.04% ncRNA, and 5.75% mRNA." [3] `(md:70)` *Note:* Drawn rounded to 94 from the sourced 93.7%.

Table 29, continued

Element	As drawn	Source
post-hoc against in-situ enrichment	`against 13-fold post hoc`	“the mRNA fraction increased from 4.6% to 60.2%” [4] `(md:71)` *Note:* 13-fold is 60.2/4.6, computed here rather than stated by the source, and it is set against microSPLiT’s 2.5-fold in-situ result to make the point that the two routes are not close.
rRNA share of yeast RNA	`about 85%`	“Total RNA per cell (~85% rRNA) 0.7-1 pg” [8] `(md:49)`